MATH 1111: Differential and Integral Calculus

Lecture Notes 1: Functions, Limits, and Continuity

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1 Foundations of Functions

1.1 Learning Outcomes

Upon successful completion of this section, students will be able to:

- **Define** a function, its domain, and its range with precision.
- **Determine** the domain and range of various elementary functions (polynomial, rational, radical, etc.).
- **Understand** the concept of a one-to-one function and apply the Horizontal Line Test.
- **Calculate** the inverse of a given one-to-one function and understand its graphical relationship to the original function.
- **Perform** the composition of functions and determine the domain of the resulting composite function.
- **Apply** these concepts to model basic engineering systems and interpret their graphical representations.

1.2 Functions: Domain, Range, and Graphs

Definition 1.1 (Function)**.** A **function** f is a rule that assigns to each element x in a set A exactly one element, called $f(x)$, in a set B. The set A is called the **domain** of the function, and the set of all possible values of $f(x)$ as x varies throughout the domain is called the **range** of the function.

In simpler terms, for every input, there is a unique output.

- **Domain**: The set of all possible input values (x).
- **Range**: The set of all possible output values ($f(x)$).

Example 1.1 (Polynomial Function)**.** Let $f(x) = x^2 + 2x - 3$.

- **Domain**: We can plug any real number into this function. So, the domain is $(-\infty, \infty)$ or $\mathbb{R}$.
- **Range**: This is a parabola opening upwards. The vertex occurs at $x = -b/(2a) = -2/2 = -1$. The value at the vertex is $f(-1) = (-1)^2 + 2(-1) - 3 = 1 - 2 - 3 = -4$. Thus, the range is $[-4, \infty)$.

Example 1.2 (Rational Function)**.** Let $g(x) = \frac{x+1}{x-2}$.

- **Domain**: The function is undefined when the denominator is zero, i.e., $x - 2 = 0 \implies x = 2$. So, the domain is all real numbers except 2, written as $(-\infty, 2) \cup (2, \infty)$ or $\mathbb{R} \setminus \{2\}$.
- **Range**: To find the range, let $y = g(x)$ and solve for x. $y = \frac{x+1}{x-2} \implies y(x-2) = x+1 \implies yx - 2y = x+1 \implies yx - x = 2y+1 \implies x(y-1) = 2y+1 \implies x = \frac{2y+1}{y-1}$. The expression for x is undefined if $y = 1$. Thus, the range is $\mathbb{R} \setminus \{1\}$.

Example 1.3 (Radical Function)**.** Let $h(x) = \sqrt{x-3}$.

- **Domain**: The expression inside a square root must be non-negative. $x-3 \geq 0 \implies x \geq 3$. The domain is $[3, \infty)$.
- **Range**: The output of a square root function is always non-negative. So, the range is $[0, \infty)$.

Example 1.4 (Logarithmic Function)**.** Let $k(x) = \ln(x+5)$.

- **Domain**: The argument of a logarithm must be strictly positive. $x+5 > 0 \implies x > -5$. The domain is $(-5, \infty)$.
- **Range**: The range of any logarithmic function is all real numbers, $(-\infty, \infty)$.

Example 1.5 (Engineering Application)**.** The voltage V across a capacitor in a simple RC circuit as it charges is given by $V(t) = V_0(1 - e^{-t/RC})$, where t is time, V_0 is the source voltage, R is resistance, and C is capacitance.

- **Domain**: Time t cannot be negative, so the physical domain is $[0, \infty)$.
- **Range**: At $t = 0$, $V(0) = 0$. As $t \to \infty$, $e^{-t/RC} \to 0$, so $V(t) \to V_0$. The range of voltages is $[0, V_0)$.

1.2.1 Figure Illustration: Graph of a Function

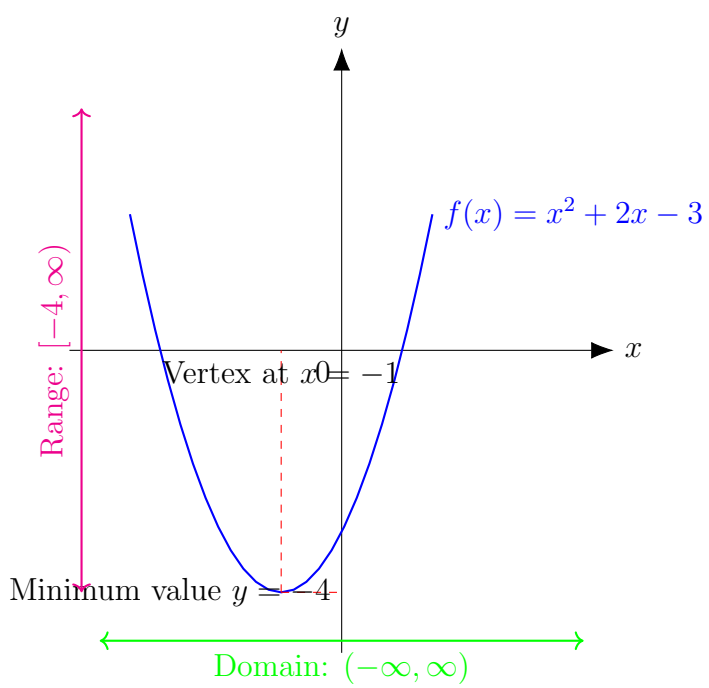


Figure 1: Graph of the function $f(x) = x^2 + 2x - 3$, illustrating its domain and range.

Description of Figure 1: The figure shows a parabola opening upwards. The **domain** is represented by the full extent of the x-axis, as the function is defined for all real numbers. The **range** is visually represented by the vertical extent of the graph, which starts at the minimum value of $y = -4$ and goes up to infinity.

1.2.2 Practice Problems

Problem 1.1. Find the domain of the function $f(x) = \frac{\sqrt{x+2}}{x-5}$.

Solution 1.1.1. We have two conditions to satisfy:

1. The term inside the square root must be non-negative: $x + 2 \geq 0 \implies x \geq -2$.
2. The denominator cannot be zero: $x - 5 \neq 0 \implies x \neq 5$.

Combining these, the domain is all numbers greater than or equal to -2, except for 5. In interval notation: $[-2, 5) \cup (5, \infty)$.

Problem 1.2. Find the domain and range of $g(x) = 5 - \sqrt{x-1}$.

Solution 1.2.1. Domain: We need $x - 1 \geq 0 \implies x \geq 1$. So, the domain is $[1, \infty)$.
Range: The term $\sqrt{x-1}$ is always ≥ 0. Therefore, $-\sqrt{x-1}$ is always ≤ 0. Adding 5 to this, we get $5 - \sqrt{x-1} \leq 5$. Thus, the range is $(-\infty, 5]$.

Problem 1.3. The power P (in Watts) dissipated in a resistor is given by $P(I) = I^2R$, where I is the current in Amperes and R is a constant resistance. If the current can vary from 0A to 5A, what is the domain and range of this function? Assume $R = 10\Omega$.

Solution 1.3.1. The function is $P(I) = 10I^2$.
Domain: The current I is restricted to $[0, 5]$. This is our domain.
Range: Since $P(I)$ is an increasing function for $I \geq 0$, we find the range by evaluating the endpoints of the domain. $P(0) = 10(0)^2 = 0$ W. $P(5) = 10(5)^2 = 250$ W. The range is $[0, 250]$.

Problem 1.4. Find the domain of $h(x) = \ln(x^2 - 4)$.

Solution 1.4.1. The argument of the natural logarithm must be strictly positive. $x^2 - 4 > 0 \implies (x-2)(x+2) > 0$. This inequality holds when both factors are positive or both are negative. Case 1: $x - 2 > 0$ and $x + 2 > 0 \implies x > 2$ and $x > -2 \implies x > 2$. Case 2: $x - 2 < 0$ and $x + 2 < 0 \implies x < 2$ and $x < -2 \implies x < -2$. The domain is $(-\infty, -2) \cup (2, \infty)$.

Problem 1.5. Find the domain of $k(x) = \frac{1}{\sqrt[4]{9-x^2}}$.

Solution 1.5.1. We need the term inside the fourth root to be strictly positive (since it's also in the denominator). $9 - x^2 > 0 \implies x^2 < 9 \implies -3 < x < 3$. The domain is $(-3, 3)$.

1.3 Inverse Functions

Definition 1.2 (One-to-One Function)**.** A function f is **one-to-one** (or injective) if it never takes on the same value twice; that is, $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$.

Theorem 1.1 (Horizontal Line Test)**.** *A function is one-to-one if and only if no horizontal line intersects its graph more than once.*

Proof. Let f be a function. ($\Rightarrow$) Assume f is one-to-one. Let's consider a horizontal line $y = c$. If this line intersects the graph of f at two distinct points (x_1, c) and (x_2, c), then $f(x_1) = c$ and $f(x_2) = c$ with $x_1 \neq x_2$. This contradicts the definition of a one-to-one function. Thus, any horizontal line can intersect the graph at most once. ($\Leftarrow$) Assume no horizontal line intersects the graph of f more than once. To show f is one-to-one, we must show that $f(x_1) = f(x_2)$ implies $x_1 = x_2$. If $f(x_1) = f(x_2) = c$, this means the points (x_1, c) and (x_2, c) are on the graph of f and on the horizontal line $y = c$. By our assumption, this line intersects the graph at most once, so the points must be identical, which means $x_1 = x_2$. Therefore, f is one-to-one. $\square$

Definition 1.3 (Inverse Function)**.** Let f be a one-to-one function with domain A and range B. Then its **inverse function** f^{-1} has domain B and range A and is defined by

$$f^{-1}(y) = x \iff f(x) = y$$

for any y in B.

Key Property: $f^{-1}(f(x)) = x$ for every x in A, and $f(f^{-1}(y)) = y$ for every y in B.

1.3.1 How to Find the Inverse of a Function

1. Write $y = f(x)$.
2. Swap the variables x and y. This gives $x = f(y)$.
3. Solve this new equation for y. The resulting expression is $f^{-1}(x)$.

Example 1.6. Find the inverse of $f(x) = 2x + 3$.

1. $y = 2x + 3$.
2. Swap variables: $x = 2y + 3$.
3. Solve for y: $x - 3 = 2y \implies y = \frac{x-3}{2}$.

So, $f^{-1}(x) = \frac{x-3}{2}$.

Example 1.7. Find the inverse of $g(x) = x^3 - 5$.

1. $y = x^3 - 5$.
2. Swap variables: $x = y^3 - 5$.
3. Solve for y: $x + 5 = y^3 \implies y = \sqrt[3]{x+5}$.

So, $g^{-1}(x) = \sqrt[3]{x+5}$.

Example 1.8 (Restricted Domain)**.** The function $h(x) = x^2$ is not one-to-one on $\mathbb{R}$. However, if we restrict its domain to $[0, \infty)$, it is one-to-one. Find its inverse.

1. $y = x^2$ for $x \geq 0$. The range is $[0, \infty)$.
2. Swap variables: $x = y^2$ for $y \geq 0$.
3. Solve for y: $y = \pm\sqrt{x}$. Since we know $y \geq 0$, we must choose the positive root. So, $y = \sqrt{x}$.

Thus, $h^{-1}(x) = \sqrt{x}$. The domain of h^{-1} is the range of h, which is $[0, \infty)$.

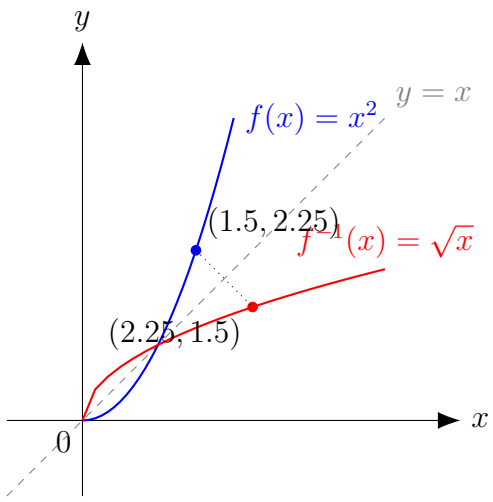


Figure 2: The graph of $f^{-1}(x) = \sqrt{x}$ (red) is a reflection of the graph of $f(x) = x^2$ for $x \geq 0$ (blue) about the line $y = x$.

1.3.2 Figure Illustration: Graph of an Inverse Function

The graph of f^{-1} is obtained by reflecting the graph of f across the line $y = x$.

1.3.3 Practice Problems

Problem 1.6. Find the inverse of the function $f(x) = \frac{4x-1}{2x+3}$.

Solution 1.6.1. 1. $y = \frac{4x-1}{2x+3}$ 2. Swap variables: $x = \frac{4y-1}{2y+3}$ 3. Solve for y:

$$\begin{aligned} x(2y+3) &= 4y-1 \\ 2xy+3x &= 4y-1 \\ 3x+1 &= 4y-2xy \\ 3x+1 &= y(4-2x) \\ y &= \frac{3x+1}{4-2x} \end{aligned}$$

So, $f^{-1}(x) = \frac{3x+1}{4-2x}$.

Problem 1.7. Let $C = \frac{5}{9}(F-32)$ be the function that converts Fahrenheit (F) to Celsius (C). Find the inverse function and explain what it represents.

Solution 1.7.1. This function gives Celsius as a function of Fahrenheit, $C(F)$. The inverse function, $F(C)$, will give Fahrenheit as a function of Celsius. 1. $C = \frac{5}{9}(F-32)$ 2. (We don't swap variables here because they have physical meaning. We just solve for F). 3. Solve for F:

$$\begin{aligned} \frac{9}{5}C &= F-32 \\ F &= \frac{9}{5}C+32 \end{aligned}$$

The inverse function is $F(C) = \frac{9}{5}C + 32$, which is the formula to convert Celsius to Fahrenheit.

Problem 1.8. Find the inverse of $g(x) = e^{2x-1}$.

Solution 1.8.1. 1. $y = e^{2x-1}$ 2. Swap variables: $x = e^{2y-1}$ 3. Solve for y by taking the natural logarithm of both sides:

$$\begin{aligned} \ln(x) &= \ln(e^{2y-1}) \\ \ln(x) &= 2y - 1 \\ \ln(x) + 1 &= 2y \\ y &= \frac{\ln(x) + 1}{2} \end{aligned}$$

So, $g^{-1}(x) = \frac{1+\ln(x)}{2}$. The domain of $g^{-1}(x)$ is $(0, \infty)$, which is the range of $g(x)$.

Problem 1.9. If $f(x)$ has domain $[0, 10]$ and range $[2, 5]$, what are the domain and range of $f^{-1}(x)$?

Solution 1.9.1. The domain of the inverse function is the range of the original function, and vice versa. Domain of $f^{-1}(x)$ = Range of $f(x) = [2, 5]$. Range of $f^{-1}(x)$ = Domain of $f(x) = [0, 10]$.

Problem 1.10. Find the inverse of $f(x) = \frac{x+5}{x-2}$ and verify that $f(f^{-1}(x)) = x$.

Solution 1.10.1. First, find the inverse. 1. $y = \frac{x+5}{x-2}$ 2. $x = \frac{y+5}{y-2}$ 3. Solve for y:

$$\begin{aligned} x(y-2) &= y + 5 \\ xy - 2x &= y + 5 \\ xy - y &= 2x + 5 \\ y(x-1) &= 2x + 5 \\ y &= \frac{2x+5}{x-1} \end{aligned}$$

So, $f^{-1}(x) = \frac{2x+5}{x-1}$. Now, verify the composition:

$$\begin{aligned} f(f^{-1}(x)) &= f\left(\frac{2x+5}{x-1}\right) \\ &= \frac{\left(\frac{2x+5}{x-1}\right) + 5}{\left(\frac{2x+5}{x-1}\right) - 2} \\ &= \frac{(2x+5) + 5(x-1)}{(2x+5) - 2(x-1)} \quad \text{(multiplying top and bottom by } x-1\text{)} \\ &= \frac{2x + 5 + 5x - 5}{2x + 5 - 2x + 2} \\ &= \frac{7x}{7} = x \end{aligned}$$

The verification is successful.

1.4 Composition of Functions

Definition 1.4 (Composition of Functions)**.** Given two functions f and g, the **composite function** $f \circ g$ (read as "f composed with g") is defined by

$$(f \circ g)(x) = f(g(x))$$

The domain of $f \circ g$ is the set of all x in the domain of g such that $g(x)$ is in the domain of f.

Example 1.9. Let $f(x) = x^2$ and $g(x) = x - 3$.

- $(f \circ g)(x) = f(g(x)) = f(x-3) = (x-3)^2$. The domain is $\mathbb{R}$.
- $(g \circ f)(x) = g(f(x)) = g(x^2) = x^2 - 3$. The domain is $\mathbb{R}$.

Note that in general, $f \circ g \neq g \circ f$.

Example 1.10. Let $f(x) = \sqrt{x}$ and $g(x) = x - 4$.

- $(f \circ g)(x) = f(g(x)) = f(x-4) = \sqrt{x-4}$. For the domain, we need x in the domain of g (which is $\mathbb{R}$) and $g(x)$ in the domain of f. The domain of f is $[0, \infty)$, so we need $g(x) \geq 0 \implies x - 4 \geq 0 \implies x \geq 4$. The domain is $[4, \infty)$.
- $(g \circ f)(x) = g(f(x)) = g(\sqrt{x}) = \sqrt{x} - 4$. For the domain, we need x in the domain of f, which is $[0, \infty)$. Then $f(x) = \sqrt{x}$ is always a real number, which is in the domain of g. So the domain is $[0, \infty)$.

Example 1.11 (Engineering Application)**.** The electrical power P in a circuit is $P(V) = V^2/R$. The voltage V across a component varies with time t as $V(t) = 5\cos(t)$. Find the power as a function of time, $P(t)$. This is a composition: $P(t) = (P \circ V)(t) = P(V(t))$.

$$P(t) = \frac{(V(t))^2}{R} = \frac{(5\cos(t))^2}{R} = \frac{25\cos^2(t)}{R}$$

This function gives the instantaneous power at any time t.

1.4.1 Practice Problems

Problem 1.11. Let $f(x) = \frac{1}{x}$ and $g(x) = x^2 + 1$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$ and their domains.

Solution 1.11.1. $(f \circ g)(x) = f(g(x)) = f(x^2+1) = \frac{1}{x^2+1}$. Domain: $g(x)$ is defined for all $x \in \mathbb{R}$. The domain of f is $x \neq 0$. We need $g(x) \neq 0$. Since $x^2 + 1 \geq 1$ for all real x, $g(x)$ is never zero. So the domain is $\mathbb{R}$.

$(g \circ f)(x) = g(f(x)) = g(\frac{1}{x}) = (\frac{1}{x})^2 + 1 = \frac{1}{x^2} + 1$. Domain: The domain of $f(x)$ is $x \neq 0$. The output $1/x$ can be any real number except 0, which is fine as an input for $g(x)$. So the domain is $x \neq 0$, or $(-\infty, 0) \cup (0, \infty)$.

Problem 1.12. Let $f(x) = \sqrt{2-x}$ and $g(x) = \sqrt{x-1}$. Find $(f \circ g)(x)$ and its domain.

Solution 1.12.1. $(f \circ g)(x) = f(g(x)) = f(\sqrt{x-1}) = \sqrt{2 - \sqrt{x-1}}$. To find the domain, we need to satisfy two conditions from the inside out: 1. From $g(x)$, we need $x - 1 \geq 0 \implies x \geq 1$. 2. From the outer function f, we need its input, which is $g(x)$, to satisfy $2 - g(x) \geq 0$. $2 - \sqrt{x-1} \geq 0 \implies 2 \geq \sqrt{x-1}$. Squaring both sides (which is safe since both are non-negative): $4 \geq x - 1 \implies 5 \geq x$ or $x \leq 5$. Combining both conditions, we have $x \geq 1$ and $x \leq 5$. The domain is $[1, 5]$.

Problem 1.13. Decompose the function $h(x) = (x^2 + 3)^4$ into two functions f and g such that $h = f \circ g$.

Solution 1.13.1. We look for an "inner" function and an "outer" function. The inner operation is squaring and adding 3. Let $g(x) = x^2 + 3$. The outer operation is raising to the 4th power. Let $f(x) = x^4$. Check: $(f \circ g)(x) = f(g(x)) = f(x^2+3) = (x^2+3)^4 = h(x)$. This is a valid decomposition.

Problem 1.14. Decompose $h(x) = \sin(\sqrt{x})$ into f and g.

Solution 1.14.1. Inner function: $g(x) = \sqrt{x}$. Outer function: $f(x) = \sin(x)$. Check: $(f \circ g)(x) = f(g(x)) = f(\sqrt{x}) = \sin(\sqrt{x}) = h(x)$.

Problem 1.15. If $f(x) = x + 1$ and $(g \circ f)(x) = x^2 + 2x + 1$, find $g(x)$.

Solution 1.15.1. We are given $g(f(x)) = x^2 + 2x + 1$. We can rewrite the right side as $(x + 1)^2$. So, $g(x + 1) = (x + 1)^2$. This equation tells us what g does to its input: it squares it. Let $u = x + 1$. Then the equation is $g(u) = u^2$. Therefore, the function is $g(x) = x^2$.

Summary of Functions

- A **function** is a rule mapping each input (domain) to a single output (range).
- The **domain** is the set of valid inputs; the **range** is the set of resulting outputs.
- A function is **one-to-one** if each output corresponds to only one input (it passes the Horizontal Line Test).
- Only one-to-one functions have an **inverse function**, $f^{-1}(x)$, which reverses the mapping. The graph of f^{-1} is a reflection of f across the line $y = x$.
- **Composition**, $(f \circ g)(x) = f(g(x))$, applies one function to the result of another. The domain of the composite function is restricted by the domains of both functions.

2 Limits and Continuity

2.1 Learning Outcomes

Upon successful completion of this section, students will be able to:

- **Explain** the intuitive concept of a limit of a function.
- **Calculate** limits using limit laws and algebraic manipulation.
- **Identify** and resolve indeterminate forms like $\frac{0}{0}$ and $\frac{\infty}{\infty}$.
- **Define** continuity at a point and on an interval.
- **Classify** different types of discontinuities (removable, jump, infinite).
- **State and apply** the Intermediate Value Theorem to solve problems in engineering, such as finding roots of equations.

2.2 Intuitive Introduction to Limits

Definition 2.1 (Intuitive Definition of a Limit)**.** We write $\lim_{x\to a} f(x) = L$ and say "the limit of $f(x)$, as x approaches a, equals L" if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a (on either side of a) but not equal to a.

The key idea is that the limit describes the behavior of a function *near* a point, not necessarily *at* the point itself.

Example 2.1. Consider the function $f(x) = \frac{x^2-1}{x-1}$. This function is not defined at $x = 1$. What happens as x gets close to 1? For $x \neq 1$, we can simplify: $f(x) = \frac{(x-1)(x+1)}{x-1} = x+1$. As x gets closer and closer to 1, $x + 1$ gets closer and closer to $1 + 1 = 2$. So, we say $\lim_{x\to 1} \frac{x^2-1}{x-1} = 2$.

2.2.1 Figure Illustration: The Concept of a Limit

2.3 Limit Laws

Suppose that c is a constant and the limits $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ exist.

Theorem 2.1 (Limit Laws)**.**

1. ***Sum Law:*** $\lim_{x\to a}[f(x) + g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$
2. ***Difference Law:*** $\lim_{x\to a}[f(x) - g(x)] = \lim_{x\to a} f(x) - \lim_{x\to a} g(x)$
3. ***Constant Multiple Law:*** $\lim_{x\to a}[cf(x)] = c \lim_{x\to a} f(x)$
4. ***Product Law:*** $\lim_{x\to a}[f(x)g(x)] = (\lim_{x\to a} f(x)) \cdot (\lim_{x\to a} g(x))$
5. ***Quotient Law:*** $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$ *if* $\lim_{x\to a} g(x) \neq 0$
6. ***Power Law:*** $\lim_{x\to a}[f(x)]^n = [\lim_{x\to a} f(x)]^n$ *where n is a positive integer.*
7. ***Root Law:*** $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)}$ *where n is a positive integer. (If n is even, we assume* $\lim_{x\to a} f(x) > 0$*).*

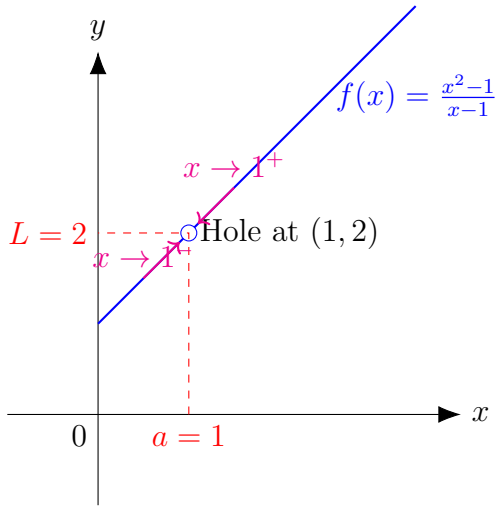


Figure 3: The graph of $f(x) = \frac{x^2-1}{x-1}$. As x approaches 1 from either the left or the right, the function value $f(x)$ approaches 2. The limit exists even though the function is not defined at $x = 1$.

Proof of the Sum Law (using $\epsilon - \delta$ definition). Let $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$. We want to show that $\lim_{x\to a}[f(x) + g(x)] = L + M$. Let $\epsilon > 0$ be given. We need to find a $\delta > 0$ such that if $0 < |x - a| < \delta$, then $|(f(x) + g(x)) - (L + M)| < \epsilon$. Using the triangle inequality, we have: $|(f(x) + g(x)) - (L + M)| = |(f(x) - L) + (g(x) - M)| \leq |f(x) - L| + |g(x) - M|$.

Since $\lim_{x\to a} f(x) = L$, we know that for any $\epsilon_1 > 0$, there exists a $\delta_1 > 0$ such that if $0 < |x - a| < \delta_1$, then $|f(x) - L| < \epsilon_1$. Let's choose $\epsilon_1 = \epsilon/2$. Similarly, since $\lim_{x\to a} g(x) = M$, there exists a $\delta_2 > 0$ such that if $0 < |x-a| < \delta_2$, then $|g(x)-M| < \epsilon/2$.

Now, let $\delta = \min(\delta_1, \delta_2)$. If $0 < |x - a| < \delta$, then both conditions are met simultaneously. Therefore, $|(f(x) + g(x)) - (L + M)| \leq |f(x) - L| + |g(x) - M| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$. This completes the proof. □

2.4 Indeterminate Forms

Indeterminate forms are expressions like $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0 \cdot \infty$, $\infty - \infty$, 1^∞, 0^0, and ∞^0. They are called "indeterminate" because the value of the limit cannot be determined from the form alone. We must use algebraic manipulation or (later) L'Hôpital's Rule to find the limit.

2.4.1 Technique 1: Factoring and Canceling

This is used for the $\frac{0}{0}$ form in rational functions.

Example 2.2. Evaluate $\lim_{x\to 3} \frac{x^2-9}{x-3}$. Plugging in $x = 3$ gives $\frac{9-9}{3-3} = \frac{0}{0}$.

$$\begin{aligned}\lim_{x\to 3} \frac{x^2 - 9}{x - 3} &= \lim_{x\to 3} \frac{(x - 3)(x + 3)}{x - 3} \\ &= \lim_{x\to 3}(x + 3) \quad (\text{Since } x \neq 3) \\ &= 3 + 3 = 6\end{aligned}$$

2.4.2 Technique 2: Rationalizing (Multiplying by the Conjugate)

This is useful for the $\frac{0}{0}$ form involving square roots.

Example 2.3. Evaluate $\lim_{x\to 0} \frac{\sqrt{x+4}-2}{x}$. Plugging in $x = 0$ gives $\frac{\sqrt{4}-2}{0} = \frac{0}{0}$.

$$\begin{aligned}
\lim_{x\to 0} \frac{\sqrt{x+4}-2}{x} &= \lim_{x\to 0} \left(\frac{\sqrt{x+4}-2}{x} \cdot \frac{\sqrt{x+4}+2}{\sqrt{x+4}+2} \right) \\
&= \lim_{x\to 0} \frac{(x+4)-4}{x(\sqrt{x+4}+2)} \\
&= \lim_{x\to 0} \frac{x}{x(\sqrt{x+4}+2)} \\
&= \lim_{x\to 0} \frac{1}{\sqrt{x+4}+2} \quad \text{(Since } x \neq 0\text{)} \\
&= \frac{1}{\sqrt{0+4}+2} = \frac{1}{2+2} = \frac{1}{4}
\end{aligned}$$

2.4.3 Practice Problems

Problem 2.1. Evaluate $\lim_{h\to 0} \frac{(2+h)^3-8}{h}$.

Solution 2.1.1. This is the $\frac{0}{0}$ form. We expand the numerator. $(2+h)^3 = 2^3 + 3(2^2)h + 3(2)h^2 + h^3 = 8 + 12h + 6h^2 + h^3$.

$$\begin{aligned}
\lim_{h\to 0} \frac{(8+12h+6h^2+h^3)-8}{h} &= \lim_{h\to 0} \frac{12h+6h^2+h^3}{h} \\
&= \lim_{h\to 0} \frac{h(12+6h+h^2)}{h} \\
&= \lim_{h\to 0} (12+6h+h^2) \\
&= 12 + 6(0) + 0^2 = 12
\end{aligned}$$

Problem 2.2. Evaluate $\lim_{x\to 4} \frac{x^2-x-12}{x^2-16}$.

Solution 2.2.1. This is the $\frac{0}{0}$ form. We factor the numerator and denominator. $x^2 - x - 12 = (x-4)(x+3)$. $x^2 - 16 = (x-4)(x+4)$.

$$\begin{aligned}
\lim_{x\to 4} \frac{(x-4)(x+3)}{(x-4)(x+4)} &= \lim_{x\to 4} \frac{x+3}{x+4} \\
&= \frac{4+3}{4+4} = \frac{7}{8}
\end{aligned}$$

Problem 2.3. Evaluate $\lim_{x\to\infty} \frac{3x^2-2x+1}{5x^2+4x-7}$.

Solution 2.3.1. This is the $\frac{\infty}{\infty}$ form. We divide the numerator and denominator by the highest power of x, which is x^2.

$$\begin{aligned}
\lim_{x\to\infty} \frac{\frac{3x^2}{x^2} - \frac{2x}{x^2} + \frac{1}{x^2}}{\frac{5x^2}{x^2} + \frac{4x}{x^2} - \frac{7}{x^2}} &= \lim_{x\to\infty} \frac{3 - \frac{2}{x} + \frac{1}{x^2}}{5 + \frac{4}{x} - \frac{7}{x^2}} \\
&= \frac{3-0+0}{5+0-0} = \frac{3}{5}
\end{aligned}$$

Problem 2.4. Evaluate $\lim_{t\to 1} \frac{1-t}{1-\sqrt{t}}$.

Solution 2.4.1. This is the $\frac{0}{0}$ form. We can rationalize the denominator or factor the numerator as a difference of squares: $1 - t = (1 - \sqrt{t})(1 + \sqrt{t})$.

$$\begin{aligned}\lim_{t\to 1} \frac{(1-\sqrt{t})(1+\sqrt{t})}{1-\sqrt{t}} &= \lim_{t\to 1}(1+\sqrt{t}) \\ &= 1 + \sqrt{1} = 2\end{aligned}$$

Problem 2.5. Evaluate $\lim_{x\to 2} \frac{\frac{1}{x}-\frac{1}{2}}{x-2}$.

Solution 2.5.1. This is the $\frac{0}{0}$ form. We simplify the complex fraction in the numerator.

$$\begin{aligned}\lim_{x\to 2} \frac{\frac{2-x}{2x}}{x-2} &= \lim_{x\to 2} \frac{-(x-2)}{2x} \cdot \frac{1}{x-2} \\ &= \lim_{x\to 2} \frac{-1}{2x} \\ &= \frac{-1}{2(2)} = -\frac{1}{4}\end{aligned}$$

2.5 Continuity

Definition 2.2 (Continuity at a Point)**.** A function f is **continuous** at a number a if

$$\lim_{x\to a} f(x) = f(a)$$

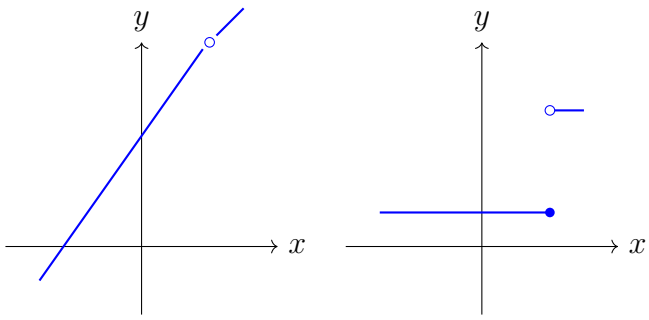
This definition implicitly requires three things:

1. $f(a)$ is defined (i.e., a is in the domain of f).
2. $\lim_{x\to a} f(x)$ exists.
3. The limit equals the function value.

Intuitively, a function is continuous if you can draw its graph without lifting your pen from the paper.

2.5.1 Types of Discontinuities

- **Removable Discontinuity**: The limit exists, but either $f(a)$ is not defined or $\lim_{x\to a} f(x) \neq f(a)$. This looks like a "hole" in the graph which could be "plugged". Example: $f(x) = \frac{x^2-1}{x-1}$ at $a = 1$.
- **Jump Discontinuity**: The left-hand limit and right-hand limit both exist but are not equal. The graph "jumps" from one value to another. Example: A step function.
- **Infinite Discontinuity**: One or both of the one-sided limits is ∞ or $-\infty$. This corresponds to a vertical asymptote. Example: $f(x) = 1/x$ at $a = 0$.



(a) Removable (b) Jump

Figure 4: Three types of discontinuities.

2.5.2 The Intermediate Value Theorem (IVT)

Theorem 2.2 (Intermediate Value Theorem)**.** *Suppose that f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number c in (a, b) such that $f(c) = N$.*

Proof. (A simplified sketch of the proof). Let's assume $f(a) < N < f(b)$. We want to find a $c \in (a, b)$ such that $f(c) = N$. Consider the set $S = \{x \in [a, b] \mid f(x) \leq N\}$. This set is non-empty (since $a \in S$) and is bounded above (by b). By the Completeness Axiom of real numbers, S must have a least upper bound (supremum). Let $c = \sup S$. We need to show that $c \in (a, b)$ and $f(c) = N$. Since $f(a) < N$ and f is continuous, there is a small interval around a where $f(x) < N$, so $c > a$. Similarly, $c < b$. Now we show $f(c) = N$ by ruling out $f(c) < N$ and $f(c) > N$. If $f(c) < N$, by continuity, there must be a small interval $(c, c + \delta)$ where $f(x)$ is still less than N. This means there are elements of S greater than c, which contradicts c being an upper bound. If $f(c) > N$, by continuity, there must be a small interval $(c - \delta, c]$ where $f(x)$ is still greater than N. This contradicts c being the *least* upper bound, as $c - \delta$ would also be an upper bound. The only remaining possibility is $f(c) = N$. □

2.5.3 Application of the IVT

The IVT is extremely useful for proving the existence of roots of equations.

Example 2.4 (Engineering Application)**.** In a circuit, the voltage equation is given by $V(t) = t^2 - e^{-t} - 3 = 0$. Show that there must be a time t_0 between $t = 1$ and $t = 2$ where the voltage is exactly zero.

Solution 2.5.2. Let $f(t) = t^2 - e^{-t} - 3$.

1. **Check Continuity**: The function $f(t)$ is a combination of a polynomial and an exponential function, both of which are continuous everywhere. So, $f(t)$ is continuous on $[1, 2]$.

2. **Check Endpoints**: $f(1) = 1^2 - e^{-1} - 3 = 1 - \frac{1}{e} - 3 = -2 - \frac{1}{e} \approx -2.368$. This is negative. $f(2) = 2^2 - e^{-2} - 3 = 4 - \frac{1}{e^2} - 3 = 1 - \frac{1}{e^2} \approx 1 - 0.135 = 0.865$. This is positive.

3. **Apply IVT**: We want to find a root, which means we are looking for a t_0 such that $f(t_0) = 0$. Let $N = 0$. Since $f(1) < 0$ and $f(2) > 0$, the value $N = 0$ is between

$f(1)$ and $f(2)$. By the Intermediate Value Theorem, there must exist a number c in the interval $(1, 2)$ such that $f(c) = 0$. This proves that a root exists between $t = 1$ and $t = 2$.

2.5.4 Practice Problems

Problem 2.6. For what value of the constant c is the function f continuous on $(-\infty, \infty)$?

$$f(x) = \begin{cases} cx^2 + 2x & \text{if } x < 2 \\ x^3 - cx & \text{if } x \geq 2 \end{cases}$$

Solution 2.6.1. The function is continuous for $x < 2$ and $x > 2$ because it is a polynomial in both cases. The only point of concern is $x = 2$. For continuity at $x = 2$, the left-hand limit must equal the right-hand limit, and this must equal the function value. $f(2) = 2^3 - c(2) = 8 - 2c$. $\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-}(cx^2 + 2x) = c(2^2) + 2(2) = 4c + 4$. $\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+}(x^3 - cx) = 2^3 - c(2) = 8 - 2c$. For continuity, we must have $4c + 4 = 8 - 2c$. $6c = 4 \implies c = \frac{4}{6} = \frac{2}{3}$.

Problem 2.7. Find the numbers at which $f(x) = \frac{x+2}{x^2-x-6}$ is discontinuous.

Solution 2.7.1. A rational function is discontinuous where its denominator is zero. $x^2 - x - 6 = 0 \implies (x - 3)(x + 2) = 0$. The function is discontinuous at $x = 3$ and $x = -2$. Let's classify them: At $x = -2$: $\lim_{x\to -2} \frac{x+2}{(x-3)(x+2)} = \lim_{x\to -2} \frac{1}{x-3} = \frac{1}{-5}$. Since the limit exists but $f(-2)$ is undefined, this is a removable discontinuity. At $x = 3$: $\lim_{x\to 3} \frac{1}{x-3}$ does not exist (it's $\pm\infty$). This is an infinite discontinuity.

Problem 2.8. Show that the equation $\cos(x) = x$ has a root in the interval $(0, 1)$.

Solution 2.8.1. We are looking for a solution to $\cos(x) - x = 0$. Let $f(x) = \cos(x) - x$. 1. **Continuity**: $f(x)$ is a combination of continuous functions, so it is continuous everywhere, including on $[0, 1]$. 2. **Endpoints**: $f(0) = \cos(0) - 0 = 1 - 0 = 1$. $f(1) = \cos(1) - 1$. Since 1 radian is about $57.3°$, $\cos(1)$ is positive but less than 1. Therefore, $f(1)$ is negative. 3. **Apply IVT**: We are looking for a root $f(c) = 0$. Let $N = 0$. Since $f(1) < 0 < f(0)$, by the IVT there must be a number c in $(0, 1)$ such that $f(c) = 0$, which means $\cos(c) = c$.

Problem 2.9. Where is the function $f(x) = \ln(x^2 - 1)$ continuous?

Solution 2.9.1. The function $f(x)$ is a composition of $\ln(u)$ and $u = x^2 - 1$. The function $\ln(u)$ is continuous for $u > 0$. So we require $x^2 - 1 > 0 \implies x^2 > 1$. This is true for $x > 1$ or $x < -1$. The function is continuous on the set $(-\infty, -1) \cup (1, \infty)$.

Problem 2.10. Is the following function continuous at $x = 0$?

$$f(x) = \begin{cases} \frac{\sin(x)}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

Solution 2.10.1. We check the three conditions for continuity at $a = 0$. 1. $f(0) = 1$. It is defined. 2. We need to find $\lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{\sin(x)}{x}$. This is a fundamental limit in calculus, and its value is 1. 3. We check if the limit equals the function value: $\lim_{x\to 0} f(x) = 1$ and $f(0) = 1$. Since $\lim_{x\to 0} f(x) = f(0)$, the function is continuous at $x = 0$.

Summary of Limits and Continuity

- A **limit** describes the value a function approaches as its input approaches some point.
- Limits can be calculated using algebraic manipulation to resolve **indeterminate forms** like $\frac{0}{0}$. Common techniques are factoring and rationalizing.
- A function is **continuous** at a point a if $\lim_{x \to a} f(x) = f(a)$. This means there are no holes, jumps, or vertical asymptotes.
- **Discontinuities** can be removable, jump, or infinite.
- The **Intermediate Value Theorem (IVT)** is a powerful result for continuous functions, guaranteeing that a function takes on all values between $f(a)$ and $f(b)$ on the interval $[a, b]$. It is widely used to prove the existence of solutions to equations.

MATH 1111: Differential and Integral Calculus

Lecture Notes 4: Ordinary Differentiation

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1 Ordinary Differentiation

This lecture introduces the cornerstone of differential calculus: the derivative. We will explore its definition, its meaning as a rate of change, techniques for its calculation, and methods for finding higher-order derivatives of complex functions.

1.1 Learning Outcomes

Upon successful completion of this section, students will be able to:

- **Define** the derivative of a function from first principles using limits.
- **Interpret** the derivative geometrically as the slope of a tangent line and physically as an instantaneous rate of change.
- **Determine** if a function is differentiable at a point and understand the relationship between differentiability and continuity.
- **Master** the rules of differentiation, including the product, quotient, and chain rules.
- **Calculate** higher-order derivatives (successive differentiation).
- **State, prove, and apply** Leibnitz's theorem to efficiently find the n-th derivative of a product of two functions.

1.2 The Derivative and Differentiability

Definition 1.1 (The Derivative)**.** The **derivative** of a function $f(x)$ with respect to the variable x is the function $f'(x)$ whose value at x is given by:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

provided this limit exists. If the limit exists, we say that f is **differentiable** at x. Other notations for the derivative include $\frac{dy}{dx}$, y', and $D(f)(x)$.

1.2.1 Geometric and Physical Interpretation

- **Geometrically**: The expression $\frac{f(x+h)-f(x)}{h}$ represents the slope of the secant line between the points $(x, f(x))$ and $(x+h, f(x+h))$. As $h \to 0$, this secant line approaches the **tangent line** to the curve at x. Therefore, $f'(x)$ is the slope of the tangent line to the graph of $y = f(x)$ at the point x.
- **Physically**: The derivative represents the **instantaneous rate of change**. For an electrical engineer, if $q(t)$ is the charge in a circuit at time t, then $q'(t) = \frac{dq}{dt}$ is the electric current $I(t)$ at that instant. If $s(t)$ is the position of an object, $s'(t)$ is its instantaneous velocity.

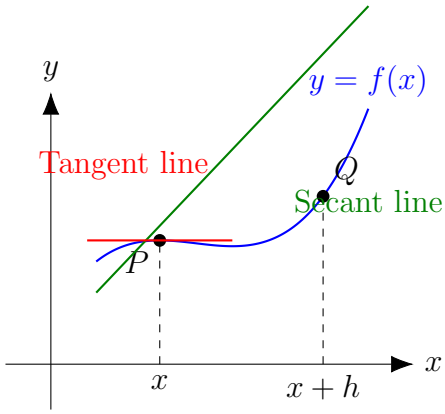

Figure 1: As $h \to 0$, the point Q slides along the curve towards P, and the green secant line rotates to become the red tangent line at P.

1.2.2 Differentiability and Continuity

An important theoretical result connects these two concepts.

Theorem 1.1. *If a function f is differentiable at a point a, then it is continuous at a.*

Proof. To show that f is continuous at a, we must show that $\lim_{x\to a} f(x) = f(a)$, which is equivalent to showing that $\lim_{x\to a}[f(x) - f(a)] = 0$. We are given that $f'(a) = \lim_{x\to a} \frac{f(x)-f(a)}{x-a}$ exists. We can write $f(x) - f(a) = \frac{f(x)-f(a)}{x-a} \cdot (x - a)$, for $x \neq a$. Now, take the limit as $x \to a$:

$$\begin{aligned}
\lim_{x\to a}[f(x) - f(a)] &= \lim_{x\to a}\left[\frac{f(x) - f(a)}{x - a} \cdot (x - a)\right] \\
&= \left(\lim_{x\to a} \frac{f(x) - f(a)}{x - a}\right) \cdot \left(\lim_{x\to a}(x - a)\right) \\
&= f'(a) \cdot 0 = 0
\end{aligned}$$

This proves that f is continuous at a. □

Note: The converse is false! Continuity does not imply differentiability. The classic counterexample is $f(x) = |x|$ at $x = 0$. The graph has a sharp corner, so there is no unique tangent, and the function is not differentiable there, yet it is continuous.

1.2.3 Practice Problems: Differentiability

Problem 1.1. Use the limit definition to find the derivative of $f(x) = x^2 - 3x$.

Solution 1.1.1.

$$\begin{aligned}
f'(x) &= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \\
&= \lim_{h\to 0} \frac{[(x+h)^2-3(x+h)]-[x^2-3x]}{h} \\
&= \lim_{h\to 0} \frac{x^2+2xh+h^2-3x-3h-x^2+3x}{h} \\
&= \lim_{h\to 0} \frac{2xh+h^2-3h}{h} \\
&= \lim_{h\to 0} \frac{h(2x+h-3)}{h} = \lim_{h\to 0}(2x+h-3) = 2x-3
\end{aligned}$$

Problem 1.2. Show that $f(x) = \sqrt[3]{x}$ is not differentiable at $x = 0$.

Solution 1.2.1. We check the limit definition for $f'(0)$:

$$f'(0) = \lim_{h\to 0} \frac{f(0+h)-f(0)}{h} = \lim_{h\to 0} \frac{\sqrt[3]{h}-0}{h} = \lim_{h\to 0} \frac{h^{1/3}}{h} = \lim_{h\to 0} h^{-2/3} = \lim_{h\to 0} \frac{1}{h^{2/3}}$$

As $h \to 0$, the denominator approaches 0, so the limit approaches $+\infty$. Since the limit does not exist as a finite number, the function is not differentiable at $x = 0$. Geometrically, this corresponds to a vertical tangent line at the origin.

1.3 Successive Differentiation (Higher-Order Derivatives)

We can differentiate a function multiple times, provided the derivatives exist.

Definition 1.2 (Higher-Order Derivatives)**.** The second derivative of f, denoted f'', is the derivative of f'. The third derivative, f''', is the derivative of f'', and so on. The n-th derivative is denoted $f^{(n)}(x)$ or $\frac{d^n y}{dx^n}$.

Example 1.1 (Physical Meaning)**.** For an object with position $s(t)$:

- First derivative: $s'(t) = v(t)$ is the **velocity**.
- Second derivative: $s''(t) = v'(t) = a(t)$ is the **acceleration**.
- Third derivative: $s'''(t) = a'(t) = j(t)$ is the **jerk** (the rate of change of acceleration).

1.3.1 The n-th Derivative of Common Functions

Finding a general formula for the n-th derivative is a key skill.

Example 1.2. Find the n-th derivative of $f(x) = e^{ax}$. $f'(x) = ae^{ax}$, $f''(x) = a^2e^{ax}$, $f'''(x) = a^3e^{ax}$. The pattern is clear: $f^{(n)}(x) = a^n e^{ax}$.

Example 1.3. Find the n-th derivative of $f(x) = \sin(x)$. $f'(x) = \cos(x) = \sin(x+\pi/2)$. $f''(x) = -\sin(x) = \sin(x+\pi)$. $f'''(x) = -\cos(x) = \sin(x+3\pi/2)$. The pattern is: $f^{(n)}(x) = \sin(x+n\pi/2)$.

Example 1.4. Find the n-th derivative of $f(x) = \frac{1}{ax+b}$. Let $f(x) = (ax+b)^{-1}$. $f'(x) = -1\cdot a(ax+b)^{-2}$. $f''(x) = (-1)(-2)\cdot a^2(ax+b)^{-3}$. $f'''(x) = (-1)(-2)(-3)\cdot a^3(ax+b)^{-4}$. The pattern is: $f^{(n)}(x) = (-1)^n n! a^n (ax+b)^{-(n+1)}$.

1.4 Leibnitz's Theorem

Finding higher-order derivatives of a product of functions can be extremely tedious. Leibnitz's Theorem provides a general and elegant formula for this task.

Theorem 1.2 (Leibnitz's Theorem)**.** *If $u(x)$ and $v(x)$ are functions that have derivatives up to the n-th order, then the n-th derivative of their product $y = uv$ is given by:*

$$y^{(n)} = (uv)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} u^{(n-k)} v^{(k)}$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is the binomial coefficient, and $u^{(0)} = u$, $v^{(0)} = v$. Explicitly:

$$(uv)^{(n)} = u^{(n)}v + nu^{(n-1)}v^{(1)} + \frac{n(n-1)}{2!}u^{(n-2)}v^{(2)} + \cdots + uv^{(n)}$$

Notice the striking similarity to the binomial expansion of $(a+b)^n$.

Proof. We prove this theorem by mathematical induction.

Base Case (n=1): $(uv)^{(1)} = \binom{1}{0}u^{(1)}v^{(0)} + \binom{1}{1}u^{(0)}v^{(1)} = 1 \cdot u'v + 1 \cdot uv' = u'v + uv'$. This is the standard product rule, so the formula holds for $n = 1$.

Inductive Step: Assume the formula is true for some integer $m \geq 1$. That is, $(uv)^{(m)} = \sum_{k=0}^{m} \binom{m}{k} u^{(m-k)} v^{(k)}$. We must show it is true for $n = m+1$. We differentiate both sides with respect to x:

$$\begin{aligned}
(uv)^{(m+1)} &= \frac{d}{dx}\left[\sum_{k=0}^{m} \binom{m}{k} u^{(m-k)} v^{(k)}\right] \\
&= \sum_{k=0}^{m} \binom{m}{k} \left[u^{(m-k+1)}v^{(k)} + u^{(m-k)}v^{(k+1)}\right] \\
&= \sum_{k=0}^{m} \binom{m}{k} u^{(m-k+1)}v^{(k)} + \sum_{k=0}^{m} \binom{m}{k} u^{(m-k)}v^{(k+1)}
\end{aligned}$$

Let's re-index the second sum by setting $j = k+1$.

$$\begin{aligned}
(uv)^{(m+1)} &= \left[\binom{m}{0}u^{(m+1)}v + \sum_{k=1}^{m} \binom{m}{k} u^{(m-k+1)}v^{(k)}\right] + \left[\sum_{j=1}^{m} \binom{m}{j-1} u^{(m-j+1)}v^{(j)} + \binom{m}{m}uv^{(m+1)}\right] \\
&= u^{(m+1)}v + \sum_{k=1}^{m} \left[\binom{m}{k} + \binom{m}{k-1}\right] u^{(m+1-k)}v^{(k)} + uv^{(m+1)}
\end{aligned}$$

Using Pascal's identity, $\binom{m}{k} + \binom{m}{k-1} = \binom{m+1}{k}$. Also, $\binom{m}{0} = 1 = \binom{m+1}{0}$ and $\binom{m}{m} = 1 = \binom{m+1}{m+1}$.

$$\begin{aligned}
(uv)^{(m+1)} &= \binom{m+1}{0}u^{(m+1)}v + \sum_{k=1}^{m} \binom{m+1}{k} u^{(m+1-k)}v^{(k)} + \binom{m+1}{m+1}uv^{(m+1)} \\
&= \sum_{k=0}^{m+1} \binom{m+1}{k} u^{(m+1-k)}v^{(k)}
\end{aligned}$$

This is the formula for $n = m+1$. Thus, by the principle of mathematical induction, the theorem is true for all positive integers n.

□

1.4.1 Practice Problems: Leibnitz's Theorem

A key strategy is to choose the function that terminates after a few derivatives (like a polynomial) as your $v(x)$.

Problem 1.3. Find the n-th derivative of $y = x^2 e^{3x}$.

Solution 1.3.1. Let $u = e^{3x}$ and $v = x^2$. We need their derivatives.

- $u^{(k)} = 3^k e^{3x}$
- $v^{(0)} = x^2$, $v^{(1)} = 2x$, $v^{(2)} = 2$, $v^{(k)} = 0$ for $k \geq 3$.

The sum in Leibnitz's theorem will only have three non-zero terms (for $k = 0, 1, 2$).

$$\begin{aligned}
y^{(n)} &= \binom{n}{0} u^{(n)} v^{(0)} + \binom{n}{1} u^{(n-1)} v^{(1)} + \binom{n}{2} u^{(n-2)} v^{(2)} \\
&= 1 \cdot (3^n e^{3x})(x^2) + n \cdot (3^{n-1} e^{3x})(2x) + \frac{n(n-1)}{2} \cdot (3^{n-2} e^{3x})(2) \\
&= e^{3x} \left[3^n x^2 + 2n \cdot 3^{n-1} x + n(n-1) \cdot 3^{n-2}\right] \\
&= 3^{n-2} e^{3x} \left[3^2 x^2 + 2n \cdot 3x + n(n-1)\right] \\
&= 3^{n-2} e^{3x} \left[9x^2 + 6nx + n(n-1)\right]
\end{aligned}$$

Problem 1.4. If $y = x\cos(x)$, find $y^{(n)}$.

Solution 1.4.1. Let $u = \cos(x)$ and $v = x$.

- $u^{(k)} = \cos(x + k\pi/2)$
- $v^{(0)} = x$, $v^{(1)} = 1$, $v^{(k)} = 0$ for $k \geq 2$.

The sum has only two non-zero terms ($k = 0, 1$).

$$\begin{aligned}
y^{(n)} &= \binom{n}{0} u^{(n)} v^{(0)} + \binom{n}{1} u^{(n-1)} v^{(1)} \\
&= 1 \cdot \cos(x + n\pi/2) \cdot x + n \cdot \cos(x + (n-1)\pi/2) \cdot 1 \\
&= x\cos(x + n\pi/2) + n\cos(x + n\pi/2 - \pi/2)
\end{aligned}$$

Using the identity $\cos(A - \pi/2) = \sin(A)$, we get:

$$y^{(n)} = x\cos(x + n\pi/2) + n\sin(x + n\pi/2)$$

Practice Problems and Solutions

Problem 1.5. Let the function $f(x)$ be defined as:

$$f(x) = \begin{cases} ax^2 + b & \text{if } x \leq 1 \\ \frac{1}{x} & \text{if } x > 1 \end{cases}$$

Find the values of the constants a and b that make the function differentiable for all real numbers.

Solution 1.5.1. For $f(x)$ to be differentiable everywhere, it must be differentiable at the point $x = 1$. This requires two conditions to be met:

1. **The function must be continuous at $x = 1$.** This means the limit from the left must equal the limit from the right, and this must equal the function value.

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x)$$

$$\lim_{x\to 1^-} (ax^2 + b) = \lim_{x\to 1^+} \frac{1}{x}$$

$$a(1)^2 + b = \frac{1}{1} \implies \mathbf{a + b = 1} \quad \text{(Equation 1)}$$

2. **The derivative from the left must equal the derivative from the right at $x = 1$.** First, we find the derivatives for each piece: For $x < 1$, $f'(x) = 2ax$. For $x > 1$, $f'(x) = -\frac{1}{x^2}$. Now, we equate the limits of the derivatives at $x = 1$:

$$\lim_{x\to 1^-} f'(x) = \lim_{x\to 1^+} f'(x)$$

$$\lim_{x\to 1^-} (2ax) = \lim_{x\to 1^+} \left(-\frac{1}{x^2}\right)$$

$$2a(1) = -\frac{1}{1^2} \implies \mathbf{2a = -1} \quad \text{(Equation 2)}$$

From Equation 2, we find $a = -\frac{1}{2}$. Substituting this value into Equation 1:

$$-\frac{1}{2} + b = 1 \implies b = 1 + \frac{1}{2} = \frac{3}{2}$$

Thus, for the function to be differentiable everywhere, the constants must be $a = -1/2$ and $b = 3/2$.

Problem 1.6. Using the first principle (the limit definition), find the derivative of $f(x) = \cos(x)$.

Solution 1.6.1. By definition, $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$.

$$f'(x) = \lim_{h\to 0} \frac{\cos(x+h) - \cos(x)}{h}$$

We use the sum-to-product trigonometric identity: $\cos(A) - \cos(B) = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$. Let $A = x + h$ and $B = x$.

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{-2\sin\left(\frac{x+h+x}{2}\right)\sin\left(\frac{x+h-x}{2}\right)}{h} \\ &= \lim_{h\to 0} \frac{-2\sin\left(x + \frac{h}{2}\right)\sin\left(\frac{h}{2}\right)}{h} \\ &= \lim_{h\to 0} \left[-\sin\left(x + \frac{h}{2}\right) \cdot \frac{\sin(h/2)}{h/2}\right] \end{aligned}$$

We split the limit into two parts. As $h \to 0$, we have $h/2 \to 0$. We use the fundamental limit $\lim_{\theta\to 0} \frac{\sin(\theta)}{\theta} = 1$.

$$\begin{aligned} f'(x) &= \left(\lim_{h\to 0} -\sin\left(x + \frac{h}{2}\right)\right) \cdot \left(\lim_{h/2\to 0} \frac{\sin(h/2)}{h/2}\right) \\ &= (-\sin(x)) \cdot (1) \\ &= -\sin(x) \end{aligned}$$

Thus, the derivative of $\cos(x)$ is $-\sin(x)$.

Problem 1.7. The position of a particle moving along a straight line is given by the function $s(t) = t^3 - 9t^2 + 24t$ for $t \geq 0$, where s is in meters and t is in seconds.

1. At what times is the particle's acceleration zero?
2. What is the particle's velocity at these times?

Solution 1.7.1. We need to find the first and second derivatives of the position function. The velocity function is the first derivative:

$$v(t) = s'(t) = 3t^2 - 18t + 24$$

The acceleration function is the second derivative:

$$a(t) = v'(t) = s''(t) = 6t - 18$$

1. To find when the acceleration is zero, we set $a(t) = 0$:

$$6t - 18 = 0 \implies 6t = 18 \implies t = 3$$

   The particle's acceleration is zero at $t = 3$ seconds.

2. To find the velocity at this time, we substitute $t = 3$ into the velocity function $v(t)$:

$$v(3) = 3(3)^2 - 18(3) + 24 = 3(9) - 54 + 24 = 27 - 54 + 24 = -3$$

   The velocity at $t = 3$ seconds is -3 m/s. This means the particle is momentarily moving in the negative direction while its velocity is changing from decreasing to increasing.

Problem 1.8. Find the n-th derivative of the function $f(x) = \frac{1}{x^2-a^2}$, where a is a non-zero constant.

Solution 1.8.1. Direct differentiation would be very complicated. The key is to first decompose the function using partial fractions.

$$\frac{1}{x^2 - a^2} = \frac{1}{(x-a)(x+a)} = \frac{A}{x-a} + \frac{B}{x+a}$$

Solving for A and B: $1 = A(x+a) + B(x-a)$. If $x = a$, then $1 = A(2a) \implies A = \frac{1}{2a}$. If $x = -a$, then $1 = B(-2a) \implies B = -\frac{1}{2a}$. So, $f(x) = \frac{1}{2a}\left(\frac{1}{x-a} - \frac{1}{x+a}\right) = \frac{1}{2a}\left((x-a)^{-1} - (x+a)^{-1}\right)$.

Now we can find the n-th derivative of each term. We use the known formula for the n-th derivative of $(cx+d)^m$: for $m=-1$, the n-th derivative is $(-1)^n n! c^n (cx+d)^{-(n+1)}$.

$$\frac{d^n}{dx^n}\left((x-a)^{-1}\right) = (-1)^n n!(1)^n (x-a)^{-(n+1)}$$
$$\frac{d^n}{dx^n}\left((x+a)^{-1}\right) = (-1)^n n!(1)^n (x+a)^{-(n+1)}$$

Combining these results:

$$\begin{aligned} f^{(n)}(x) &= \frac{1}{2a}\left[(-1)^n n!(x-a)^{-(n+1)} - (-1)^n n!(x+a)^{-(n+1)}\right] \\ &= \frac{(-1)^n n!}{2a}\left[\frac{1}{(x-a)^{n+1}} - \frac{1}{(x+a)^{n+1}}\right] \end{aligned}$$

Problem 1.9. If $y = e^{a\arcsin(x)}$, prove that $(1-x^2)y_2 - xy_1 - a^2 y = 0$, where $y_1 = \frac{dy}{dx}$ and $y_2 = \frac{d^2y}{dx^2}$. Then, using Leibnitz's theorem, show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0$.

Solution 1.9.1. First, we find the first two derivatives of $y = e^{a\arcsin(x)}$.

$$y_1 = e^{a\arcsin(x)} \cdot \frac{a}{\sqrt{1-x^2}} = \frac{ay}{\sqrt{1-x^2}}$$

Rearrange to eliminate the square root: $y_1\sqrt{1-x^2} = ay$. Squaring both sides: $y_1^2(1-x^2) = a^2y^2$.

Now, differentiate this equation with respect to x using the product rule:

$$(2y_1y_2)(1-x^2) + y_1^2(-2x) = a^2(2yy_1)$$

Since y_1 is not identically zero, we can divide the entire equation by $2y_1$:

$$y_2(1-x^2) - xy_1 = a^2y$$

Rearranging gives the desired differential equation: $\mathbf{(1-x^2)y_2 - xy_1 - a^2y = 0}$.

Next, we apply Leibnitz's theorem to find the n-th derivative of this equation. We differentiate each term n times.

1. $D^n[(1-x^2)y_2]$: Let $u = y_2, v = 1-x^2$. $v' = -2x, v'' = -2, v''' = 0$.

$$\binom{n}{0}u^{(n)}v + \binom{n}{1}u^{(n-1)}v' + \binom{n}{2}u^{(n-2)}v'' = y_{n+2}(1-x^2) - 2nxy_{n+1} - n(n-1)y_n$$

2. $D^n[-xy_1]$: Let $u = y_1, v = -x$. $v' = -1, v'' = 0$.

$$\binom{n}{0}u^{(n)}v + \binom{n}{1}u^{(n-1)}v' = y_{n+1}(-x) - ny_n(1) = -xy_{n+1} - ny_n$$

3. $D^n[-a^2y]$: This is simply $-a^2y_n$.

Summing these results and setting them to zero:

$$[y_{n+2}(1-x^2) - 2nxy_{n+1} - n(n-1)y_n] + [-xy_{n+1} - ny_n] - a^2y_n = 0$$

Now, we group terms by the order of the derivative:

$$(1-x^2)y_{n+2} + (-2nx - x)y_{n+1} + (-n(n-1) - n - a^2)y_n = 0$$

Simplifying the coefficients:

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 - n + n + a^2)y_n = 0$$

$$\mathbf{(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0}$$

This is the required recurrence relation.

Problem 1.10. Find the n-th derivative of $y = x^2\sin(x)$.

Solution 1.10.1. We use Leibnitz's Theorem with $u = \sin(x)$ and $v = x^2$. The derivatives of v will terminate.

- $u^{(k)} = \sin(x + k\pi/2)$
- $v = x^2$, $v^{(1)} = 2x$, $v^{(2)} = 2$, $v^{(k)} = 0$ for $k \geq 3$.

The Leibnitz formula will have only three non-zero terms:

$$y^{(n)} = \binom{n}{0}u^{(n)}v + \binom{n}{1}u^{(n-1)}v^{(1)} + \binom{n}{2}u^{(n-2)}v^{(2)}$$

Substituting the derivatives:

$$y^{(n)} = 1\cdot\sin(x+n\pi/2)\cdot x^2 + n\cdot\sin(x+(n-1)\pi/2)\cdot(2x) + \frac{n(n-1)}{2}\cdot\sin(x+(n-2)\pi/2)\cdot(2)$$

$$y^{(n)} = x^2\sin(x+n\pi/2) + 2nx\sin(x+n\pi/2-\pi/2) + n(n-1)\sin(x+n\pi/2-\pi)$$

Using the identities $\sin(A-\pi/2) = -\cos(A)$ and $\sin(A-\pi) = -\sin(A)$:

$$y^{(n)} = x^2\sin(x+n\pi/2) - 2nx\cos(x+n\pi/2) - n(n-1)\sin(x+n\pi/2)$$

$$y^{(n)} = (x^2 - n(n-1))\sin(x+n\pi/2) - 2nx\cos(x+n\pi/2)$$

Problem 1.11. Explain conceptually why the function $f(x) = x^{2/3}$ is continuous at $x = 0$ but not differentiable at $x = 0$. Describe the geometric feature of the graph at this point.

Solution 1.11.1. Continuity at x=0: A function is continuous at a point a if $\lim_{x\to a} f(x) = f(a)$. Here, $f(0) = 0^{2/3} = 0$. The limit is $\lim_{x\to 0} x^{2/3} = 0$. Since the limit equals the function value, $f(x)$ is continuous at $x = 0$.

Non-differentiability at x=0: We check the limit definition of the derivative at $x = 0$:

$$f'(0) = \lim_{h\to 0}\frac{f(0+h)-f(0)}{h} = \lim_{h\to 0}\frac{h^{2/3}-0}{h} = \lim_{h\to 0}\frac{h^{2/3}}{h^1} = \lim_{h\to 0} h^{-1/3} = \lim_{h\to 0}\frac{1}{\sqrt[3]{h}}$$

This limit does not exist as a finite number.

- As $h \to 0^+$ (from the right), $\frac{1}{\sqrt[3]{h}} \to +\infty$.
- As $h \to 0^-$ (from the left), $\frac{1}{\sqrt[3]{h}} \to -\infty$.

Since the left-hand and right-hand limits of the slope do not agree (and are not finite), the function is not differentiable at $x = 0$.

Geometric Feature: The behavior of the derivative's limit indicates that the tangent line to the curve becomes vertical as x approaches 0. Because the slope approaches $+\infty$ from one side and $-\infty$ from the other, the graph forms a sharp point called a **cusp** at the origin. A function cannot have a unique, non-vertical tangent line at a cusp, which is the geometric reason for its non-differentiability.

Summary of Ordinary Differentiation

- **The Derivative** is defined as the limit of the average rate of change, representing the instantaneous rate of change or the slope of the tangent line.
- **Differentiability** is a "smoothness" condition that is stronger than continuity; a function must be continuous to be differentiable, but not vice-versa.
- **Successive Differentiation** involves finding higher-order derivatives, which have important physical interpretations like acceleration and jerk.
- **Leibnitz's Theorem** is a powerful and essential tool for finding the n-th derivative of a product of two functions, mirroring the structure of the binomial theorem. It simplifies what would otherwise be a prohibitively complex calculation.

MATH 1111: Differential and Integral Calculus

Lecture Notes 2: Expansions, Extrema, and Approximations

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1 The Mean Value Theorem and Its Consequences

1.1 Learning Outcomes

Upon successful completion of this section, students will be able to:

- **State and prove** Rolle's Theorem and the Mean Value Theorem (MVT).
- **Interpret** the geometric meaning of both theorems, relating instantaneous rates of change to average rates of change.
- **Verify** that a given function satisfies the hypotheses of these theorems on a given interval.
- **Find** the value(s) of 'c' guaranteed to exist by the theorems.
- **Apply** the MVT to solve practical problems, such as estimating changes in a function and analyzing vehicle motion.
- **Use** the MVT as a theoretical tool to prove fundamental properties of functions, such as showing that a function with a zero derivative is constant.

1.2 Rolle's Theorem

Rolle's Theorem is a foundational result that serves as a special case of the more general Mean Value Theorem. It provides a simple condition under which a function must have a horizontal tangent.

Theorem 1.1 (Rolle's Theorem)**.** *Let f be a function that satisfies the following three hypotheses:*

1. *f is continuous on the closed interval $[a, b]$.*
2. *f is differentiable on the open interval (a, b).*
3. *$f(a) = f(b)$.*

Then there is a number c in (a, b) such that $f'(c) = 0$.

Proof. There are two cases to consider.

Case 1: f is a constant function. If $f(x) = k$ for all x in $[a, b]$, then $f'(x) = 0$ for all x in (a, b). We can choose any number c in (a, b) and we will have $f'(c) = 0$.

Case 2: f is not a constant function. By the Extreme Value Theorem, since f is continuous on the closed interval $[a, b]$, it must attain an absolute maximum value and an absolute minimum value on $[a, b]$. Since f is not constant, at least one of these extreme values must occur at a point c inside the open interval (a, b). (If both occurred at the endpoints a and b, then since $f(a) = f(b)$, the maximum and minimum values would be the same, making the function constant, which contradicts our assumption).

By Fermat's Theorem, if a local extremum occurs at a point c in (a, b) and $f'(c)$ exists, then $f'(c) = 0$. Since f is differentiable on (a, b), $f'(c)$ must exist. Therefore, there is a number c in (a, b) such that $f'(c) = 0$.

□

1.2.1 Geometric Interpretation of Rolle's Theorem

Geometrically, Rolle's Theorem states that if a smooth curve starts and ends at the same height, there must be at least one point between the start and end where the tangent line is horizontal.

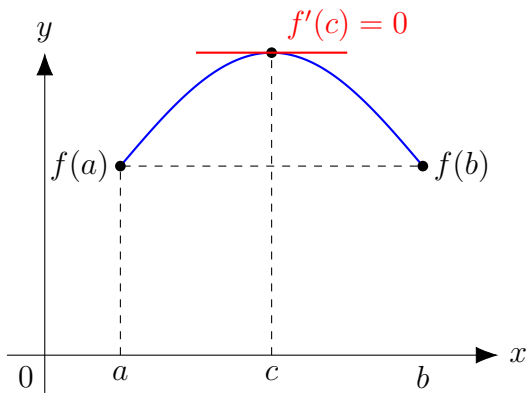

Figure 1: An illustration of Rolle's Theorem. The function satisfies $f(a) = f(b)$, and there exists a point $c \in (a, b)$ where the tangent is horizontal.

Example 1.1 (Application in Physics)**.** A projectile is fired from the ground. It goes up and then falls back to the ground. Let its height at time t be given by $h(t)$. If it leaves the ground at $t = 0$ and returns at $t = T$, then $h(0) = h(T) = 0$. The function $h(t)$ is continuous and differentiable. By Rolle's Theorem, there must be some time $t_0 \in (0, T)$ where $h'(t_0) = 0$. Since $h'(t)$ represents the vertical velocity, this means the projectile's vertical velocity is zero at some point—its highest point.

1.2.2 Practice Problems on Rolle's Theorem

Problem 1.1. Verify that the function $f(x) = x^3 - 4x$ satisfies the hypotheses of Rolle's Theorem on the interval $[0, 2]$, and find all numbers c that satisfy the conclusion.

Solution 1.1.1. 1. **Continuity**: $f(x)$ is a polynomial, so it is continuous everywhere, including on $[0, 2]$.
2. **Differentiability**: $f'(x) = 3x^2 - 4$, which exists for all x, so f is differentiable on $(0, 2)$.
3. **Endpoint Values**: $f(0) = 0^3 - 4(0) = 0$. $f(2) = 2^3 - 4(2) = 8 - 8 = 0$. Thus, $f(0) = f(2)$. All hypotheses are satisfied. We now find $c \in (0, 2)$ such that $f'(c) = 0$. $f'(c) = 3c^2 - 4 = 0 \implies 3c^2 = 4 \implies c^2 = \frac{4}{3}$. $c = \pm\sqrt{\frac{4}{3}} = \pm\frac{2}{\sqrt{3}}$. We must choose the value that lies in the interval $(0, 2)$. The value $c = \frac{2}{\sqrt{3}} \approx 1.155$ is in $(0, 2)$. The negative value is not. So, the required number is $c = \frac{2}{\sqrt{3}}$.

Problem 1.2. Does Rolle's Theorem apply to $f(x) = |x - 1|$ on $[0, 2]$?

Solution 1.2.1. 1. **Continuity**: The function is continuous on $[0, 2]$.
2. **Differentiability**: The function $f(x) = |x - 1|$ is not differentiable at $x = 1$, which is in the open interval $(0, 2)$.
3. **Endpoint Values**: $f(0) = |0 - 1| = 1$, $f(2) = |2 - 1| = 1$. So $f(0) = f(2)$. Since the differentiability hypothesis fails, Rolle's Theorem does not apply. We cannot conclude

that there is a point c where $f'(c) = 0$. Indeed, the graph is a 'V' shape with its vertex at $x = 1$, and there is no point with a horizontal tangent.

1.3 The Mean Value Theorem (MVT)

The Mean Value Theorem is one of the most important theorems in calculus. It is a generalization of Rolle's Theorem and connects the average rate of change of a function over an interval to its instantaneous rate of change at a specific point.

Theorem 1.2 (The Mean Value Theorem)**.** *Let f be a function that satisfies the following two hypotheses:*

1. *f is continuous on the closed interval $[a, b]$.*
2. *f is differentiable on the open interval (a, b).*

Then there is a number c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof. The proof cleverly uses Rolle's Theorem. We define an auxiliary function $h(x)$ that measures the vertical distance between the graph of $f(x)$ and the secant line passing through $(a, f(a))$ and $(b, f(b))$.
The equation of the secant line is $y - f(a) = \frac{f(b)-f(a)}{b-a}(x - a)$, which can be written as $y = f(a) + \frac{f(b)-f(a)}{b-a}(x - a)$. Let $h(x) = f(x) - \left[f(a) + \frac{f(b)-f(a)}{b-a}(x - a)\right]$. Now we check if $h(x)$ satisfies the hypotheses of Rolle's Theorem on $[a, b]$.
1. **Continuity**: $h(x)$ is the difference between $f(x)$ and a linear function, both of which are continuous on $[a, b]$. Thus, $h(x)$ is continuous on $[a, b]$.
2. **Differentiability**: Similarly, $h(x)$ is differentiable on (a, b), and its derivative is $h'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}$.
3. **Endpoint Values**: $h(a) = f(a) - \left[f(a) + \frac{f(b)-f(a)}{b-a}(a - a)\right] = f(a) - f(a) = 0$. $h(b) = f(b) - \left[f(a) + \frac{f(b)-f(a)}{b-a}(b - a)\right] = f(b) - [f(a) + f(b) - f(a)] = 0$. So, $h(a) = h(b) = 0$.
Since $h(x)$ satisfies all hypotheses of Rolle's Theorem, there exists a number c in (a, b) such that $h'(c) = 0$. $h'(c) = f'(c) - \frac{f(b)-f(a)}{b-a} = 0$. Rearranging gives the desired result: $f'(c) = \frac{f(b)-f(a)}{b-a}$. □

1.3.1 Geometric Interpretation of the MVT

The MVT guarantees that for any smooth curve, there is at least one point where the tangent line is parallel to the secant line connecting the endpoints of the curve.

Example 1.2 (Application in Transportation)**.** A car enters a highway at 1:00 PM and exits 150 km away at 3:00 PM. The average speed is $\frac{150 \text{ km}}{2 \text{ hr}} = 75$ km/h. Let $s(t)$ be the car's distance from the entrance at time t. The function $s(t)$ is continuous and differentiable. By the MVT, there must be some time t_0 between 1:00 PM and 3:00 PM at which the instantaneous velocity, $s'(t_0)$, was exactly 75 km/h. This is the principle behind average speed cameras; if your average speed exceeds the limit, you must have been speeding at some point.

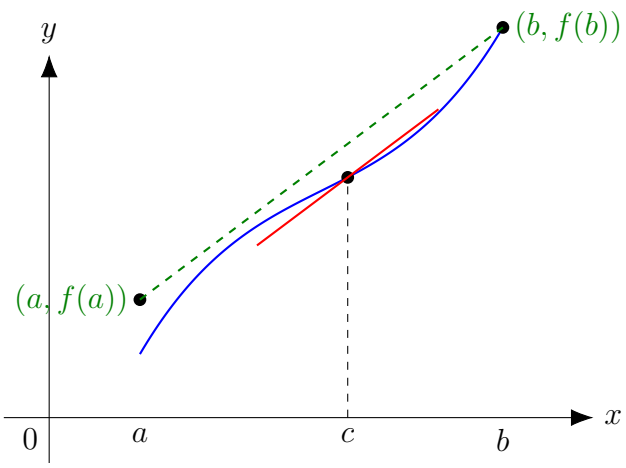

Figure 2: An illustration of the Mean Value Theorem. The tangent line at c (red) is parallel to the secant line connecting the endpoints (green dashed).

1.3.2 Practice Problems on the MVT

Problem 1.3. Verify that the function $f(x) = x^3 - x$ satisfies the MVT on $[0, 2]$, and find all numbers c that satisfy the conclusion.

Solution 1.3.1. 1. **Continuity**: $f(x)$ is a polynomial, so it's continuous on $[0, 2]$. 2. **Differentiability**: $f'(x) = 3x^2 - 1$, which exists on $(0, 2)$. The hypotheses are satisfied. We need to find $c \in (0, 2)$ such that $f'(c) = \frac{f(2)-f(0)}{2-0}$. $f(2) = 2^3 - 2 = 6$. $f(0) = 0^3 - 0 = 0$. The average slope is $\frac{6-0}{2-0} = 3$. We set $f'(c) = 3$: $3c^2 - 1 = 3 \implies 3c^2 = 4 \implies c^2 = \frac{4}{3}$. $c = \pm\sqrt{\frac{4}{3}} = \pm\frac{2}{\sqrt{3}}$. The value in the interval $(0, 2)$ is $c = \frac{2}{\sqrt{3}}$.

Problem 1.4. Use the MVT to prove that if $f'(x) = 0$ for all x in an interval (a, b), then f is constant on (a, b).

Solution 1.4.1. Let x_1 and x_2 be any two points in (a, b) with $x_1 < x_2$. We want to show $f(x_1) = f(x_2)$. The function f is continuous and differentiable on $[x_1, x_2]$ (since it is on all of (a, b)). By the MVT, there exists a number $c \in (x_1, x_2)$ such that:

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

But we are given that $f'(x) = 0$ for all x in (a, b), so $f'(c) = 0$.

$$0 = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

Since $x_1 \neq x_2$, the denominator is non-zero. This implies the numerator must be zero: $f(x_2) - f(x_1) = 0 \implies f(x_2) = f(x_1)$. Since x_1 and x_2 were arbitrary points in the interval, the function must be constant.

Summary of Rolle's Theorem and the MVT

- **Rolle's Theorem**: If a differentiable function starts and ends at the same value over an interval, its derivative must be zero somewhere in between. It guarantees a horizontal tangent.

- **Mean Value Theorem**: If a function is differentiable over an interval, its instantaneous rate of change must equal its average rate of change at some point within that interval. It guarantees a tangent parallel to the secant line.
- **Applications**: These theorems are crucial for proving many other results in calculus. The MVT, in particular, provides a powerful link between the derivative of a function and its behavior over an interval.

2 Applications: Optimization and Approximation

2.1 Learning Outcomes

Upon successful completion of this section, students will be able to:

- **Define** critical points, local and global extrema.
- **Apply** the First and Second Derivative Tests to find and classify local extrema of a function.
- **Solve** optimization problems by modeling a real-world scenario with a function and finding its absolute maximum or minimum.
- **Derive** and state Taylor's and Maclaurin's formulae.
- **Compute** Taylor and Maclaurin polynomials for elementary functions like $e^x, \sin(x), \cos(x)$, and $\ln(1+x)$.
- **Use** Taylor polynomials to approximate function values and understand the role of the remainder term in error estimation.
- **Apply** Taylor expansions in engineering contexts, such as the small-angle approximation.

2.2 Maximum and Minimum Values (Extrema)

Definition 2.1 (Extrema and Critical Points).

- Let c be a number in the domain of a function f. Then $f(c)$ is the **absolute maximum** value of f if $f(c) \geq f(x)$ for all x in the domain.
- $f(c)$ is the **absolute minimum** value if $f(c) \leq f(x)$ for all x.
- $f(c)$ is a **local maximum** if $f(c) \geq f(x)$ for all x near c.
- $f(c)$ is a **local minimum** if $f(c) \leq f(x)$ for all x near c.
- A **critical point** of a function f is a number c in its domain such that either $f'(c) = 0$ or $f'(c)$ does not exist.

Theorem 2.1 (Fermat's Theorem for extrema). *If f has a local maximum or minimum at c, and if $f'(c)$ exists, then $f'(c) = 0$.*

This theorem tells us that local extrema can only occur at critical points. This is why we search for them!

2.2.1 Finding and Classifying Local Extrema

Theorem 2.2 (The First Derivative Test). *Suppose that c is a critical point of a continuous function f.*

- *If f' changes from positive to negative at c, then f has a local maximum at c.*
- *If f' changes from negative to positive at c, then f has a local minimum at c.*

- *If f' does not change sign at c, then f has no local extremum at c.*

Theorem 2.3 (The Second Derivative Test)**.** *Suppose f'' is continuous near c.*

- *If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c. (Concave up $\implies$ minimum)*
- *If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c. (Concave down $\implies$ maximum)*
- *If $f'(c) = 0$ and $f''(c) = 0$, the test is inconclusive.*

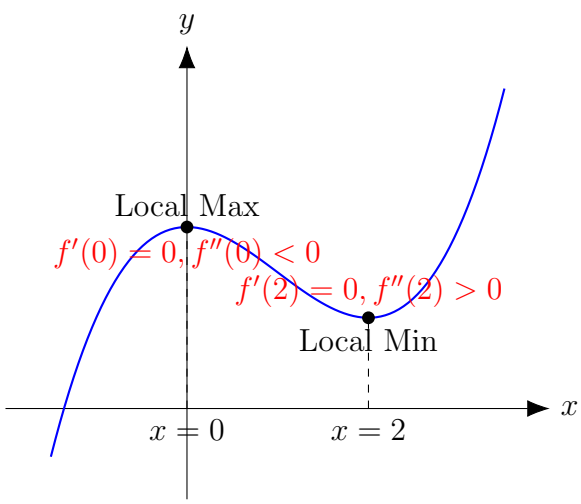

Figure 3: A function with a local maximum (concave down) and a local minimum (concave up).

2.2.2 Practice Problems on Optimization

Problem 2.1. Find the local maximum and minimum values of $f(x) = x^3 - 6x^2 + 5$.

Solution 2.1.1. First, find the critical points by setting $f'(x) = 0$. $f'(x) = 3x^2 - 12x = 3x(x-4)$. $f'(x) = 0$ when $x = 0$ or $x = 4$. These are our critical points.

Now, use the Second Derivative Test to classify them. $f''(x) = 6x - 12$.

- At $x = 0$: $f''(0) = 6(0) - 12 = -12 < 0$. Since the second derivative is negative, this is a **local maximum**. The value is $f(0) = 5$.
- At $x = 4$: $f''(4) = 6(4) - 12 = 12 > 0$. Since the second derivative is positive, this is a **local minimum**. The value is $f(4) = 4^3 - 6(4^2) + 5 = 64 - 96 + 5 = -27$.

Problem 2.2 (Engineering Application: Optimization)**.** A company needs to design a cylindrical can that holds 1 Liter (1000 cm^3) of oil. Find the dimensions (radius and height) that will minimize the cost of the metal to manufacture the can.

Solution 2.2.1. 1. **Identify Quantities**: Let the radius be r and height be h. The volume is $V = \pi r^2 h = 1000$. The surface area (which we want to minimize) is $A = 2\pi r^2$ (top and bottom) $+2\pi rh$ (side).

2. **Express as a Function of One Variable**: From the volume constraint, $h = \frac{1000}{\pi r^2}$. Substitute this into the area formula:

$$A(r) = 2\pi r^2 + 2\pi r\left(\frac{1000}{\pi r^2}\right) = 2\pi r^2 + \frac{2000}{r}$$

3. **Find the Minimum**: Find the critical points by taking the derivative with respect to r and setting it to zero.

$$A'(r) = 4\pi r - \frac{2000}{r^2}$$

Set $A'(r) = 0$: $4\pi r = \frac{2000}{r^2} \implies 4\pi r^3 = 2000 \implies r^3 = \frac{2000}{4\pi} = \frac{500}{\pi}$. So, $r = \sqrt[3]{\frac{500}{\pi}} \approx 5.42$ cm.

4. **Verify it is a Minimum**: Use the Second Derivative Test. $A''(r) = 4\pi + \frac{4000}{r^3}$. Since r must be positive, $A''(r)$ is always positive. Thus, our critical point corresponds to a local (and in this case, absolute) minimum.

5. **Find the Dimensions**: Radius: $r = \sqrt[3]{\frac{500}{\pi}} \approx 5.42$ cm. Height: $h = \frac{1000}{\pi r^2} = \frac{1000}{\pi(\frac{500}{\pi})^{2/3}} = \frac{1000}{\pi^{1/3}500^{2/3}} = \frac{2\cdot 500}{\pi^{1/3}500^{2/3}} = 2\frac{500^{1/3}}{\pi^{1/3}} = 2r$. So, the optimal can has a height equal to its diameter ($h = 2r$). $h \approx 2(5.42) = 10.84$ cm.

2.3 Taylor and Maclaurin Expansions

The central idea of Taylor expansions is to approximate a complex, non-polynomial function near a specific point with a simpler polynomial. This is one of the most powerful techniques in applied mathematics and engineering.

2.3.1 Derivation of the Taylor Polynomial

Suppose we want to approximate a function $f(x)$ near $x = a$ with a polynomial $P_n(x)$ of degree n. For the approximation to be good, we require that the polynomial and its derivatives match the function and its derivatives at the point $x = a$.
Let $P_n(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n$. We impose the conditions:

- $P_n(a) = f(a) \implies c_0 = f(a)$.
- $P_n'(a) = f'(a)$. Since $P_n'(x) = c_1 + 2c_2(x-a) + \ldots$, we get $P_n'(a) = c_1$. So, $c_1 = f'(a)$.
- $P_n''(a) = f''(a)$. Since $P_n''(x) = 2c_2 + 6c_3(x-a) + \ldots$, we get $P_n''(a) = 2c_2$. So, $c_2 = \frac{f''(a)}{2}$.
- In general, for the k-th derivative, $P_n^{(k)}(x) = k!c_k + \ldots$. So $P_n^{(k)}(a) = k!c_k$, which gives $c_k = \frac{f^{(k)}(a)}{k!}$.

Substituting these coefficients back gives us the Taylor polynomial.

Definition 2.2 (Taylor and Maclaurin Polynomials)**.** The **Taylor polynomial of degree** n for a function f centered at a is:

$$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

When the center is $a = 0$, this is called the **Maclaurin polynomial**.

Theorem 2.4 (Taylor's Theorem with Remainder)**.** *If a function f has $n+1$ derivatives in an interval I containing a, then for any x in I,*

$$f(x) = T_n(x) + R_n(x)$$

where $R_n(x)$ is the remainder (or error) term. The Lagrange form of the remainder is given by:

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$$

for some number c between a and x.

2.3.2 Common Maclaurin Series

Here are some of the most important series, which you should memorize.

1. $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ (for all x)
2. $\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$ (for all x)
3. $\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$ (for all x)
4. $\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$ (for $-1 < x \leq 1$)
5. $(1+x)^k = 1 + kx + \frac{k(k-1)}{2!}x^2 + \dots$ (Binomial Series, for $|x| < 1$)

Example 2.1 (Application: Small-Angle Approximation)**.** In physics and engineering, for small angles θ, we often use the approximation $\sin(\theta) \approx \theta$. Why is this valid? The Maclaurin series for $\sin(\theta)$ is $\theta - \frac{\theta^3}{6} + \frac{\theta^5}{120} - \dots$. If θ is small (e.g., $\theta = 0.1$ radians, about $5.7°$), then $\theta^3 = 0.001$. The second term is tiny compared to the first. So, for small angles, we can truncate the series after the first term: $\sin(\theta) \approx \theta$. This linear approximation simplifies many complex equations, such as the equation of motion for a pendulum.

2.3.3 Practice Problems on Taylor Expansions

Problem 2.3. Find the third-degree Taylor polynomial for $f(x) = \sqrt{x}$ centered at $a = 4$.

Solution 2.3.1. We need the first three derivatives of $f(x) = x^{1/2}$ evaluated at $a = 4$.

- $f(x) = x^{1/2} \implies f(4) = \sqrt{4} = 2$
- $f'(x) = \frac{1}{2}x^{-1/2} \implies f'(4) = \frac{1}{2\sqrt{4}} = \frac{1}{4}$
- $f''(x) = -\frac{1}{4}x^{-3/2} \implies f''(4) = -\frac{1}{4(4^{3/2})} = -\frac{1}{4(8)} = -\frac{1}{32}$
- $f'''(x) = \frac{3}{8}x^{-5/2} \implies f'''(4) = \frac{3}{8(4^{5/2})} = \frac{3}{8(32)} = \frac{3}{256}$

Now, plug these into the Taylor formula: $T_3(x) = f(4) + f'(4)(x-4) + \frac{f''(4)}{2!}(x-4)^2 + \frac{f'''(4)}{3!}(x-4)^3$ $T_3(x) = 2 + \frac{1}{4}(x-4) - \frac{1/32}{2}(x-4)^2 + \frac{3/256}{6}(x-4)^3$ $T_3(x) = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2 + \frac{1}{512}(x-4)^3$.

Problem 2.4. Use the polynomial from the previous problem to approximate $\sqrt{4.1}$ and estimate the error.

Solution 2.4.1. We use $T_3(x)$ with $x = 4.1$. $T_3(4.1) = 2 + \frac{1}{4}(0.1) - \frac{1}{64}(0.1)^2 + \frac{1}{512}(0.1)^3$ $T_3(4.1) = 2 + 0.025 - \frac{0.01}{64} + \frac{0.001}{512}$ $T_3(4.1) = 2.025 - 0.00015625 + 0.000001953 \approx 2.0248457$. The actual value is $\sqrt{4.1} \approx 2.02484567$. The approximation is excellent.

To estimate the error, we use the remainder term $R_3(x) = \frac{f^{(4)}(c)}{4!}(x-4)^4$ for some $c \in (4, 4.1)$. $f^{(4)}(x) = -\frac{15}{16}x^{-7/2}$. The error is $|R_3(4.1)| = \left|\frac{-15/16c^{-7/2}}{24}(0.1)^4\right| = \frac{15}{16 \cdot 24}\frac{1}{c^{7/2}}(0.0001)$. To get an upper bound, we choose the smallest value for c, which is $c = 4$, to make the denominator as small as possible (and thus the fraction as large as possible). $|R_3(4.1)| \leq \frac{15}{384}\frac{1}{4^{7/2}}(0.0001) = \frac{15}{384}\frac{1}{128}(0.0001) \approx 3 \times 10^{-8}$. This shows that the error is extremely small.

Summary of Optimization and Approximation

- **Optimization**: We can find the maximum or minimum values of a function by analyzing its derivatives. Critical points (where $f' = 0$ or DNE) are the only candidates for local extrema. The First and Second Derivative Tests are used to classify these points.
- **Taylor Series**: Provide a way to represent a function as an infinite polynomial.
- **Taylor Polynomials**: Finite truncations of Taylor series that serve as excellent approximations of the function near the center point a.
- **Remainder Term**: The error in a Taylor approximation can be precisely bounded using the remainder formula, which is crucial for applications where the degree of accuracy must be known.

3 The Remainder Term and Taylor's Theorem

In our previous lecture, we learned to approximate functions with Taylor polynomials. While powerful, an approximation is incomplete without an understanding of its accuracy. This lecture focuses entirely on the error of this approximation.

3.1 Learning Outcomes

Upon successful completion of this section, students will be able to:

- **Define** the remainder term $R_n(x)$ in Taylor's formula.
- **State and prove** Taylor's Theorem with the Lagrange form of the remainder.
- **Interpret** the remainder term as the precise error of the n-th degree Taylor approximation.
- **Calculate** an explicit upper bound for the error on a given interval using Taylor's Inequality.
- **Determine** the number of terms (degree n) of a Taylor polynomial required to approximate a function to a specified accuracy.
- **Apply** error analysis to practical problems in science and engineering, such as evaluating the validity of physical approximations.

3.2 Defining the Error of Approximation

Recall that the n-th degree Taylor polynomial of a function $f(x)$ centered at $x = a$ is given by

$$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$$

This polynomial, $T_n(x)$, is our approximation of $f(x)$. The approximation is not exact unless $f(x)$ itself is a polynomial of degree at most n. The difference between the true value and the approximated value is the error.

Definition 3.1 (The Remainder Term)**.** If $f(x)$ is approximated by its n-th degree Taylor polynomial $T_n(x)$, the **remainder term**, denoted $R_n(x)$, is the difference between the function and the polynomial:

$$R_n(x) = f(x) - T_n(x)$$

Therefore, the exact value of the function can be expressed by **Taylor's Formula**:

$$f(x) = T_n(x) + R_n(x)$$

The term $R_n(x)$ represents the **exact error** of the approximation $f(x) \approx T_n(x)$.

Our main goal is not to find the exact value of $R_n(x)$ (which would require knowing $f(x)$ exactly), but rather to find a **bound** on its magnitude, i.e., to find a number M such that $|R_n(x)| \leq M$.

3.3 Taylor's Theorem and the Lagrange Remainder

The following theorem gives us a powerful explicit formula for the remainder term.

Theorem 3.1 (Taylor's Theorem with Lagrange Remainder)**.** *Suppose a function f has $n+1$ continuous derivatives on an open interval I containing a. Then for any x in I, there exists a number c strictly between a and x such that*

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$$

This is known as the **Lagrange form of the remainder**. Notice its striking similarity to the next term in the Taylor series, but with the $(n+1)$-th derivative evaluated at an unknown point c instead of at a.

Proof. This proof is a clever application of Rolle's Theorem. Let x be a fixed value in the interval I. We want to find a formula for $R_n(x) = f(x) - T_n(x)$. Let's define a new function $g(t)$ as:

$$g(t) = f(x) - \left[\sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}(x-t)^k\right] - K(x-t)^{n+1}$$

Here, t is the variable, and x is treated as a constant. We choose the constant K such that $g(a) = 0$. Let's check the value of $g(x)$: $g(x) = f(x) - \left[\frac{f^{(0)}(x)}{0!}(x-x)^0 + \dots\right] - K(x-x)^{n+1} = f(x) - f(x) - 0 = 0$. So we have $g(x) = 0$.

Our choice of K makes $g(a) = 0$. We defined $g(t)$ such that $g(a) = f(x) - T_n(x) - K(x-a)^{n+1} = 0$. This means $R_n(x) = K(x-a)^{n+1}$. Our goal is to find K.

We have $g(a) = 0$ and $g(x) = 0$. The function $g(t)$ is continuous and differentiable on the interval between a and x. By Rolle's Theorem, there must be a number c between a and x such that $g'(c) = 0$.

Let's compute the derivative of $g(t)$ with respect to t. This requires careful use of the product rule on each term in the sum. $g'(t) = 0 - \frac{d}{dt}\left[f(t) + f'(t)(x-t) + \frac{f''(t)}{2!}(x-t)^2 + \dots + \frac{f^{(n)}(t)}{n!}(x - (n+1)K(x-t)^n$. Let's look at the derivative of a single term $\frac{f^{(k)}(t)}{k!}(x-t)^k$:

$$\frac{d}{dt}\left[\frac{f^{(k)}(t)}{k!}(x-t)^k\right] = \frac{f^{(k+1)}(t)}{k!}(x-t)^k - \frac{f^{(k)}(t)}{(k-1)!}(x-t)^{k-1}$$

When we sum this up, we get a "telescoping sum": $g'(t) = -\left[(f'(t) - f'(t)) + \left(\frac{f''(t)}{1!}(x-t)^1 - \frac{f''(t)}{1!}(x -\right.\right.$ All terms cancel except for the very last part of the derivative of the n-th term. The sum's derivative simplifies to $-\frac{f^{(n+1)}(t)}{n!}(x-t)^n$. So, $g'(t) = -\frac{f^{(n+1)}(t)}{n!}(x-t)^n + (n+1)K(x-t)^n = \left[(n+1)K - \frac{f^{(n+1)}(t)}{n!}\right](x-t)^n$.

By Rolle's theorem, there is a c between a and x where $g'(c) = 0$: $\left[(n+1)K - \frac{f^{(n+1)}(c)}{n!}\right](x-c)^n = 0$. Since $c \neq x$, we must have $(n+1)K - \frac{f^{(n+1)}(c)}{n!} = 0$, which gives $K = \frac{f^{(n+1)}(c)}{(n+1)!}$. Substituting this K back into our expression for the remainder, $R_n(x) = K(x-a)^{n+1}$, we get the desired result. □

3.4 Bounding the Remainder: Taylor's Inequality

The practical power of Taylor's Theorem comes from finding an upper bound for the error. We don't know c exactly, but we can often find a maximum value for $|f^{(n+1)}|$ on the interval of interest.

Corollary 3.2 (Taylor's Inequality)**.** *If $|f^{(n+1)}(x)| \leq M$ for all x in an interval containing a, then the remainder $R_n(x)$ satisfies the inequality:*

$$|R_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1}$$

for all x in that interval.

3.4.1 Visualizing the Error and the Bound

The figure below shows the function $f(x) = \cos(x)$ and its second-degree Maclaurin polynomial $T_2(x) = 1 - x^2/2$. The red shaded region is the actual error, $|R_2(x)|$. The green curves represent the error bound given by Taylor's Inequality.

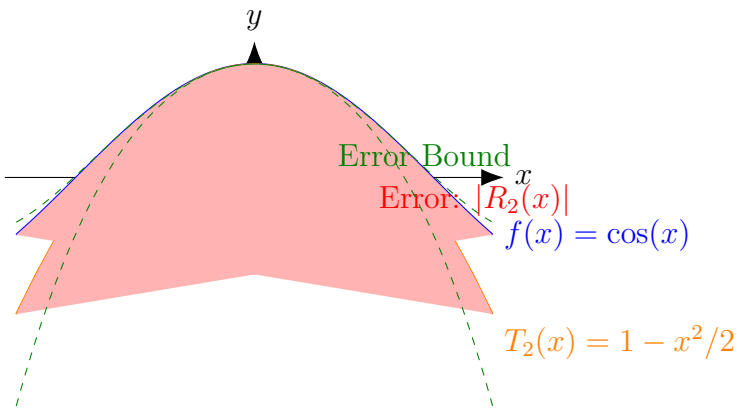

Figure 4: The error $|R_2(x)|$ between $\cos(x)$ and its approximation $T_2(x)$ is always contained within the error bound calculated from Taylor's Inequality.

3.4.2 Examples and Applications

Example 3.1. Approximate $f(x) = e^x$ with its 3rd-degree Maclaurin polynomial on the interval $[-1, 1]$ and find the error bound.

- $T_3(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}$.
- Error term is $R_3(x) = \frac{f^{(4)}(c)}{4!}x^4$. Here $f^{(4)}(x) = e^x$.
- We need to bound $|f^{(4)}(c)| = |e^c|$ for $c \in [-1, 1]$. Since e^x is an increasing function, its maximum value on $[-1, 1]$ occurs at $x = 1$, so $|e^c| \leq e^1 < 3$. Let's use $M = 3$ as a safe upper bound.
- By Taylor's Inequality, $|R_3(x)| \leq \frac{3}{4!}|x|^4 = \frac{3}{24}|x|^4 = \frac{|x|^4}{8}$.
- For any x in $[-1, 1]$, the error is at most $\frac{1^4}{8} = 0.125$.

Example 3.2 (Determining Required Degree)**.** What degree Maclaurin polynomial is required to approximate $\cos(0.1)$ to an accuracy of 10^{-7}?

- We want $|R_n(0.1)| \leq 10^{-7}$.
- The derivatives of $\cos(x)$ are $\pm\sin(x)$ or $\pm\cos(x)$. In all cases, $|f^{(n+1)}(c)| \leq 1$. So we can always use $M = 1$.
- $|R_n(0.1)| \leq \frac{1}{(n+1)!}|0.1|^{n+1}$. We need to find n such that $\frac{(0.1)^{n+1}}{(n+1)!} < 10^{-7}$.
- Let's test values of n:
  - $n = 2$: $|R_2(0.1)| \leq \frac{(0.1)^3}{3!} = \frac{0.001}{6} \approx 1.6 \times 10^{-4}$. Not good enough.
  - $n = 3$: For cosine, $T_3 = T_2$. We check R_3: $|R_3(0.1)| \leq \frac{(0.1)^4}{4!} = \frac{0.0001}{24} \approx 4.1 \times 10^{-6}$. Not good enough.
  - $n = 4$: $T_4 = T_5$. We check R_5: $|R_5(0.1)| \leq \frac{(0.1)^6}{6!} = \frac{10^{-6}}{720} \approx 1.3 \times 10^{-9}$. This is less than 10^{-7}.
- We need a polynomial up to the 5th derivative's effect on the remainder, which corresponds to the 4th-degree polynomial $T_4(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!}$ (since the x^5 term is zero).

Example 3.3 (Engineering Application: Nonlinear Sensor)**.** A sensor's output voltage is $V_{out} = \sqrt{4 + V_{in}}$. We want to use a linear approximation around $V_{in} = 0$. For what range of input voltages is this approximation accurate to within ± 0.01 V?

- Let $f(x) = \sqrt{4 + x}$ centered at $a = 0$.
- $f(0) = 2$. $f'(x) = \frac{1}{2\sqrt{4+x}}$, so $f'(0) = \frac{1}{4}$.
- Linear approximation: $T_1(x) = f(0) + f'(0)x = 2 + \frac{1}{4}x$.
- Error: $|R_1(x)| \leq \frac{M}{2!}|x|^2$. We need to find $M \geq |f''(c)|$.
- $f''(x) = -\frac{1}{4(4+x)^{3/2}}$. So $|f''(x)| = \frac{1}{4(4+x)^{3/2}}$. This function is largest when its denominator is smallest. Let's assume we are testing for a small range, say $|x| \leq d$. The denominator is smallest when $x = -d$. Let's test a known safe range, e.g., for $x \in [-1, 1]$, the denominator is smallest at $x = -1$, so $|f''(c)| \leq \frac{1}{4(3)^{3/2}} \approx 0.048$. Let's use $M = 0.05$.
- We want $|R_1(x)| \leq \frac{0.05}{2}x^2 \leq 0.01$.
- $x^2 \leq \frac{0.02}{0.05} = 0.4$.
- This means $|x| \leq \sqrt{0.4} \approx 0.632$.
- So, the linear model is accurate to within 0.01 V for input voltages between approximately -0.632 V and $+0.632$ V.

3.5 Practice Problems

Problem 3.1. Let $f(x) = \sin(x)$. Approximate $\sin(0.2)$ using the third-degree Maclaurin polynomial, $T_3(x)$, and find an upper bound for the error.

Solution 3.1.1. The Maclaurin series for $\sin(x)$ is $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$. The third-degree polynomial is $T_3(x) = x - \frac{x^3}{6}$. Approximation: $\sin(0.2) \approx T_3(0.2) = 0.2 - \frac{(0.2)^3}{6} = 0.2 - \frac{0.008}{6} \approx 0.2 - 0.001333 = 0.198667$. The actual value is $\sin(0.2) \approx 0.198669$.

Error Bound: The remainder is $R_3(x)$. For $\sin(x)$, the odd polynomial is the same as the next even one ($T_3 = T_4$). So we can use the bound for $R_4(x)$. $|R_4(x)| \leq \frac{M}{5!}|x|^5$. The fifth derivative of $\sin(x)$ is $\cos(x)$. For any c, $|\cos(c)| \leq 1$, so we can use $M = 1$. $|R_4(0.2)| \leq \frac{1}{5!}(0.2)^5 = \frac{1}{120}(0.00032) \approx 2.67 \times 10^{-6}$. Our actual error was about 2×10^{-6}, which is well within this bound.

Problem 3.2. Use a Taylor polynomial to approximate $\int_0^{0.5} \cos(x^2)dx$ and ensure your error is less than 10^{-3}.

Solution 3.2.1. The function $\cos(u)$ has the series $1 - \frac{u^2}{2!} + \frac{u^4}{4!} - \dots$. Let $u = x^2$. Then $\cos(x^2) = 1 - \frac{(x^2)^2}{2!} + \frac{(x^2)^4}{4!} - \dots = 1 - \frac{x^4}{2} + \frac{x^8}{24} - \dots$. We can integrate this polynomial term-by-term: $\int_0^{0.5} \cos(x^2)dx \approx \int_0^{0.5} \left(1 - \frac{x^4}{2}\right) dx = \left[x - \frac{x^5}{10}\right]_0^{0.5} = (0.5 - \frac{(0.5)^5}{10}) - 0 = 0.5 - \frac{0.03125}{10} = 0.5 - 0.003125 = 0.496875$.

This is an alternating series. The error in stopping is less than the magnitude of the next term. Error $\leq \left|\int_0^{0.5} \frac{x^8}{24}dx\right| = \left[\frac{x^9}{9 \cdot 24}\right]_0^{0.5} = \frac{(0.5)^9}{216} = \frac{1/512}{216} \approx 9 \times 10^{-6}$. This error is much smaller than 10^{-3}, so our approximation is sufficient.

Problem 3.3. Prove that the Maclaurin series for e^x converges to e^x for all real numbers x.

Solution 3.3.1. We need to show that the remainder term $R_n(x) \to 0$ as $n \to \infty$ for any fixed x. The remainder is $|R_n(x)| \leq \frac{M}{(n+1)!}|x|^{n+1}$. The $(n+1)$-th derivative is $f^{(n+1)}(t) = e^t$. Let's fix x. We need to find a bound M for $|e^t|$ on the interval between 0 and x. If $x > 0$, $t \in [0, x]$, so $e^t \leq e^x$. We can take $M = e^x$. If $x < 0$, $t \in [x, 0]$, so $e^t \leq e^0 = 1$. We can take $M = 1$. In either case, M is a constant with respect to n. So we need to show that $\lim_{n\to\infty} \frac{M|x|^{n+1}}{(n+1)!} = 0$. This is a standard limit result. For any fixed x, the factorial $(n+1)!$ grows much faster than the exponential $|x|^{n+1}$. Therefore, the limit is zero. Since the remainder goes to zero, the series converges to the function for all x.

Summary of the Remainder Term

- **Exact Error**: The remainder $R_n(x) = f(x) - T_n(x)$ is the precise error when approximating $f(x)$ with $T_n(x)$.
- **Lagrange Formula**: Taylor's Theorem gives us an explicit, though not fully determined, formula for the error: $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$.
- **Error Bound**: The primary application of the theorem is to find a practical upper bound on the error's magnitude, $|R_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1}$, where M is the maximum value of $|f^{(n+1)}|$ on the interval.

- **Engineering Significance**: Error analysis is not just an academic exercise. It allows engineers to confidently use simplified models, know the limits of their validity, and guarantee that calculations meet required design tolerances.

MATH 1111: Differential and Integral Calculus

Lecture Notes 5: Euler's theorem, Tangents and normal. Asymptotes

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1 Partial Differentiation

In single-variable calculus, we study functions of the form $y = f(x)$. However, most real-world phenomena depend on multiple variables. For instance, the voltage in a circuit may depend on resistance and capacitance, $V = f(R, C)$. Partial differentiation allows us to understand how such a function changes as we vary one independent variable while holding the others constant.

1.1 Learning Outcomes

Upon successful completion of this section, students will be able to:

- **Define** a function of several variables and understand its domain.
- **Calculate** first-order and higher-order partial derivatives of a multivariable function.
- **Interpret** the geometric meaning of partial derivatives as slopes of tangent lines to surface traces.
- **Define** homogeneous functions and verify if a function is homogeneous.
- **State, prove, and apply** Euler's theorem for homogeneous functions to solve problems and verify results.

1.2 Functions of Several Variables and Partial Derivatives

Definition 1.1 (Partial Derivative)**.** Let $z = f(x, y)$ be a function of two variables.

1. The **partial derivative of f with respect to x** is the function f_x or $\frac{\partial f}{\partial x}$ defined by:
$$\frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x+h, y) - f(x, y)}{h}$$
To calculate it, we treat y as a constant and differentiate $f(x, y)$ with respect to x.
2. The **partial derivative of f with respect to y** is the function f_y or $\frac{\partial f}{\partial y}$ defined by:
$$\frac{\partial f}{\partial y} = \lim_{h \to 0} \frac{f(x, y+h) - f(x, y)}{h}$$
To calculate it, we treat x as a constant and differentiate $f(x, y)$ with respect to y.

Example 1.1. If $f(x, y) = x^3 + x^2y^3 - 2y^2$, find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

- To find $\frac{\partial f}{\partial x}$, treat y as a constant: $\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^3) + y^3\frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial x}(2y^2) = 3x^2 + y^3(2x) - 0 = 3x^2 + 2xy^3$.
- To find $\frac{\partial f}{\partial y}$, treat x as a constant: $\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x^3) + x^2\frac{\partial}{\partial y}(y^3) - \frac{\partial}{\partial y}(2y^2) = 0 + x^2(3y^2) - 4y = 3x^2y^2 - 4y$.

Example 1.2 (EEE Application)**.** The impedance Z of a series RLC circuit is $Z(R, \omega) = \sqrt{R^2 + (\omega L - 1/(\omega C))^2}$. Find $\frac{\partial Z}{\partial R}$, the rate of change of impedance with respect to resistance at a fixed frequency ω.
$$\frac{\partial Z}{\partial R} = \frac{1}{2\sqrt{R^2 + (\dots)^2}} \cdot (2R) = \frac{R}{\sqrt{R^2 + (\omega L - 1/(\omega C))^2}} = \frac{R}{Z}$$

1.2.1 Geometric Interpretation

The graph of $z = f(x, y)$ is a surface.

- The partial derivative $f_x(a, b)$ is the slope of the tangent line to the curve formed by intersecting the surface with the plane $y = b$. It measures the rate of change in the x-direction.
- The partial derivative $f_y(a, b)$ is the slope of the tangent line to the curve formed by intersecting the surface with the plane $x = a$. It measures the rate of change in the y-direction.

Geometric Interpretation of Partial Derivatives

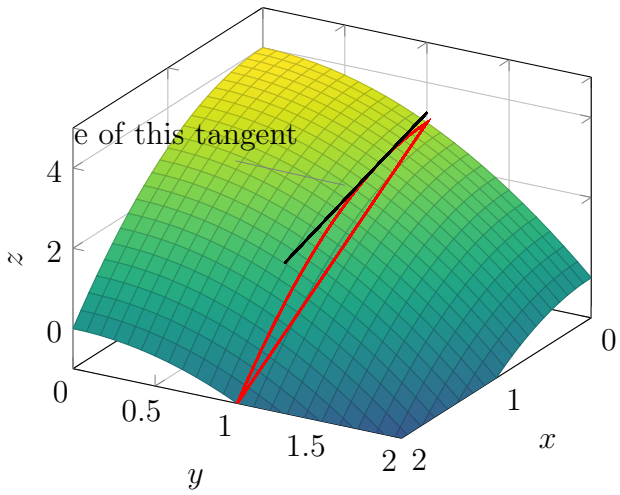

Figure 1: The surface $z = 4 - x^2 - y^2$. The red curve is the trace in the plane $y = 1$. The partial derivative f_x at a point on this curve gives the slope of the black tangent line.

1.2.2 Higher-Order Partial Derivatives

We can differentiate multiple times, leading to second-order partial derivatives:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) \qquad f_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)$$

$$f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) \qquad f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)$$

Clairaut's Theorem: For most well-behaved functions encountered in engineering, the mixed partial derivatives are equal: $f_{xy} = f_{yx}$.

1.3 Euler's Theorem on Homogeneous Functions

Definition 1.2 (Homogeneous Function)**.** A function $f(x, y)$ is a **homogeneous function of degree** n if for any real number $t > 0$,

$$f(tx, ty) = t^n f(x, y)$$

The number n can be any real number.

Example 1.3. Is $f(x,y) = x^2y - 4y^3$ homogeneous? $f(tx,ty) = (tx)^2(ty) - 4(ty)^3 = t^2x^2ty - 4t^3y^3 = t^3x^2y - 4t^3y^3 = t^3(x^2y - 4y^3) = t^3f(x,y)$. Yes, it is homogeneous of degree 3.

Example 1.4. Is $g(x,y) = \frac{x}{y^2} + \frac{y}{x^2}$ homogeneous? $g(tx,ty) = \frac{tx}{(ty)^2} + \frac{ty}{(tx)^2} = \frac{tx}{t^2y^2} + \frac{ty}{t^2x^2} = \frac{1}{t}\frac{x}{y^2} + \frac{1}{t}\frac{y}{x^2} = t^{-1}g(x,y)$. Yes, it is homogeneous of degree -1.

Example 1.5. Is $h(x,y) = x^2 + \sin(y)$ homogeneous? $h(tx,ty) = (tx)^2 + \sin(ty) = t^2x^2 + \sin(ty)$. We cannot factor out a power of t. No, it is not homogeneous.

Theorem 1.1 (Euler's Theorem)**.** *If $f(x,y)$ is a homogeneous function of degree n with continuous partial derivatives, then*

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = nf(x,y)$$

Proof. Since f is homogeneous of degree n, we have $f(tx,ty) = t^nf(x,y)$. Let's define a new function $g(t) = f(tx,ty)$. Then $g(t) = t^nf(x,y)$. Differentiating both sides with respect to t: The right side is simple: $g'(t) = nt^{n-1}f(x,y)$. For the left side, we use the chain rule for multivariable functions. Let $u = tx$ and $v = ty$. Then $g(t) = f(u,v)$. $g'(t) = \frac{\partial f}{\partial u}\frac{du}{dt} + \frac{\partial f}{\partial v}\frac{dv}{dt}$. Since $\frac{du}{dt} = x$ and $\frac{dv}{dt} = y$, we have $g'(t) = \frac{\partial f}{\partial u} \cdot x + \frac{\partial f}{\partial v} \cdot y$. Equating the two expressions for $g'(t)$:

$$x\frac{\partial f}{\partial u} + y\frac{\partial f}{\partial v} = nt^{n-1}f(x,y)$$

This equation holds for any t. Let's set $t = 1$. When $t = 1$, we have $u = x$ and $v = y$. The equation becomes:

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = n(1)^{n-1}f(x,y) = nf(x,y)$$

This completes the proof. □

Example 1.6 (Verification of Euler's Theorem)**.** Let $f(x,y) = x^2y - 4y^3$. We know it's homogeneous of degree $n = 3$. Let's verify. $\frac{\partial f}{\partial x} = 2xy$. $\frac{\partial f}{\partial y} = x^2 - 12y^2$. $x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = x(2xy) + y(x^2 - 12y^2) = 2x^2y + x^2y - 12y^3 = 3x^2y - 12y^3 = 3(x^2y - 4y^3) = 3f(x,y)$. The theorem holds, as $n = 3$.

Practice Problems: Partial Derivatives and Euler's Theorem

Problem 1.1. Find all second-order partial derivatives of $f(x,y) = \ln(x^2 + y^2)$.

Solution 1.1.1. $f_x = \frac{1}{x^2+y^2} \cdot (2x) = \frac{2x}{x^2+y^2}$. $f_y = \frac{1}{x^2+y^2} \cdot (2y) = \frac{2y}{x^2+y^2}$. Now for the second derivatives: $f_{xx} = \frac{\partial}{\partial x}\left(\frac{2x}{x^2+y^2}\right) = \frac{2(x^2+y^2)-2x(2x)}{(x^2+y^2)^2} = \frac{2y^2-2x^2}{(x^2+y^2)^2}$. $f_{yy} = \frac{\partial}{\partial y}\left(\frac{2y}{x^2+y^2}\right) = \frac{2(x^2+y^2)-2y(2y)}{(x^2+y^2)^2} = \frac{2x^2-2y^2}{(x^2+y^2)^2}$. $f_{xy} = \frac{\partial}{\partial y}\left(\frac{2x}{x^2+y^2}\right) = 2x \cdot \frac{-1}{(x^2+y^2)^2} \cdot (2y) = \frac{-4xy}{(x^2+y^2)^2}$. By Clairaut's theorem, f_{yx} should be the same. Let's check: $f_{yx} = \frac{\partial}{\partial x}\left(\frac{2y}{x^2+y^2}\right) = 2y \cdot \frac{-1}{(x^2+y^2)^2} \cdot (2x) = \frac{-4xy}{(x^2+y^2)^2}$. They match.

Problem 1.2. If $u = \arctan(y/x)$, verify Euler's theorem.

Solution 1.2.1. First, determine the degree of homogeneity. $u(tx, ty) = \arctan(ty/tx) = \arctan(y/x) = t^0 u(x, y)$. The function is homogeneous of degree $n = 0$. Euler's theorem predicts $xu_x + yu_y = 0 \cdot u = 0$. Let's verify this by calculating the partial derivatives. $u_x = \frac{1}{1+(y/x)^2} \cdot \left(-\frac{y}{x^2}\right) = \frac{x^2}{x^2+y^2} \cdot \left(-\frac{y}{x^2}\right) = \frac{-y}{x^2+y^2}$. $u_y = \frac{1}{1+(y/x)^2} \cdot \left(\frac{1}{x}\right) = \frac{x^2}{x^2+y^2} \cdot \left(\frac{1}{x}\right) = \frac{x}{x^2+y^2}$. Now, compute $xu_x + yu_y$: $x\left(\frac{-y}{x^2+y^2}\right) + y\left(\frac{x}{x^2+y^2}\right) = \frac{-xy}{x^2+y^2} + \frac{xy}{x^2+y^2} = 0$. The theorem is verified.

Summary of Partial Differentiation

- **Partial Derivatives** measure the rate of change of a multivariable function with respect to one variable, holding all others constant. Geometrically, they are slopes of tangent lines to traces on the function's surface.
- **Homogeneous Functions** are functions that scale in a predictable way: $f(tx, ty) = t^n f(x, y)$.
- **Euler's Theorem** provides a simple and direct relationship between a homogeneous function, its partial derivatives, and its degree of homogeneity: $xf_x + yf_y = nf$.

2 Geometric Applications and Asymptotes

We now explore two geometric applications. The first, tangent planes, is a direct application of partial derivatives. The second, asymptotes, is a topic from single-variable calculus essential for understanding the end behavior of curves.

2.1 Learning Outcomes

Upon successful completion of this section, students will be able to:

- **Derive and find** the equation of the tangent plane and the normal line to a surface at a given point.
- **Define** vertical, horizontal, and oblique asymptotes of a function.
- **Develop and apply** a systematic method to find all asymptotes of a given rational function.
- **Sketch** a curve, including its asymptotes, to visualize its behavior.

2.2 Tangent Planes and Normal Lines

For a surface $z = f(x, y)$, the tangent plane at a point $P(x_0, y_0, z_0)$ is the plane that "just touches" the surface at P. It is the multivariable analogue of a tangent line.

2.2.1 Derivation of the Tangent Plane Equation

A plane is defined by a point on the plane, (x_0, y_0, z_0), and a vector normal to the plane, $\vec{n} = \langle a, b, c\rangle$. The equation is $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$.
To find the normal vector, we can rewrite the surface as an implicit function $F(x, y, z) = f(x, y) - z = 0$. It is a fundamental result of vector calculus that the gradient vector, $\nabla F = \langle F_x, F_y, F_z\rangle$, is orthogonal (normal) to the level surface $F(x, y, z) = k$. Here, $\nabla F = \langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, -1\rangle$. So, our normal vector is $\vec{n} = \langle f_x(x_0, y_0), f_y(x_0, y_0), -1\rangle$. Plugging this into the plane equation gives: $f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - 1(z - z_0) = 0$.

Definition 2.1 (Tangent Plane and Normal Line)**.** The equation of the **tangent plane** to the surface $z = f(x, y)$ at the point (x_0, y_0, z_0) is:

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

The **normal line** is the line passing through (x_0, y_0, z_0) that is parallel to the normal vector. Its symmetric equations are:

$$\frac{x - x_0}{f_x(x_0, y_0)} = \frac{y - y_0}{f_y(x_0, y_0)} = \frac{z - z_0}{-1}$$

Example 2.1. Find the tangent plane and normal line to the surface $z = 2x^2 + y^2$ at the point $(1, 1, 3)$. $f_x = 4x \implies f_x(1, 1) = 4$. $f_y = 2y \implies f_y(1, 1) = 2$. The point is $(x_0, y_0, z_0) = (1, 1, 3)$. **Tangent Plane**: $z - 3 = 4(x - 1) + 2(y - 1) \implies z - 3 = 4x - 4 + 2y - 2 \implies 4x + 2y - z = 3$. **Normal Line**: $\frac{x-1}{4} = \frac{y-1}{2} = \frac{z-3}{-1}$.

2.3 Asymptotes

Asymptotes are lines that a curve approaches as it heads towards infinity.

Definition 2.2 (Types of Asymptotes).

1. **Vertical Asymptote**: The line $x = a$ is a vertical asymptote of $y = f(x)$ if $\lim_{x\to a^+} f(x) = \pm\infty$ or $\lim_{x\to a^-} f(x) = \pm\infty$. For rational functions, these occur where the denominator is zero (and the numerator is non-zero).
2. **Horizontal Asymptote**: The line $y = L$ is a horizontal asymptote if $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$.
3. **Oblique (Slant) Asymptote**: The line $y = mx + c$ is an oblique asymptote if $\lim_{x\to\pm\infty}[f(x) - (mx + c)] = 0$. For a rational function $f(x) = P(x)/Q(x)$, an oblique asymptote exists if the degree of P is exactly one greater than the degree of Q.

2.3.1 Finding Oblique Asymptotes

The slope m and intercept c of an oblique asymptote $y = mx + c$ can be found by:

$$m = \lim_{x\to\infty} \frac{f(x)}{x} \quad \text{and} \quad c = \lim_{x\to\infty} [f(x) - mx]$$

Example 2.2. Find all asymptotes of $f(x) = \frac{2x^2-x-3}{x-2}$.

- **Vertical Asymptote**: Denominator is zero at $x = 2$. The numerator at $x = 2$ is $2(4) - 2 - 3 = 3 \neq 0$. So, $x = 2$ is a vertical asymptote.
- **Horizontal Asymptote**: Degree of numerator (2) is greater than degree of denominator (1), so there is no horizontal asymptote.
- **Oblique Asymptote**: Since deg(num) = deg(den) + 1, an oblique asymptote exists. We can find it by polynomial long division:

$$\frac{2x^2 - x - 3}{x - 2} = (2x + 3) + \frac{3}{x - 2}$$

  As $x \to \infty$, the remainder term $\frac{3}{x-2} \to 0$. So the curve approaches the line $y = 2x + 3$. The oblique asymptote is $y = 2x + 3$.

Practice Problems: Geometry and Asymptotes

Problem 2.1. Find the equation of the tangent plane to the elliptic paraboloid $z = 3x^2 + y^2 + 1$ at the point $(-1, 2, 8)$.

Solution 2.1.1. Let $f(x, y) = 3x^2 + y^2 + 1$. The point is $(x_0, y_0, z_0) = (-1, 2, 8)$. First, calculate the partial derivatives at the point. $f_x = 6x \implies f_x(-1, 2) = 6(-1) = -6$. $f_y = 2y \implies f_y(-1, 2) = 2(2) = 4$. Using the tangent plane formula $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$: $z - 8 = -6(x - (-1)) + 4(y - 2)$ $z - 8 = -6(x + 1) + 4(y - 2)$ $z - 8 = -6x - 6 + 4y - 8$ $6x - 4y + z = -6$.

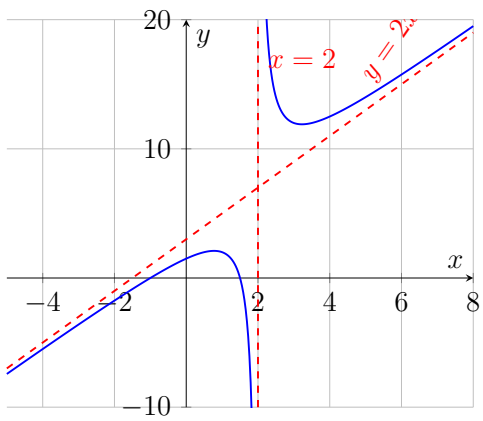

Figure 2: The graph of $f(x) = \frac{2x^2-x-3}{x-2}$ (blue) with its vertical asymptote at $x = 2$ and oblique asymptote $y = 2x + 3$ (red dashed).

Problem 2.2. Find all asymptotes of the curve $y = \frac{x^3}{x^2-1}$.

Solution 2.2.1. Vertical Asymptotes: The denominator is $x^2-1 = (x-1)(x+1) = 0$, so $x = 1$ and $x = -1$ are vertical asymptotes. **Horizontal Asymptote**: The degree of the numerator (3) is greater than the degree of the denominator (2), so there is no horizontal asymptote. **Oblique Asymptote**: Since deg(num) = deg(den) + 1, there is an oblique asymptote. We use long division:

$$\frac{x^3}{x^2-1} = x + \frac{x}{x^2-1}$$

As $x \to \infty$, the term $\frac{x}{x^2-1} \to 0$. So, the oblique asymptote is the line $y = x$.

Summary of Geometric Applications

- **Tangent Planes**: Using partial derivatives f_x and f_y as slopes in the x and y directions, we can define a plane that is tangent to a surface at a point. This is the best linear approximation to the surface near that point.
- **Normal Lines**: The line perpendicular to the tangent plane is the normal line. Its direction vector is given by the gradient.
- **Asymptotes**: These are lines that a curve approaches for large values of x or at points of discontinuity. They are essential for understanding the global behavior and sketching the graph of a function.

Differential and Integral Calculus

Class Notes on Integration Theory

B.Sc (Engineering) in EEE
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1 Indefinite Integrals

1.1 Introduction to Indefinite Integrals

Learning Outcomes

Learning Outcomes:

1. Understand the concept of antiderivatives and indefinite integrals
2. Master the method of substitution for integration
3. Apply integration by parts effectively
4. Evaluate integrals involving special trigonometric functions
5. Integrate rational fractions using partial fraction decomposition
6. Develop problem-solving skills through diverse applications

Definition 1.1 (Indefinite Integral)**.** Let $f(x)$ be a function defined on an interval I. A function $F(x)$ is called an **antiderivative** or **primitive** of $f(x)$ on I if

$$F'(x) = f(x) \quad \text{for all } x \in I$$

The family of all antiderivatives of $f(x)$ is called the **indefinite integral** of $f(x)$ and is denoted by

$$\int f(x)\,dx = F(x) + C$$

where C is an arbitrary constant called the **constant of integration**.

Example 1.1. Find $\int x^2\,dx$.

Solution: We seek a function whose derivative is x^2. Since $\frac{d}{dx}\left(\frac{x^3}{3}\right) = x^2$, we have

$$\int x^2\,dx = \frac{x^3}{3} + C$$

1.1.1 Basic Integration Formulas

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1 \tag{1}$$

$$\int \frac{1}{x} \, dx = \ln|x| + C \tag{2}$$

$$\int e^x \, dx = e^x + C \tag{3}$$

$$\int a^x \, dx = \frac{a^x}{\ln a} + C, \quad a > 0, a \neq 1 \tag{4}$$

$$\int \sin x \, dx = -\cos x + C \tag{5}$$

$$\int \cos x \, dx = \sin x + C \tag{6}$$

$$\int \sec^2 x \, dx = \tan x + C \tag{7}$$

$$\int \csc^2 x \, dx = -\cot x + C \tag{8}$$

$$\int \sec x \tan x \, dx = \sec x + C \tag{9}$$

$$\int \csc x \cot x \, dx = -\csc x + C \tag{10}$$

$$\int \frac{1}{\sqrt{1-x^2}} \, dx = \arcsin x + C = -\arccos x + C' \tag{11}$$

$$\int \frac{1}{1+x^2} \, dx = \arctan x + C = -\text{arccot}\, x + C' \tag{12}$$

1.2 Method of Substitution

Definition 1.2 (Integration by Substitution)**.** The method of substitution (also called u-substitution) is based on the chain rule for differentiation. If $u = g(x)$ is a differentiable function, then

$$\int f(g(x))g'(x) \, dx = \int f(u) \, du = F(u) + C = F(g(x)) + C$$

where $F'(u) = f(u)$.

Procedure for Substitution:

1. Choose an appropriate substitution $u = g(x)$
2. Compute $du = g'(x) \, dx$
3. Rewrite the integral in terms of u
4. Integrate with respect to u
5. Substitute back to express the answer in terms of x

Example 1.2. Evaluate $\int 2x\cos(x^2)\,dx$.

Solution: Let $u = x^2$, then $du = 2x\,dx$. The integral becomes:

$$\begin{aligned}\int 2x\cos(x^2)\,dx &= \int \cos(u)\,du \\ &= \sin(u) + C \\ &= \sin(x^2) + C\end{aligned}$$

Example 1.3. Evaluate $\int \frac{e^{\sqrt{x}}}{\sqrt{x}}\,dx$.

Solution: Let $u = \sqrt{x}$, then $x = u^2$ and $dx = 2u\,du$. Also, $\sqrt{x} = u$. Thus:

$$\begin{aligned}\int \frac{e^{\sqrt{x}}}{\sqrt{x}}\,dx &= \int \frac{e^u}{u} \cdot 2u\,du \\ &= 2\int e^u\,du \\ &= 2e^u + C \\ &= 2e^{\sqrt{x}} + C\end{aligned}$$

Theorem 1.1 (Substitution Theorem)**.** *Let g be a differentiable function on an interval I and let f be continuous on the range of g. If F is an antiderivative of f, then*

$$\int f(g(x))g'(x)\,dx = F(g(x)) + C$$

Proof. By the chain rule, if $F'(u) = f(u)$, then

$$\frac{d}{dx}[F(g(x))] = F'(g(x)) \cdot g'(x) = f(g(x)) \cdot g'(x)$$

Therefore, $F(g(x))$ is an antiderivative of $f(g(x))g'(x)$, which proves the theorem. □

1.3 Integration by Parts

Definition 1.3 (Integration by Parts Formula)**.** If u and v are differentiable functions of x, then

$$\int u\,dv = uv - \int v\,du$$

This formula is derived from the product rule for differentiation:

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$$

Integrating both sides gives the integration by parts formula.

Theorem 1.2 (Integration by Parts)**.** *If $u(x)$ and $v(x)$ are continuously differentiable functions, then*

$$\int u(x)v'(x)\,dx = u(x)v(x) - \int u'(x)v(x)\,dx$$

Proof. From the product rule: $(uv)' = u'v + uv'$
Integrating both sides:

$$\int (uv)'\,dx = \int u'v\,dx + \int uv'\,dx$$

$$uv = \int u'v\,dx + \int uv'\,dx$$

Rearranging: $\int uv'\,dx = uv - \int u'v\,dx$ □

LIATE Rule for choosing u:

- **L**ogarithmic functions
- **I**nverse trigonometric functions
- **A**lgebraic functions (polynomials)
- **T**rigonometric functions
- **E**xponential functions

Choose u as the function that appears first in this list.

Example 1.4. Evaluate $\int x \sin x\,dx$.
Solution: Let $u = x$ (algebraic) and $dv = \sin x\,dx$
Then $du = dx$ and $v = -\cos x$
Using integration by parts:

$$\begin{aligned}\int x\sin x\,dx &= x(-\cos x) - \int (-\cos x)\,dx \\ &= -x\cos x + \int \cos x\,dx \\ &= -x\cos x + \sin x + C\end{aligned}$$

Example 1.5. Evaluate $\int \ln x\,dx$.
Solution: Let $u = \ln x$ and $dv = dx$
Then $du = \frac{1}{x}\,dx$ and $v = x$
Using integration by parts:

$$\begin{aligned}\int \ln x\,dx &= x\ln x - \int x \cdot \frac{1}{x}\,dx \\ &= x\ln x - \int 1\,dx \\ &= x\ln x - x + C\end{aligned}$$

Example 1.6 (Repeated Integration by Parts)**.** Evaluate $\int x^2 e^x\,dx$.
Solution: First application: Let $u = x^2$, $dv = e^x\,dx$, so $du = 2x\,dx$, $v = e^x$

$$\int x^2 e^x\,dx = x^2 e^x - \int 2xe^x\,dx = x^2 e^x - 2\int xe^x\,dx$$

Second application: For $\int xe^x\,dx$, let $u = x$, $dv = e^x\,dx$, so $du = dx$, $v = e^x$

$$\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x$$

Combining:

$$\begin{aligned}\int x^2e^x\,dx &= x^2e^x - 2(xe^x - e^x) + C\\ &= x^2e^x - 2xe^x + 2e^x + C\\ &= e^x(x^2 - 2x + 2) + C\end{aligned}$$

1.4 Special Trigonometric Integrals

1.4.1 Integrals of the form $\int \sin^m x \cos^n x\,dx$

Case 1: If m is odd, let $u = \cos x$
Case 2: If n is odd, let $u = \sin x$
Case 3: If both m and n are even, use half-angle formulas:

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

Example 1.7. Evaluate $\int \sin^3 x \cos^2 x\,dx$.
Solution: Since the power of sine is odd, write $\sin^3 x = \sin^2 x \cdot \sin x = (1 - \cos^2 x)\sin x$
Let $u = \cos x$, then $du = -\sin x\,dx$

$$\begin{aligned}\int \sin^3 x \cos^2 x\,dx &= \int (1 - \cos^2 x)\cos^2 x \sin x\,dx\\ &= -\int (1 - u^2)u^2\,du\\ &= -\int (u^2 - u^4)\,du\\ &= -\left(\frac{u^3}{3} - \frac{u^5}{5}\right) + C\\ &= -\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C\end{aligned}$$

Example 1.8. Evaluate $\int \sin^2 x \cos^2 x\,dx$.
Solution: Using the identity $\sin^2 x \cos^2 x = \frac{1}{4}\sin^2 2x = \frac{1}{4} \cdot \frac{1 - \cos 4x}{2}$:

$$\begin{aligned}\int \sin^2 x \cos^2 x\,dx &= \int \frac{1 - \cos 4x}{8}\,dx\\ &= \frac{1}{8}\left(x - \frac{\sin 4x}{4}\right) + C\\ &= \frac{x}{8} - \frac{\sin 4x}{32} + C\end{aligned}$$

1.4.2 Integrals of the form $\int \tan^m x \sec^n x \, dx$

Case 1: If n is even, let $u = \tan x$
Case 2: If m is odd, let $u = \sec x$
Case 3: If m is even and n is odd, use $\tan^2 x = \sec^2 x - 1$

Example 1.9. Evaluate $\int \tan^2 x \sec^4 x \, dx$.
Solution: Let $u = \tan x$, then $du = \sec^2 x \, dx$
Also, $\sec^2 x = 1 + \tan^2 x = 1 + u^2$

$$\begin{aligned}
\int \tan^2 x \sec^4 x \, dx &= \int \tan^2 x \sec^2 x \cdot \sec^2 x \, dx \\
&= \int u^2(1+u^2) \, du \\
&= \int (u^2 + u^4) \, du \\
&= \frac{u^3}{3} + \frac{u^5}{5} + C \\
&= \frac{\tan^3 x}{3} + \frac{\tan^5 x}{5} + C
\end{aligned}$$

1.4.3 Trigonometric Substitution

For integrals involving $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, or $\sqrt{x^2 - a^2}$:

Expression	Substitution	Identity
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$\sec^2 \theta - 1 = \tan^2 \theta$

Example 1.10. Evaluate $\int \frac{dx}{\sqrt{9-x^2}}$.
Solution: Let $x = 3 \sin \theta$, then $dx = 3 \cos \theta \, d\theta$
$\sqrt{9 - x^2} = \sqrt{9 - 9 \sin^2 \theta} = 3 \cos \theta$

$$\begin{aligned}
\int \frac{dx}{\sqrt{9-x^2}} &= \int \frac{3 \cos \theta \, d\theta}{3 \cos \theta} \\
&= \int d\theta \\
&= \theta + C \\
&= \arcsin\left(\frac{x}{3}\right) + C
\end{aligned}$$

1.5 Integration of Rational Fractions

Definition 1.4 (Rational Function)**.** A **rational function** is a function of the form

$$R(x) = \frac{P(x)}{Q(x)}$$

where $P(x)$ and $Q(x)$ are polynomials and $Q(x) \neq 0$.

Steps for Integrating Rational Functions:

1. If degree of $P(x) \geq$ degree of $Q(x)$, perform polynomial long division
2. Factor $Q(x)$ completely
3. Decompose into partial fractions
4. Integrate each term separately

1.5.1 Partial Fraction Decomposition

Type 1: Distinct Linear Factors
If $Q(x) = (x - a_1)(x - a_2) \cdots (x - a_n)$ with distinct a_i, then

$$\frac{P(x)}{Q(x)} = \frac{A_1}{x - a_1} + \frac{A_2}{x - a_2} + \cdots + \frac{A_n}{x - a_n}$$

Type 2: Repeated Linear Factors
If $Q(x)$ contains $(x - a)^k$, then include terms

$$\frac{A_1}{x - a} + \frac{A_2}{(x - a)^2} + \cdots + \frac{A_k}{(x - a)^k}$$

Type 3: Irreducible Quadratic Factors
If $Q(x)$ contains $x^2 + bx + c$ (irreducible), include

$$\frac{Ax + B}{x^2 + bx + c}$$

Example 1.11. Evaluate $\int \frac{2x+1}{x^2-x-2}\, dx$.
Solution: Factor: $x^2 - x - 2 = (x - 2)(x + 1)$
Partial fractions: $\frac{2x+1}{(x-2)(x+1)} = \frac{A}{x-2} + \frac{B}{x+1}$
Multiply by $(x - 2)(x + 1)$: $2x + 1 = A(x + 1) + B(x - 2)$
Setting $x = 2$: $5 = 3A \Rightarrow A = \frac{5}{3}$
Setting $x = -1$: $-1 = -3B \Rightarrow B = \frac{1}{3}$
Therefore:

$$\begin{aligned}\int \frac{2x + 1}{x^2 - x - 2}\, dx &= \int \left(\frac{5/3}{x - 2} + \frac{1/3}{x + 1} \right) dx \\ &= \frac{5}{3} \ln |x - 2| + \frac{1}{3} \ln |x + 1| + C\end{aligned}$$

Example 1.12. Evaluate $\int \frac{x^2+2x+3}{(x+1)(x^2+1)}\, dx$.
Solution: Partial fractions: $\frac{x^2+2x+3}{(x+1)(x^2+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$
Multiply by $(x + 1)(x^2 + 1)$: $x^2 + 2x + 3 = A(x^2 + 1) + (Bx + C)(x + 1)$
Setting $x = -1$: $2 = 2A \Rightarrow A = 1$
Expanding: $x^2 + 2x + 3 = Ax^2 + A + Bx^2 + Bx + Cx + C$
Comparing coefficients:

- x^2: $1 = A + B \Rightarrow B = 0$
- x^1: $2 = B + C \Rightarrow C = 2$

- x^0: $3 = A + C$ (checks)

Therefore:

$$\begin{aligned}\int \frac{x^2+2x+3}{(x+1)(x^2+1)}\,dx &= \int \left(\frac{1}{x+1} + \frac{2}{x^2+1}\right) dx \\ &= \ln|x+1| + 2\arctan x + C\end{aligned}$$

1.6 Practice Problems: Indefinite Integrals

Mathematical Questions Set 1

Problem 1: Evaluate $\int \frac{x^3}{\sqrt{1+x^2}}\,dx$

Solution: Let $u = 1 + x^2$, then $du = 2x\,dx$ and $x^2 = u - 1$

$$\begin{aligned}\int \frac{x^3}{\sqrt{1+x^2}}\,dx &= \int \frac{x^2 \cdot x}{\sqrt{1+x^2}}\,dx \\ &= \frac{1}{2}\int \frac{u-1}{\sqrt{u}}\,du \\ &= \frac{1}{2}\int (u^{1/2} - u^{-1/2})\,du \\ &= \frac{1}{2}\left(\frac{2u^{3/2}}{3} - 2u^{1/2}\right) + C \\ &= \frac{u^{3/2}}{3} - u^{1/2} + C \\ &= \frac{(1+x^2)^{3/2}}{3} - \sqrt{1+x^2} + C\end{aligned}$$

Problem 2: Evaluate $\int x^2 \ln x\,dx$

Solution: Using integration by parts with $u = \ln x$ and $dv = x^2\,dx$:
$du = \frac{1}{x}\,dx$ and $v = \frac{x^3}{3}$

$$\begin{aligned}\int x^2 \ln x\,dx &= \frac{x^3}{3}\ln x - \int \frac{x^3}{3}\cdot\frac{1}{x}\,dx \\ &= \frac{x^3 \ln x}{3} - \frac{1}{3}\int x^2\,dx \\ &= \frac{x^3 \ln x}{3} - \frac{x^3}{9} + C\end{aligned}$$

Problem 3: Evaluate $\int \sin^5 x\,dx$

Solution: Write $\sin^5 x = \sin^4 x \cdot \sin x = (1 - \cos^2 x)^2 \sin x$
Let $u = \cos x$, then $du = -\sin x\,dx$

$$\begin{aligned}\int \sin^5 x\,dx &= -\int (1-u^2)^2\,du \\ &= -\int (1 - 2u^2 + u^4)\,du \\ &= -\left(u - \frac{2u^3}{3} + \frac{u^5}{5}\right) + C \\ &= -\cos x + \frac{2\cos^3 x}{3} - \frac{\cos^5 x}{5} + C\end{aligned}$$

Problem 4: Evaluate $\int \frac{3x+5}{x^2+4x+3}\,dx$
Solution: Factor: $x^2+4x+3=(x+1)(x+3)$
Partial fractions: $\frac{3x+5}{(x+1)(x+3)}=\frac{A}{x+1}+\frac{B}{x+3}$
$3x+5=A(x+3)+B(x+1)$
$x=-1$: $2=2A\Rightarrow A=1$
$x=-3$: $-4=-2B\Rightarrow B=2$

$$\begin{aligned}\int \frac{3x+5}{x^2+4x+3}\,dx &= \int\left(\frac{1}{x+1}+\frac{2}{x+3}\right)dx\\ &= \ln|x+1|+2\ln|x+3|+C\end{aligned}$$

Problem 5: Evaluate $\int e^{2x}\cos 3x\,dx$
Solution: Let $I=\int e^{2x}\cos 3x\,dx$
First integration by parts: $u=\cos 3x$, $dv=e^{2x}\,dx$
$du=-3\sin 3x\,dx$, $v=\frac{e^{2x}}{2}$

$$I=\frac{e^{2x}\cos 3x}{2}+\frac{3}{2}\int e^{2x}\sin 3x\,dx$$

Second integration by parts: $u=\sin 3x$, $dv=e^{2x}\,dx$
$du=3\cos 3x\,dx$, $v=\frac{e^{2x}}{2}$

$$I=\frac{e^{2x}\cos 3x}{2}+\frac{3}{2}\left[\frac{e^{2x}\sin 3x}{2}-\frac{3}{2}\int e^{2x}\cos 3x\,dx\right]$$

$$\begin{aligned}I&=\frac{e^{2x}\cos 3x}{2}+\frac{3e^{2x}\sin 3x}{4}-\frac{9}{4}I\\ I+\frac{9}{4}I&=\frac{e^{2x}(2\cos 3x+3\sin 3x)}{4}\\ \frac{13}{4}I&=\frac{e^{2x}(2\cos 3x+3\sin 3x)}{4}\\ I&=\frac{e^{2x}(2\cos 3x+3\sin 3x)}{13}+C\end{aligned}$$

Application-Oriented Problems Set 1

Problem 1 (Engineering - Velocity to Displacement): A particle moves along a straight line with velocity $v(t)=3t^2-4t+2$ m/s. If the particle is at position $s=5$ m when $t=0$, find the position function $s(t)$.

Solution: Since $v(t)=\frac{ds}{dt}$, we have $s(t)=\int v(t)\,dt$

$$\begin{aligned}s(t)&=\int(3t^2-4t+2)\,dt\\ &=t^3-2t^2+2t+C\end{aligned}$$

Using initial condition $s(0)=5$: $5=0-0+0+C\Rightarrow C=5$

Therefore: $s(t)=t^3-2t^2+2t+5$ meters

Problem 2 (Electrical Engineering - Charge from Current): The current in an electrical circuit is given by $i(t)=0.5e^{-2t}\sin 4t$ amperes. Find the charge $Q(t)$ that has flowed through the circuit from $t=0$ to time t, given that $Q(0)=0$.

Solution: Since $i=\frac{dQ}{dt}$, we have $Q=\int i\,dt$

Let $I = \int e^{-2t} \sin 4t \, dt$
Using integration by parts twice (similar to Problem 5 above):
First: $u = \sin 4t$, $dv = e^{-2t}\, dt$, gives $du = 4\cos 4t\, dt$, $v = -\frac{e^{-2t}}{2}$
Second: $u = \cos 4t$, $dv = e^{-2t}\, dt$
After solving: $I = \frac{e^{-2t}(-2\sin 4t - 4\cos 4t)}{20} + C = \frac{-e^{-2t}(\sin 4t + 2\cos 4t)}{10} + C$
Therefore: $Q(t) = 0.5 \cdot \frac{-e^{-2t}(\sin 4t + 2\cos 4t)}{10} + C$
Using $Q(0) = 0$: $0 = 0.5 \cdot \frac{-(1)(0+2)}{10} + C \Rightarrow C = 0.1$
$Q(t) = 0.1 - \frac{e^{-2t}(\sin 4t + 2\cos 4t)}{20}$ coulombs

Problem 3 (Signal Processing - Power Calculation): In signal processing, the power of a signal $x(t) = A\sin(\omega t)$ over one period is proportional to $\int_0^T x^2(t)\, dt$ where $T = \frac{2\pi}{\omega}$. Find this integral.

Solution:

$$\int_0^T A^2 \sin^2(\omega t)\, dt = A^2 \int_0^{2\pi/\omega} \sin^2(\omega t)\, dt$$

Using $\sin^2 u = \frac{1-\cos 2u}{2}$:

$$\begin{aligned}
&= A^2 \int_0^{2\pi/\omega} \frac{1-\cos(2\omega t)}{2}\, dt \\
&= \frac{A^2}{2}\left[t - \frac{\sin(2\omega t)}{2\omega}\right]_0^{2\pi/\omega} \\
&= \frac{A^2}{2}\left[\frac{2\pi}{\omega} - 0\right] \\
&= \frac{\pi A^2}{\omega}
\end{aligned}$$

Problem 4 (Mechanical Engineering - Work Done): A force $F(x) = \frac{k}{x^2}$ acts on an object moving from position $x = a$ to $x = b$. Express the work done as an indefinite integral and evaluate it.

Solution: Work $= \int F(x)\, dx = \int \frac{k}{x^2}\, dx = k \int x^{-2}\, dx$

$$\begin{aligned}
W &= k\left[\frac{x^{-1}}{-1}\right] + C \\
&= -\frac{k}{x} + C
\end{aligned}$$

For motion from a to b: $W = \left[-\frac{k}{x}\right]_a^b = -\frac{k}{b} + \frac{k}{a} = k\left(\frac{1}{a} - \frac{1}{b}\right)$

Problem 5 (Control Systems - Transfer Function): In control theory, finding the inverse Laplace transform involves partial fractions. Decompose $\frac{5s+3}{s^2+3s+2}$ into partial fractions.

Solution: Factor: $s^2 + 3s + 2 = (s+1)(s+2)$
$\frac{5s+3}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$
$5s + 3 = A(s+2) + B(s+1)$
$s = -1$: $-2 = A \Rightarrow A = -2$
$s = -2$: $-7 = -B \Rightarrow B = 7$
Therefore: $\frac{5s+3}{s^2+3s+2} = \frac{-2}{s+1} + \frac{7}{s+2}$

Problem 6 (Thermodynamics - Heat Transfer): The rate of heat transfer through a rod is $\frac{dQ}{dt} = k(T_1 - T_0)e^{-\alpha t}$. Find the total heat transferred.

Solution:

$$Q = \int k(T_1 - T_0)e^{-\alpha t}\,dt = k(T_1 - T_0)\int e^{-\alpha t}\,dt$$

$$Q = k(T_1 - T_0)\cdot\frac{e^{-\alpha t}}{-\alpha} + C = -\frac{k(T_1 - T_0)}{\alpha}e^{-\alpha t} + C$$

Problem 7 (Probability - Cumulative Distribution): A probability density function is given by $f(x) = \frac{2x}{9}$ for $0 \le x \le 3$. Find the cumulative distribution function $F(x) = \int_0^x f(t)\,dt$.

Solution:

$$F(x) = \int_0^x \frac{2t}{9}\,dt = \frac{2}{9}\left[\frac{t^2}{2}\right]_0^x = \frac{2}{9}\cdot\frac{x^2}{2} = \frac{x^2}{9}$$

For $0 \le x \le 3$: $F(x) = \frac{x^2}{9}$
Verification: $F(3) = \frac{9}{9} = 1$

Summary

Key Points - Indefinite Integrals:

- The indefinite integral represents a family of antiderivatives differing by a constant
- **Substitution method** is effective when the integrand contains a function and its derivative
- **Integration by parts** follows from the product rule; use LIATE to choose u
- For **trigonometric integrals**, use identities and appropriate substitutions based on powers
- **Partial fractions** decompose rational functions into simpler fractions that can be integrated term by term
- Always verify your answer by differentiation
- Integration requires practice and pattern recognition

2 Definite Integrals

2.1 Introduction to Definite Integrals

Learning Outcomes

Learning Outcomes:

1. Understand the concept of definite integrals and their geometric interpretation
2. Master the Fundamental Theorem of Calculus
3. Apply properties of definite integrals effectively
4. Evaluate definite integrals using various techniques
5. Derive and apply reduction formulas
6. Solve engineering problems involving definite integrals

Definition 2.1 (Definite Integral)**.** Let f be a function defined on $[a, b]$. The **definite integral** of f from a to b is denoted by

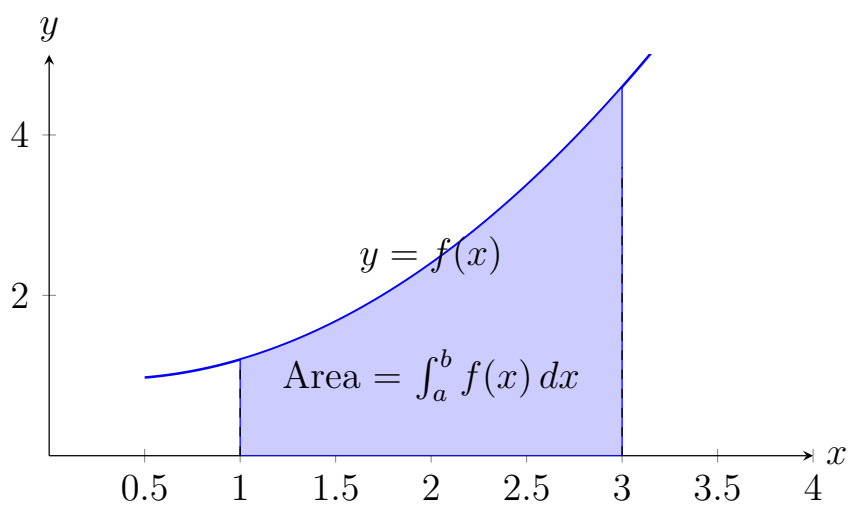
$$\int_a^b f(x)\,dx$$

and represents the signed area between the curve $y = f(x)$ and the x-axis from $x = a$ to $x = b$.

More precisely, partition $[a, b]$ into n subintervals of width $\Delta x = \frac{b-a}{n}$. The definite integral is the limit of Riemann sums:

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$$

where x_i^* is a sample point in the i-th subinterval.



2.2 Fundamental Theorem of Calculus

Theorem 2.1 (Fundamental Theorem of Calculus - Part 1)**.** *If f is continuous on $[a, b]$ and $F(x) = \int_a^x f(t)\,dt$, then F is differentiable on (a, b) and*

$$F'(x) = f(x)$$

Proof. By definition of the derivative:

$$F'(x) = \lim_{h\to 0} \frac{F(x+h) - F(x)}{h} = \lim_{h\to 0} \frac{1}{h}\left(\int_a^{x+h} f(t)\,dt - \int_a^x f(t)\,dt\right)$$

Using additivity of integrals:

$$F'(x) = \lim_{h\to 0} \frac{1}{h}\int_x^{x+h} f(t)\,dt$$

By the Mean Value Theorem for Integrals, there exists $c \in [x, x+h]$ such that:

$$\int_x^{x+h} f(t)\,dt = f(c)\cdot h$$

Therefore:

$$F'(x) = \lim_{h\to 0} \frac{f(c)\cdot h}{h} = \lim_{h\to 0} f(c)$$

As $h \to 0$, $c \to x$, and by continuity of f: $\lim_{c\to x} f(c) = f(x)$
Thus, $F'(x) = f(x)$. □

Theorem 2.2 (Fundamental Theorem of Calculus - Part 2)**.** *If f is continuous on $[a, b]$ and F is any antiderivative of f on $[a, b]$, then*

$$\int_a^b f(x)\,dx = F(b) - F(a) = [F(x)]_a^b$$

Proof. Let $G(x) = \int_a^x f(t)\,dt$. By Part 1, $G'(x) = f(x)$.
Since F is also an antiderivative of f, we have $F'(x) = f(x)$.
Therefore, $G'(x) = F'(x)$, which means $G(x) = F(x) + C$ for some constant C.
Evaluating at $x = a$:

$$G(a) = \int_a^a f(t)\,dt = 0 = F(a) + C$$

Thus, $C = -F(a)$, so $G(x) = F(x) - F(a)$.
Therefore:

$$\int_a^b f(x)\,dx = G(b) = F(b) - F(a)$$

□

Example 2.1. Evaluate $\int_0^2 (3x^2 + 2x)\,dx$.
Solution: An antiderivative is $F(x) = x^3 + x^2$
By FTC Part 2:

$$\begin{aligned}\int_0^2 (3x^2 + 2x)\,dx &= [x^3 + x^2]_0^2 \\ &= (2^3 + 2^2) - (0^3 + 0^2) \\ &= 8 + 4 - 0 \\ &= 12\end{aligned}$$

2.3 Properties of Definite Integrals

Theorem 2.3 (Properties of Definite Integrals)**.** *Let f and g be continuous functions on $[a, b]$, and let c be a constant. Then:*

1. ***Linearity:***

$$\int_a^b [cf(x) + dg(x)]\,dx = c\int_a^b f(x)\,dx + d\int_a^b g(x)\,dx$$

2. ***Additivity:*** *For $a < c < b$,*

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

3. ***Reversal of Limits:***

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$

4. ***Zero Width:***

$$\int_a^a f(x)\,dx = 0$$

5. ***Comparison:*** *If $f(x) \leq g(x)$ on $[a, b]$, then*

$$\int_a^b f(x)\,dx \leq \int_a^b g(x)\,dx$$

6. ***Absolute Value:***

$$\left|\int_a^b f(x)\,dx\right| \leq \int_a^b |f(x)|\,dx$$

7. ***Even and Odd Functions:***

$$\text{If } f(-x) = f(x) \text{ (even):} \quad \int_{-a}^a f(x)\,dx = 2\int_0^a f(x)\,dx$$

$$\text{If } f(-x) = -f(x) \text{ (odd):} \quad \int_{-a}^a f(x)\,dx = 0$$

Example 2.2. Evaluate $\int_{-2}^2 (x^3 - 3x)\,dx$.

Solution: Notice that $f(x) = x^3 - 3x$ is an odd function since

$$f(-x) = (-x)^3 - 3(-x) = -x^3 + 3x = -(x^3 - 3x) = -f(x)$$

By the property of odd functions:

$$\int_{-2}^2 (x^3 - 3x)\,dx = 0$$

Example 2.3. Evaluate $\int_{-1}^1 (x^4 + \cos x)\,dx$.

Solution: $f(x) = x^4 + \cos x$ is even since $f(-x) = (-x)^4 + \cos(-x) = x^4 + \cos x = f(x)$

Therefore:

$$\begin{aligned}\int_{-1}^{1}(x^4+\cos x)\,dx &= 2\int_0^1 (x^4+\cos x)\,dx\\ &= 2\left[\frac{x^5}{5}+\sin x\right]_0^1\\ &= 2\left(\frac{1}{5}+\sin 1-0\right)\\ &= \frac{2}{5}+2\sin 1\end{aligned}$$

2.4 Evaluation of Definite Integrals

2.4.1 Direct Application of FTC

Example 2.4. Evaluate $\int_1^4 \sqrt{x}\,dx$.

Solution:

$$\int_1^4 \sqrt{x}\,dx = \int_1^4 x^{1/2}\,dx = \left[\frac{x^{3/2}}{3/2}\right]_1^4 = \left[\frac{2x^{3/2}}{3}\right]_1^4$$

$$= \frac{2(4)^{3/2}}{3} - \frac{2(1)^{3/2}}{3} = \frac{2\cdot 8}{3} - \frac{2}{3} = \frac{16-2}{3} = \frac{14}{3}$$

2.4.2 Substitution in Definite Integrals

Theorem 2.4 (Substitution for Definite Integrals)**.** *If $u = g(x)$ where g' is continuous on $[a,b]$ and f is continuous on the range of g, then*

$$\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$$

Example 2.5. Evaluate $\int_0^{\pi/2} \sin^2 x\cos x\,dx$.

Solution: Let $u = \sin x$, then $du = \cos x\,dx$

When $x = 0$: $u = \sin 0 = 0$

When $x = \pi/2$: $u = \sin(\pi/2) = 1$

Therefore:

$$\begin{aligned}\int_0^{\pi/2} \sin^2 x\cos x\,dx &= \int_0^1 u^2\,du\\ &= \left[\frac{u^3}{3}\right]_0^1\\ &= \frac{1}{3}-0 = \frac{1}{3}\end{aligned}$$

Example 2.6. Evaluate $\int_0^1 \frac{x}{(1+x^2)^2}\,dx$.

Solution: Let $u = 1+x^2$, then $du = 2x\,dx$, so $x\,dx = \frac{1}{2}\,du$

When $x = 0$: $u = 1$

When $x = 1$: $u = 2$

Therefore:

$$\begin{aligned}\int_0^1 \frac{x}{(1+x^2)^2}\,dx &= \int_1^2 \frac{1}{u^2}\cdot\frac{1}{2}\,du \\ &= \frac{1}{2}\int_1^2 u^{-2}\,du \\ &= \frac{1}{2}\left[-u^{-1}\right]_1^2 \\ &= \frac{1}{2}\left(-\frac{1}{2}+1\right) \\ &= \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}\end{aligned}$$

2.4.3 Integration by Parts for Definite Integrals

For definite integrals:

$$\int_a^b u\,dv = [uv]_a^b - \int_a^b v\,du$$

Example 2.7. Evaluate $\int_1^e x\ln x\,dx$.
Solution: Let $u = \ln x$ and $dv = x\,dx$
Then $du = \frac{1}{x}\,dx$ and $v = \frac{x^2}{2}$

$$\begin{aligned}\int_1^e x\ln x\,dx &= \left[\frac{x^2\ln x}{2}\right]_1^e - \int_1^e \frac{x^2}{2}\cdot\frac{1}{x}\,dx \\ &= \left[\frac{x^2\ln x}{2}\right]_1^e - \frac{1}{2}\int_1^e x\,dx \\ &= \left[\frac{x^2\ln x}{2}\right]_1^e - \frac{1}{2}\left[\frac{x^2}{2}\right]_1^e \\ &= \left(\frac{e^2\cdot 1}{2} - \frac{1\cdot 0}{2}\right) - \frac{1}{4}(e^2-1) \\ &= \frac{e^2}{2} - \frac{e^2}{4} + \frac{1}{4} \\ &= \frac{e^2+1}{4}\end{aligned}$$

2.5 Reduction Formulas

Definition 2.2 (Reduction Formula)**.** A **reduction formula** is a formula that expresses an integral involving a parameter n in terms of an integral involving a smaller value of n. These formulas are useful for evaluating integrals with integer powers.

2.5.1 Reduction Formula for $\int \sin^n x\,dx$

Theorem 2.5. *For $n \geq 2$:*

$$I_n = \int \sin^n x\,dx = -\frac{\sin^{n-1} x\cos x}{n} + \frac{n-1}{n} I_{n-2}$$

Proof. Write $\sin^n x = \sin^{n-1} x \cdot \sin x$
Let $u = \sin^{n-1} x$ and $dv = \sin x\, dx$
Then $du = (n-1)\sin^{n-2} x \cos x\, dx$ and $v = -\cos x$

$$\begin{aligned}
I_n &= -\sin^{n-1} x \cos x + \int (n-1)\sin^{n-2} x \cos^2 x\, dx \\
&= -\sin^{n-1} x \cos x + (n-1)\int \sin^{n-2} x(1-\sin^2 x)\, dx \\
&= -\sin^{n-1} x \cos x + (n-1)\int \sin^{n-2} x\, dx - (n-1)\int \sin^n x\, dx \\
&= -\sin^{n-1} x \cos x + (n-1)I_{n-2} - (n-1)I_n
\end{aligned}$$

Solving for I_n:

$$\begin{aligned}
I_n + (n-1)I_n &= -\sin^{n-1} x \cos x + (n-1)I_{n-2} \\
nI_n &= -\sin^{n-1} x \cos x + (n-1)I_{n-2} \\
I_n &= -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}
\end{aligned}$$

□

2.5.2 Reduction Formula for $\int \cos^n x\, dx$

Theorem 2.6. *For $n \geq 2$:*

$$J_n = \int \cos^n x\, dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} J_{n-2}$$

Proof is similar to the sine case.

2.5.3 Reduction Formula for $\int x^n e^x\, dx$

Theorem 2.7. *For $n \geq 1$:*

$$K_n = \int x^n e^x\, dx = x^n e^x - nK_{n-1}$$

Proof. Let $u = x^n$ and $dv = e^x\, dx$
Then $du = nx^{n-1}\, dx$ and $v = e^x$

$$\begin{aligned}
K_n &= x^n e^x - \int e^x \cdot nx^{n-1}\, dx \\
&= x^n e^x - n\int x^{n-1} e^x\, dx \\
&= x^n e^x - nK_{n-1}
\end{aligned}$$

□

Example 2.8. Use the reduction formula to evaluate $\int_0^{\pi/2} \sin^4 x\, dx$.

Solution: Using $I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}$ with $n = 4$:

$$I_4 = -\frac{\sin^3 x \cos x}{4} + \frac{3}{4} I_2$$

For I_2:

$$I_2 = -\frac{\sin x \cos x}{2} + \frac{1}{2} I_0$$

where $I_0 = \int dx = x$

Therefore:

$$I_2 = -\frac{\sin x \cos x}{2} + \frac{x}{2}$$

Substituting back:

$$I_4 = -\frac{\sin^3 x \cos x}{4} + \frac{3}{4}\left(-\frac{\sin x \cos x}{2} + \frac{x}{2}\right)$$

For the definite integral:

$$\begin{aligned}\int_0^{\pi/2} \sin^4 x\, dx &= \left[-\frac{\sin^3 x \cos x}{4}\right]_0^{\pi/2} + \frac{3}{4}\left[-\frac{\sin x \cos x}{2} + \frac{x}{2}\right]_0^{\pi/2} \\ &= 0 + \frac{3}{4}\left(0 + \frac{\pi}{4}\right) \\ &= \frac{3\pi}{16}\end{aligned}$$

2.5.4 Wallis's Formula

Theorem 2.8 (Wallis's Formula)**.** *For $n \geq 2$:*

$$\int_0^{\pi/2} \sin^n x\, dx = \int_0^{\pi/2} \cos^n x\, dx = \begin{cases} \frac{(n-1)(n-3)\cdots 2}{n(n-2)\cdots 3} \cdot \frac{\pi}{2} & \textit{if } n \textit{ is even} \\ \frac{(n-1)(n-3)\cdots 2}{n(n-2)\cdots 3} \cdot 1 & \textit{if } n \textit{ is odd} \end{cases}$$

Example 2.9. Evaluate $\int_0^{\pi/2} \sin^6 x\, dx$ using Wallis's formula.

Solution: Since $n = 6$ is even:

$$\int_0^{\pi/2} \sin^6 x\, dx = \frac{5 \cdot 3 \cdot 1}{6 \cdot 4 \cdot 2} \cdot \frac{\pi}{2} = \frac{15}{48} \cdot \frac{\pi}{2} = \frac{5\pi}{32}$$

2.6 Practice Problems: Definite Integrals

Mathematical Questions Set 2

Problem 1: Evaluate $\int_0^1 x^2\sqrt{1-x^2}\,dx$
Solution: Let $x = \sin\theta$, then $dx = \cos\theta\,d\theta$
When $x = 0$: $\theta = 0$; When $x = 1$: $\theta = \pi/2$
$\sqrt{1-x^2} = \cos\theta$

$$\begin{aligned}
\int_0^1 x^2\sqrt{1-x^2}\,dx &= \int_0^{\pi/2} \sin^2\theta\cos\theta\cdot\cos\theta\,d\theta \\
&= \int_0^{\pi/2} \sin^2\theta\cos^2\theta\,d\theta \\
&= \int_0^{\pi/2} \frac{1}{4}\sin^2 2\theta\,d\theta \\
&= \frac{1}{4}\int_0^{\pi/2} \frac{1-\cos 4\theta}{2}\,d\theta \\
&= \frac{1}{8}\left[\theta - \frac{\sin 4\theta}{4}\right]_0^{\pi/2} \\
&= \frac{1}{8}\cdot\frac{\pi}{2} = \frac{\pi}{16}
\end{aligned}$$

Problem 2: Evaluate $\int_0^{\ln 2} e^{2x}\sqrt{1+e^{2x}}\,dx$
Solution: Let $u = 1+e^{2x}$, then $du = 2e^{2x}\,dx$, so $e^{2x}\,dx = \frac{1}{2}\,du$
When $x = 0$: $u = 2$; When $x = \ln 2$: $u = 1+4 = 5$

$$\begin{aligned}
\int_0^{\ln 2} e^{2x}\sqrt{1+e^{2x}}\,dx &= \int_2^5 \sqrt{u}\cdot\frac{1}{2}\,du \\
&= \frac{1}{2}\int_2^5 u^{1/2}\,du \\
&= \frac{1}{2}\left[\frac{2u^{3/2}}{3}\right]_2^5 \\
&= \frac{1}{3}[5^{3/2} - 2^{3/2}] \\
&= \frac{1}{3}[5\sqrt{5} - 2\sqrt{2}]
\end{aligned}$$

Problem 3: Evaluate $\int_0^{\pi/4} \frac{\sin x}{\cos^3 x}\,dx$
Solution: Let $u = \cos x$, then $du = -\sin x\,dx$
When $x = 0$: $u = 1$; When $x = \pi/4$: $u = \frac{1}{\sqrt{2}}$

$$\begin{aligned}
\int_0^{\pi/4} \frac{\sin x}{\cos^3 x}\,dx &= -\int_1^{1/\sqrt{2}} \frac{1}{u^3}\,du \\
&= \int_{1/\sqrt{2}}^1 u^{-3}\,du \\
&\overset{22}{=} \left[-\frac{u^{-2}}{2}\right]_{1/\sqrt{2}}^1 \\
&= -\frac{1}{\ }(1-2) = \frac{1}{\ }
\end{aligned}$$

Application-Oriented Problems Set 2

Problem 1 (Area Under Curve): Find the area bounded by the curve $y = x^2$, the x-axis, and the lines $x = 0$ and $x = 2$.
Solution: The area is given by:

$$A = \int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3} - 0 = \frac{8}{3} \text{ square units}$$

Problem 2 (Displacement from Velocity): A particle moves with velocity $v(t) = t^2 - 4t + 3$ m/s for $0 \le t \le 4$. Find the total displacement and total distance traveled.
Solution: Displacement:

$$s = \int_0^4 (t^2 - 4t + 3)\,dt = \left[\frac{t^3}{3} - 2t^2 + 3t\right]_0^4 = \frac{64}{3} - 32 + 12 = \frac{4}{3} \text{ m}$$

Distance: First find when $v(t) = 0$:

$$t^2 - 4t + 3 = 0 \Rightarrow (t-1)(t-3) = 0 \Rightarrow t = 1, 3$$

$v(t) > 0$ on $(0, 1)$ and $(3, 4)$; $v(t) < 0$ on $(1, 3)$

$$\begin{aligned}
\text{Distance} &= \int_0^1 (t^2 - 4t + 3)\,dt + \left|\int_1^3 (t^2 - 4t + 3)\,dt\right| + \int_3^4 (t^2 - 4t + 3)\,dt \\
&= \left[\frac{t^3}{3} - 2t^2 + 3t\right]_0^1 - \left[\frac{t^3}{3} - 2t^2 + 3t\right]_1^3 + \left[\frac{t^3}{3} - 2t^2 + 3t\right]_3^4 \\
&= \frac{4}{3} - \left(-\frac{4}{3}\right) + \frac{4}{3} = 4 \text{ m}
\end{aligned}$$

Problem 3 (Work Done by Variable Force): A force $F(x) = 10 + 3x^2$ N acts on an object moving from $x = 0$ to $x = 4$ m. Calculate the work done.
Solution:

$$W = \int_0^4 (10 + 3x^2)\,dx = [10x + x^3]_0^4 = 40 + 64 = 104 \text{ J}$$

Problem 4 (Average Value): Find the average value of the function $f(x) = \sin x$ on the interval $[0, \pi]$.
Solution: The average value is:

$$f_{avg} = \frac{1}{\pi - 0}\int_0^\pi \sin x\,dx = \frac{1}{\pi}[-\cos x]_0^\pi = \frac{1}{\pi}[1 - (-1)] = \frac{2}{\pi}$$

Problem 5 (Center of Mass): A thin rod of length L has linear density $\rho(x) = kx$ where k is constant. Find the x-coordinate of the center of mass.
Solution: The center of mass is:

$$\bar{x} = \frac{\int_0^L x\rho(x)\,dx}{\int_0^L \rho(x)\,dx} = \frac{\int_0^L kx^2\,dx}{\int_0^L kx\,dx} = \frac{k[x^3/3]_0^L}{k[x^2/2]_0^L} = \frac{L^3/3}{L^2/2} = \frac{2L}{3}$$

Problem 6 (Volume of Revolution - Disk Method): Find the volume obtained by rotating the region under $y = \sqrt{x}$ from $x = 0$ to $x = 4$ about the x-axis.

Summary

Key Points - Definite Integrals:

- The definite integral represents the signed area between a curve and the x-axis
- The **Fundamental Theorem of Calculus** connects differentiation and integration
- Use **properties of definite integrals** (linearity, additivity, symmetry) to simplify calculations
- For **even functions**: $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$
- For **odd functions**: $\int_{-a}^{a} f(x)\,dx = 0$
- When using substitution, change the limits of integration accordingly
- **Reduction formulas** are powerful for evaluating powers of trigonometric and exponential functions
- Definite integrals have numerous applications in physics, engineering, and probability

3 Comprehensive Problem Set

Challenge Problems

Problem 1: Evaluate $\int_0^{\pi/2} \frac{\sin x}{\sin x+\cos x}\,dx$
Hint: Use the property that $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$
Solution: Let $I = \int_0^{\pi/2} \frac{\sin x}{\sin x+\cos x}\,dx$
Using the substitution property with $u = \frac{\pi}{2} - x$: $I = \int_0^{\pi/2} \frac{\sin(\pi/2-x)}{\sin(\pi/2-x)+\cos(\pi/2-x)}\,dx = \int_0^{\pi/2} \frac{\cos x}{\cos x+\sin x}\,dx$
Adding both expressions: $2I = \int_0^{\pi/2} \frac{\sin x+\cos x}{\sin x+\cos x}\,dx = \int_0^{\pi/2} 1\,dx = \frac{\pi}{2}$
Therefore: $I = \frac{\pi}{4}$
Problem 2: Find $\int \frac{dx}{(x^2+1)(x^2+4)}$
Solution: Using partial fractions with irreducible quadratics: $\frac{1}{(x^2+1)(x^2+4)} = \frac{Ax+B}{x^2+1} + \frac{Cx+D}{x^2+4}$
Multiplying: $1 = (Ax+B)(x^2+4) + (Cx+D)(x^2+1)$
Comparing coefficients:

- x^3: $0 = A + C$
- x^2: $0 = B + D$
- x^1: $0 = 4A + C$
- x^0: $1 = 4B + D$

From equations: $A = 0$, $C = 0$, $B = \frac{1}{3}$, $D = -\frac{1}{3}$
Therefore: $\int \frac{dx}{(x^2+1)(x^2+4)} = \frac{1}{3}\int \frac{dx}{x^2+1} - \frac{1}{3}\int \frac{dx}{x^2+4} = \frac{1}{3}\arctan x - \frac{1}{6}\arctan \frac{x}{2} + C$
Problem 3: Evaluate $\int_0^{\pi/2} x\sin x\,dx$
Solution: Using integration by parts with $u = x$ and $dv = \sin x\,dx$:
$du = dx$ and $v = -\cos x$

$$\begin{aligned}\int_0^{\pi/2} x\sin x\,dx &= [x(-\cos x)]_0^{\pi/2} - \int_0^{\pi/2} (-\cos x)\,dx \\ &= [0-0] + \int_0^{\pi/2} \cos x\,dx \\ &= [\sin x]_0^{\pi/2} \\ &= 1 - 0 = 1\end{aligned}$$

Problem 4: Prove that $\int_0^{\pi} xf(\sin x)\,dx = \frac{\pi}{2}\int_0^{\pi} f(\sin x)\,dx$
Solution: Let $I = \int_0^{\pi} xf(\sin x)\,dx$
Using substitution $u = \pi - x$, $du = -dx$:
When $x = 0$: $u = \pi$; When $x = \pi$: $u = 0$
Also, $\sin(\pi - u) = \sin u$

$$\begin{aligned}I &= -\int_{\pi}^{0} (\pi-u)f(\sin(\pi-u))\,du \\ &= \int_0^{\pi} (\pi-u)f(\sin u)\,du \\ &= \pi\int_0^{\pi} f(\sin u)\,du - \int_0^{\pi} uf(\sin u)\,du \\ &= \int_0^{\pi}\end{aligned}$$

Additional Advanced Problems

Problem 6: Evaluate $\int_0^1 \frac{\ln(1+x)}{1+x^2}\,dx$
Solution: This is known as a challenging integral. Using integration by parts with $u = \ln(1+x)$ and $dv = \frac{dx}{1+x^2}$:
$du = \frac{dx}{1+x}$ and $v = \arctan x$

$$\begin{aligned}\int_0^1 \frac{\ln(1+x)}{1+x^2}\,dx &= [\ln(1+x)\arctan x]_0^1 - \int_0^1 \frac{\arctan x}{1+x}\,dx \\ &= \ln 2 \cdot \frac{\pi}{4} - \int_0^1 \frac{\arctan x}{1+x}\,dx\end{aligned}$$

The remaining integral can be evaluated using advanced techniques (series or special functions). The final answer is: $\int_0^1 \frac{\ln(1+x)}{1+x^2}\,dx = \frac{\pi \ln 2}{8}$
Problem 7: Find the area between $y = \sin x$ and $y = \cos x$ from $x = 0$ to $x = \pi/2$
Solution: First find where the curves intersect: $\sin x = \cos x \Rightarrow \tan x = 1 \Rightarrow x = \pi/4$
For $0 \le x \le \pi/4$: $\cos x \ge \sin x$
For $\pi/4 \le x \le \pi/2$: $\sin x \ge \cos x$

$$\begin{aligned}A &= \int_0^{\pi/4} (\cos x - \sin x)\,dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x)\,dx \\ &= [\sin x + \cos x]_0^{\pi/4} + [-\cos x - \sin x]_{\pi/4}^{\pi/2} \\ &= \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} - 0 - 1\right) + \left(0 - 1 + \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right) \\ &= \sqrt{2} - 1 + \sqrt{2} - 1 \\ &= 2\sqrt{2} - 2 = 2(\sqrt{2} - 1) \text{ square units}\end{aligned}$$

4 Advanced Topics and Extensions

4.1 Improper Integrals (Brief Introduction)

Definition 4.1 (Improper Integral - Type I)**.** If f is continuous on $[a, \infty)$, then $\int_a^\infty f(x)\,dx = \lim_{t\to\infty} \int_a^t f(x)\,dx$ if this limit exists.

Example 4.1. Evaluate $\int_1^\infty \frac{1}{x^2}\,dx$
Solution:

$$\begin{aligned}\int_1^\infty \frac{1}{x^2}\,dx &= \lim_{t\to\infty} \int_1^t x^{-2}\,dx \\ &= \lim_{t\to\infty} \left[-x^{-1}\right]_1^t \\ &= \lim_{t\to\infty} \left(-\frac{1}{t} + 1\right) \\ &= 0 + 1 = 1\end{aligned}$$

This integral converges to 1.

4.2 Numerical Integration Methods

When integrals cannot be evaluated analytically, numerical methods are used:

4.2.1 Trapezoidal Rule

$\int_a^b f(x)\,dx \approx \frac{b-a}{2n}[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)]$
where $x_i = a + i\Delta x$ and $\Delta x = \frac{b-a}{n}$.

4.2.2 Simpson's Rule

$\int_a^b f(x)\,dx \approx \frac{b-a}{3n}[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 4f(x_{n-1}) + f(x_n)]$
where n must be even.

Example 4.2. Approximate $\int_0^1 e^{-x^2}\,dx$ using the Trapezoidal Rule with $n = 4$.
Solution: $\Delta x = \frac{1-0}{4} = 0.25$
$x_0 = 0, x_1 = 0.25, x_2 = 0.5, x_3 = 0.75, x_4 = 1$

$$\begin{aligned}\int_0^1 e^{-x^2}\,dx &\approx \frac{0.25}{2}[e^0 + 2e^{-0.0625} + 2e^{-0.25} + 2e^{-0.5625} + e^{-1}] \\ &\approx 0.125[1 + 2(0.9394) + 2(0.7788) + 2(0.5698) + 0.3679] \\ &\approx 0.125[1 + 1.8788 + 1.5576 + 1.1396 + 0.3679] \\ &\approx 0.125 \times 5.9439 \\ &\approx 0.743\end{aligned}$$

Technique	**When to Use**
Direct Integration	Standard forms, basic functions
Substitution	Integrand contains $f(g(x))g'(x)$
Integration by Parts	Products of different function types
Trigonometric Identities	Powers of trig functions
Trigonometric Substitution	$\sqrt{a^2 \pm x^2}$ or $\sqrt{x^2 - a^2}$
Partial Fractions	Rational functions
Reduction Formulas	High powers of functions

5 Summary and Study Guide

5.1 Integration Techniques Summary

5.2 Key Formulas to Remember

Integration by Parts: $\int u\,dv = uv - \int v\,du$

Substitution: $\int f(g(x))g'(x)\,dx = \int f(u)\,du, \quad u = g(x)$

Fundamental Theorem of Calculus: $\int_a^b f(x)\,dx = F(b) - F(a), \quad$ where $F'(x) = f(x)$

Properties of Definite Integrals:

- Linearity: $\int_a^b [cf(x) + dg(x)]\,dx = c\int_a^b f(x)\,dx + d\int_a^b g(x)\,dx$
- Additivity: $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$
- Even function: $\int_{-a}^a f(x)\,dx = 2\int_0^a f(x)\,dx$
- Odd function: $\int_{-a}^a f(x)\,dx = 0$

5.3 Common Mistakes to Avoid

1. **Forgetting the constant of integration** in indefinite integrals
2. **Not changing limits** when using substitution in definite integrals
3. **Incorrect choice of** u **and** dv in integration by parts
4. **Algebraic errors** in partial fraction decomposition
5. **Sign errors** when evaluating definite integrals
6. **Not checking** if a function is odd or even before integrating
7. **Forgetting absolute value** in $\int \frac{1}{x}\,dx = \ln|x| + C$

5.4 Problem-Solving Strategy

1. **Identify** the type of integral
2. **Check** for obvious simplifications (odd/even, symmetry)
3. **Choose** the appropriate technique

4. **Execute** the technique carefully
5. **Verify** your answer by differentiation
6. For definite integrals, **check** reasonableness (sign, magnitude)

6 Applications in Electrical Engineering

6.1 RMS Value of AC Signals

The Root Mean Square (RMS) value of a periodic signal $v(t)$ with period T is: $V_{RMS} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt}$

Example 6.1. Find the RMS value of $v(t) = V_0 \sin(\omega t)$ over one period $T = \frac{2\pi}{\omega}$.
Solution:

$$\begin{aligned} V_{RMS}^2 &= \frac{1}{T}\int_0^T V_0^2 \sin^2(\omega t)\,dt \\ &= \frac{V_0^2}{T}\int_0^T \frac{1-\cos(2\omega t)}{2}\,dt \\ &= \frac{V_0^2}{2T}\left[t - \frac{\sin(2\omega t)}{2\omega}\right]_0^T \\ &= \frac{V_0^2}{2T}\cdot T = \frac{V_0^2}{2} \end{aligned}$$

Therefore: $V_{RMS} = \frac{V_0}{\sqrt{2}} = \frac{V_0}{1.414} \approx 0.707V_0$

6.2 Energy Stored in Capacitors and Inductors

Capacitor: $E = \int_0^t vi\,dt = \int_0^Q V\,dq = \frac{1}{2}CV^2$
Inductor: $E = \int_0^t vi\,dt = \int_0^I Li\,di = \frac{1}{2}LI^2$

6.3 Fourier Series

The Fourier coefficients involve definite integrals: $a_n = \frac{2}{T}\int_0^T f(t)\cos(n\omega t)\,dt$ $b_n = \frac{2}{T}\int_0^T f(t)\sin(n\omega t)\,dt$

7 Exam Preparation Tips

7.1 Topic Checklist

Before the exam, ensure you can:

- ☐ Evaluate basic integrals from memory
- ☐ Apply substitution method correctly
- ☐ Use integration by parts (including repeated applications)
- ☐ Integrate trigonometric functions using identities

- ☐ Perform partial fraction decomposition
- ☐ Apply the Fundamental Theorem of Calculus
- ☐ Use properties of definite integrals (odd/even functions)
- ☐ Derive and apply reduction formulas
- ☐ Solve application problems (area, volume, work, etc.)

7.2 Practice Recommendations

1. Work through **all example problems** in these notes
2. Solve **at least 5 problems** from each section
3. Practice **mixed problems** to improve technique recognition
4. **Time yourself** on practice problems
5. **Review mistakes** and understand why they occurred
6. Form **study groups** to discuss difficult problems

8 Final Comprehensive Exercises

Mixed Practice Problems

1. Evaluate: $\int \frac{x+2}{\sqrt{x^2+4x+5}}\,dx$
2. Evaluate: $\int_0^{\pi/3} \tan^2 x\,dx$
3. Find: $\int x^3 \ln x\,dx$
4. Evaluate: $\int \frac{x^2+1}{x^3+3x^2+2x}\,dx$
5. Calculate: $\int_0^1 xe^{-x}\,dx$
6. Evaluate: $\int \sin^3 x \cos^4 x\,dx$
7. Find: $\int_0^{\pi/4} \sec^2 x \tan x\,dx$
8. Evaluate: $\int \frac{dx}{\sqrt{4-9x^2}}$
9. Calculate: $\int_{-2}^{2} (x^3 + x^2)\,dx$
10. Evaluate: $\int_1^e \frac{(\ln x)^2}{x}\,dx$

"Practice makes perfect in integration!"

Acknowledgments

These lecture notes have been prepared to support the learning objectives of the B.Sc (Engineering) in Electrical and Electronic Engineering program at Jamalpur Science & Technology University. The content has been carefully structured to provide a comprehensive understanding of integration theory with emphasis on both theoretical foundations and practical applications.

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END OF LECTURE NOTES

"The only way to learn mathematics is to do mathematics."
— Paul Halmos

Department of Mathematics
Jamalpur Science & Technology University
Academic Year 2024-2025

Differential and Integral Calculus

Integration of Special Functions

Complete Class Notes

Course Teacher: Dr. M Shahjalal
Department of Mathematics
Jamalpur Science & Technology University

Prepared for B.Sc. (Eng) in EEE Students

Academic Year 2025

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1 Introduction to Special Functions

Special functions are particular mathematical functions that have established names and notations due to their importance in mathematical analysis, functional analysis, geometry, physics, and other applications. In engineering, particularly in Electrical and Electronic Engineering, these functions appear frequently in signal processing, communications, control systems, and electromagnetic theory.

1.1 Course Learning Outcomes

Upon completion of this course on Integration of Special Functions, students will be able to:

1. Define and evaluate Beta and Gamma functions using various representations
2. Apply properties of Beta and Gamma functions to simplify complex integrals
3. Solve engineering problems involving special functions in signal processing and probability theory
4. Evaluate Gaussian integrals and apply them to communication systems
5. Use special functions in analyzing electrical circuits and electromagnetic systems
6. Apply special functions to solve differential equations arising in engineering
7. Implement numerical methods for evaluating special functions
8. Recognize and apply special functions in Laplace and Fourier transforms

2 Gamma Function

2.1 Learning Outcomes

After studying this section, students will be able to:

- Define the Gamma function and understand its convergence
- Evaluate Gamma function for various arguments
- Apply the recurrence relation $\Gamma(n+1) = n\Gamma(n)$
- Prove fundamental properties of the Gamma function
- Use Gamma function in engineering applications

2.2 Definition and Basic Concepts

Definition 2.1: Gamma Function

The **Gamma function** is defined by the improper integral:

$$\Gamma(n) = \int_0^\infty e^{-x} x^{n-1}\, dx, \quad n > 0 \tag{1}$$

This integral converges for all $n > 0$ and defines the Gamma function.

2.2.1 Key Terminology

- **Improper Integral**: An integral with one or both limits of integration being infinity, or where the integrand becomes unbounded.
- **Convergence**: The improper integral converges if the limit exists and is finite.
- **Domain**: The set of values for which the function is defined. For $\Gamma(n)$, the domain is $(0, \infty)$.
- **Extension**: The Gamma function can be extended to negative values (except negative integers) using analytic continuation.

Example 2.1. Evaluate $\Gamma(1)$.
Solution:

$$\begin{aligned}
\Gamma(1) &= \int_0^\infty e^{-x} x^{1-1}\, dx \\
&= \int_0^\infty e^{-x}\, dx \\
&= \left[-e^{-x}\right]_0^\infty \\
&= 0 - (-1) = 1
\end{aligned}$$

Therefore, $\Gamma(1) = 1$.

2.3 Visual Representation of Gamma Function

Description of Figure 1: The graph shows that the Gamma function is continuous and positive for $x > 0$. It has vertical asymptotes at $x = 0, -1, -2, -3, \ldots$ (non-positive integers). The function increases rapidly as x increases. For positive integers, $\Gamma(n) = (n-1)!$.

2.4 Fundamental Properties of Gamma Function

Theorem 2.1: Recurrence Relation

For all $n > 0$, the Gamma function satisfies:

$$\Gamma(n+1) = n\Gamma(n) \tag{2}$$

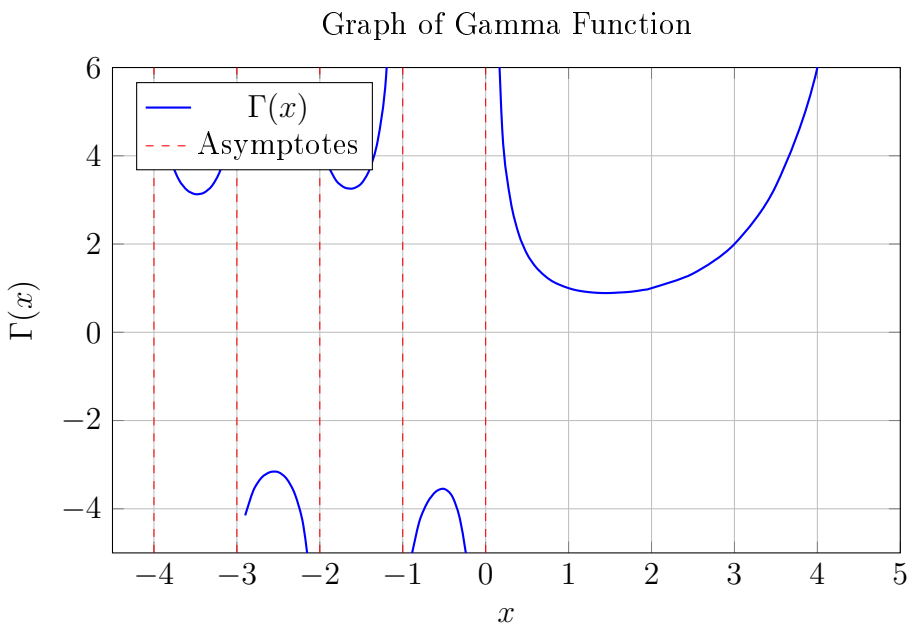

Figure 1: The Gamma function showing vertical asymptotes at non-positive integers

Proof. Using integration by parts with $u = x^n$ and $dv = e^{-x}dx$:

$$\begin{aligned}
\Gamma(n+1) &= \int_0^\infty e^{-x}x^n\,dx \\
&= \left[-e^{-x}x^n\right]_0^\infty + \int_0^\infty e^{-x}\cdot nx^{n-1}\,dx \\
&= 0 + n\int_0^\infty e^{-x}x^{n-1}\,dx \\
&= n\Gamma(n)
\end{aligned}$$

The boundary term vanishes because $\lim_{x\to\infty} x^n e^{-x} = 0$ and $\lim_{x\to 0^+} x^n e^{-x} = 0$. □

Corollary 2.1: Gamma Function for Positive Integers

For any positive integer n:

$$\Gamma(n) = (n-1)! \tag{3}$$

Proof. By repeated application of the recurrence relation:

$$\begin{aligned}
\Gamma(n) &= (n-1)\Gamma(n-1) \\
&= (n-1)(n-2)\Gamma(n-2) \\
&= (n-1)(n-2)(n-3)\cdots 2\cdot 1\cdot\Gamma(1) \\
&= (n-1)!\cdot 1 = (n-1)!
\end{aligned}$$

□

Theorem 2.2: Special Value

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \tag{4}$$

Proof. We need to evaluate $\Gamma(1/2) = \int_0^\infty e^{-x} x^{-1/2}\, dx$.

Let $x = t^2$, then $dx = 2t\, dt$:

$$\begin{aligned}\Gamma\left(\frac{1}{2}\right) &= \int_0^\infty e^{-t^2}(t^2)^{-1/2} \cdot 2t\, dt \\ &= 2\int_0^\infty e^{-t^2}\, dt\end{aligned}$$

Let $I = \int_0^\infty e^{-t^2}\, dt$. Then:

$$\begin{aligned}I^2 &= \left(\int_0^\infty e^{-x^2}\, dx\right)\left(\int_0^\infty e^{-y^2}\, dy\right) \\ &= \int_0^\infty \int_0^\infty e^{-(x^2+y^2)}\, dx\, dy\end{aligned}$$

Converting to polar coordinates ($x = r\cos\theta$, $y = r\sin\theta$):

$$\begin{aligned}I^2 &= \int_0^{\pi/2} \int_0^\infty e^{-r^2} r\, dr\, d\theta \\ &= \frac{\pi}{2} \cdot \frac{1}{2} = \frac{\pi}{4}\end{aligned}$$

Therefore, $I = \sqrt{\pi}/2$ and $\Gamma(1/2) = 2I = \sqrt{\pi}$. □

2.5 Additional Important Properties

1. **Legendre Duplication Formula:**

$$\Gamma(n)\Gamma\left(n + \frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2n-1}}\Gamma(2n) \tag{5}$$

2. **Reflection Formula (Euler's Reflection Formula):**

$$\Gamma(n)\Gamma(1-n) = \frac{\pi}{\sin(\pi n)}, \quad n \notin \mathbb{Z} \tag{6}$$

3. **Stirling's Approximation:** For large n:

$$\Gamma(n) \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n \tag{7}$$

2.6 Problems Applying Gamma Function Theorems

Problem 2.1. Evaluate $\Gamma(7)$ using the recurrence relation.

Solution: Using $\Gamma(n+1) = n\Gamma(n)$ repeatedly:

$$\begin{aligned}\Gamma(7) &= 6\Gamma(6) = 6 \cdot 5\Gamma(5) = 6 \cdot 5 \cdot 4\Gamma(4) \\ &= 6 \cdot 5 \cdot 4 \cdot 3\Gamma(3) = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2\Gamma(2) \\ &= 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1\Gamma(1) = 720 \cdot 1 = 720\end{aligned}$$

Problem 2.2. Prove that $\Gamma(3/2) = \frac{\sqrt{\pi}}{2}$.

Solution: Using the recurrence relation:

$$\begin{aligned}\Gamma\left(\frac{3}{2}\right) &= \Gamma\left(\frac{1}{2} + 1\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) \\ &= \frac{1}{2} \cdot \sqrt{\pi} = \frac{\sqrt{\pi}}{2}\end{aligned}$$

2.7 Five Mathematical Problems with Detailed Solutions

Problem 2.3. Evaluate $\int_0^\infty x^5 e^{-x}\,dx$.

Solution: Comparing with the Gamma function definition:

$$\begin{aligned}\int_0^\infty x^5 e^{-x}\,dx &= \int_0^\infty e^{-x} x^{6-1}\,dx = \Gamma(6)\\ &= 5! = 120\end{aligned}$$

Problem 2.4. Evaluate $\int_0^\infty x^3 e^{-2x}\,dx$.

Solution: Let $u = 2x$, then $x = u/2$ and $dx = du/2$:

$$\begin{aligned}\int_0^\infty x^3 e^{-2x}\,dx &= \int_0^\infty \left(\frac{u}{2}\right)^3 e^{-u} \cdot \frac{du}{2}\\ &= \frac{1}{16}\int_0^\infty u^3 e^{-u}\,du\\ &= \frac{1}{16}\Gamma(4) = \frac{1}{16}\cdot 6 = \frac{3}{8}\end{aligned}$$

Problem 2.5. Show that $\Gamma(5/2) = \frac{3\sqrt{\pi}}{4}$.

Solution: Using the recurrence relation:

$$\begin{aligned}\Gamma\left(\frac{5}{2}\right) &= \frac{3}{2}\Gamma\left(\frac{3}{2}\right) = \frac{3}{2}\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right)\\ &= \frac{3}{4}\sqrt{\pi} = \frac{3\sqrt{\pi}}{4}\end{aligned}$$

Problem 2.6. Evaluate $\int_0^\infty \sqrt{x}\,e^{-x}\,dx$.

Solution: Rewrite $\sqrt{x} = x^{1/2}$:

$$\begin{aligned}\int_0^\infty \sqrt{x}\,e^{-x}\,dx &= \int_0^\infty x^{1/2} e^{-x}\,dx\\ &= \int_0^\infty e^{-x} x^{3/2-1}\,dx = \Gamma\left(\frac{3}{2}\right)\\ &= \frac{\sqrt{\pi}}{2}\end{aligned}$$

Problem 2.7. Evaluate $\int_0^\infty x^n e^{-ax}\,dx$ where $a > 0$ and $n > -1$.

Solution: Let $u = ax$, then $x = u/a$ and $dx = du/a$:

$$\begin{aligned}\int_0^\infty x^n e^{-ax}\,dx &= \int_0^\infty \left(\frac{u}{a}\right)^n e^{-u} \cdot \frac{du}{a}\\ &= \frac{1}{a^{n+1}}\int_0^\infty u^n e^{-u}\,du\\ &= \frac{\Gamma(n+1)}{a^{n+1}}\end{aligned}$$

2.8 Ten Application-Oriented Problems for EEE Students

Problem 2.8. (Signal Processing) The impulse response of a system is given by $h(t) = t^2 e^{-3t}$ for $t \geq 0$. Find the total energy of the impulse response, given by $E = \int_0^\infty |h(t)|^2 \, dt$.

Solution:

$$E = \int_0^\infty (t^2 e^{-3t})^2 \, dt = \int_0^\infty t^4 e^{-6t} \, dt$$

Let $u = 6t$, then $t = u/6$ and $dt = du/6$:

$$\begin{aligned} E &= \int_0^\infty \left(\frac{u}{6}\right)^4 e^{-u} \cdot \frac{du}{6} \\ &= \frac{1}{6^5} \int_0^\infty u^4 e^{-u} \, du = \frac{\Gamma(5)}{7776} \\ &= \frac{4!}{7776} = \frac{24}{7776} = \frac{1}{324} \text{ units} \end{aligned}$$

Problem 2.9. (Probability Theory) In a Poisson process with rate λ, the probability density function of the waiting time for the n-th event is:

$$f(t) = \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!}, \quad t \geq 0$$

Verify that this is a valid probability density function by showing $\int_0^\infty f(t) \, dt = 1$.

Solution:

$$\begin{aligned} \int_0^\infty f(t) \, dt &= \int_0^\infty \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!} \, dt \\ &= \frac{\lambda^n}{(n-1)!} \int_0^\infty t^{n-1} e^{-\lambda t} \, dt \end{aligned}$$

Let $u = \lambda t$, then $t = u/\lambda$ and $dt = du/\lambda$:

$$\begin{aligned} &= \frac{\lambda^n}{(n-1)!} \int_0^\infty \left(\frac{u}{\lambda}\right)^{n-1} e^{-u} \cdot \frac{du}{\lambda} \\ &= \frac{\lambda^n}{(n-1)! \cdot \lambda^n} \int_0^\infty u^{n-1} e^{-u} \, du \\ &= \frac{1}{(n-1)!} \Gamma(n) = \frac{(n-1)!}{(n-1)!} = 1 \end{aligned}$$

Thus, the function is a valid PDF.

Problem 2.10. (Communication Systems) The error function, used in analyzing bit error rates, is defined as:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} \, dt$$

Express $\int_0^\infty e^{-t^2} \, dt$ in terms of the Gamma function.

Solution: Let $u = t^2$, then $t = \sqrt{u}$ and $dt = \frac{du}{2\sqrt{u}}$:

$$\begin{aligned}\int_0^\infty e^{-t^2}\,dt &= \int_0^\infty e^{-u} \cdot \frac{du}{2\sqrt{u}} \\ &= \frac{1}{2}\int_0^\infty e^{-u}u^{-1/2}\,du \\ &= \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}\end{aligned}$$

Problem 2.11. (Power Electronics) The average power dissipated in a resistor with current $i(t) = I_0 t e^{-\alpha t}$ for $t \geq 0$ through resistance R is:

$$P_{avg} = \frac{R}{T}\int_0^T i^2(t)\,dt$$

For $T \to \infty$ and $\alpha = 2\ \mathrm{s}^{-1}$, $I_0 = 1$ A, $R = 10\ \Omega$, find the total energy dissipated.

Solution:

$$E_{total} = R\int_0^\infty i^2(t)\,dt = RI_0^2\int_0^\infty t^2 e^{-2\alpha t}\,dt$$

Using the formula:

$$\int_0^\infty t^2 e^{-2\alpha t}\,dt = \frac{\Gamma(3)}{(2\alpha)^3} = \frac{2!}{8\alpha^3} = \frac{2}{8\alpha^3} = \frac{1}{4\alpha^3}$$

With $\alpha = 2$, $I_0 = 1$, $R = 10$:

$$E_{total} = 10 \cdot 1 \cdot \frac{1}{4 \cdot 8} = \frac{10}{32} = 0.3125\ \mathrm{J}$$

Problem 2.12. (Control Systems) The response of a second-order system to a unit step input involves the integral:

$$y(t) = \int_0^t \tau e^{-\zeta\omega_n\tau}\,d\tau$$

Evaluate this for $\zeta = 0.5$, $\omega_n = 2$ rad/s, and $t \to \infty$.

Solution:

$$\begin{aligned}y(\infty) &= \int_0^\infty \tau e^{-\zeta\omega_n\tau}\,d\tau \\ &= \int_0^\infty \tau e^{-\tau}\,d\tau \quad (\text{with } \zeta\omega_n = 1) \\ &= \Gamma(2) = 1! = 1\end{aligned}$$

Problem 2.13. (Electromagnetic Theory) The radiation pattern of an antenna involves the integral:

$$I = \int_0^{\pi/2} \sin^3\theta\,d\theta$$

Evaluate this using the Gamma function relation.

Solution: Using the formula for $\int_0^{\pi/2} \sin^{2m+1}\theta\, d\theta$:

$$\int_0^{\pi/2} \sin^{2m+1}\theta\, d\theta = \frac{\Gamma(m+1)\Gamma(1/2)}{\Gamma(m+3/2)}$$

For $m = 1$ (since $2m + 1 = 3$):

$$I = \frac{\Gamma(2)\Gamma(1/2)}{\Gamma(5/2)} = \frac{1 \cdot \sqrt{\pi}}{\frac{3\sqrt{\pi}}{4}} = \frac{4}{3}$$

Problem 2.14. (Filter Design) The frequency response of a Butterworth low-pass filter involves evaluating:

$$H(\omega) = \frac{1}{1 + (\omega/\omega_c)^{2n}}$$

The 3-dB bandwidth calculation requires $\int_0^\infty \frac{1}{1+x^{2n}}\, dx$. Express this in terms of Gamma functions for $n = 1$.

Solution: For $n = 1$:

$$\int_0^\infty \frac{1}{1+x^2}\, dx = [\arctan(x)]_0^\infty = \frac{\pi}{2}$$

This can be expressed using Gamma functions:

$$\int_0^\infty \frac{1}{1+x^2}\, dx = \frac{\pi}{2\sin(\pi/2)} = \frac{\pi}{2}$$

using the Beta-Gamma relation.

Problem 2.15. (Digital Signal Processing) The Gaussian window function used in filter design is:

$$w(n) = e^{-\frac{1}{2}(\alpha n/N)^2}$$

The energy of this window over $n = -\infty$ to ∞ involves:

$$E = \sum_{n=-\infty}^{\infty} w^2(n) \approx \int_{-\infty}^{\infty} e^{-(\alpha x/N)^2}\, dx$$

Evaluate this integral for $\alpha = 2.5$, $N = 10$.

Solution:

$$\begin{aligned} E &= \int_{-\infty}^{\infty} e^{-(\alpha x/N)^2}\, dx \\ &= 2\int_0^{\infty} e^{-(\alpha x/N)^2}\, dx \end{aligned}$$

Let $u = (\alpha x/N)^2$, then $x = \frac{N}{\alpha}\sqrt{u}$ and $dx = \frac{N}{2\alpha\sqrt{u}}\, du$:

$$\begin{aligned} E &= 2 \cdot \frac{N}{2\alpha}\int_0^{\infty} e^{-u} u^{-1/2}\, du \\ &= \frac{N}{\alpha}\Gamma\left(\frac{1}{2}\right) = \frac{N\sqrt{\pi}}{\alpha} \\ &= \frac{10\sqrt{\pi}}{2.5} = 4\sqrt{\pi} \approx 7.09 \end{aligned}$$

Problem 2.16. (Semiconductor Physics) The Fermi-Dirac distribution in semiconductors involves integrals of the form:

$$F_n(E) = \int_0^\infty \frac{x^n}{1 + e^{x - E_F/kT}}\, dx$$

For the special case where $E_F \gg kT$, this approximates to:

$$F_n \approx \int_0^\infty x^n e^{-x}\, dx$$

Evaluate F_2.

Solution:

$$F_2 = \int_0^\infty x^2 e^{-x}\, dx = \Gamma(3) = 2! = 2$$

Problem 2.17. (Radar Systems) The matched filter output for a Gaussian pulse is:

$$y(t) = \int_{-\infty}^\infty e^{-a(t-\tau)^2} e^{-b\tau^2}\, d\tau$$

For $a = b = 1$, evaluate this integral.

Solution: Expanding the exponents:

$$\begin{aligned} -a(t-\tau)^2 - b\tau^2 &= -a(t^2 - 2t\tau + \tau^2) - b\tau^2 \\ &= -(a+b)\tau^2 + 2at\tau - at^2 \end{aligned}$$

Completing the square:

$$\begin{aligned} y(t) &= e^{-at^2} \int_{-\infty}^\infty e^{-(a+b)\left(\tau - \frac{at}{a+b}\right)^2 + \frac{a^2t^2}{a+b}}\, d\tau \\ &= e^{-at^2} e^{\frac{a^2t^2}{a+b}} \sqrt{\frac{\pi}{a+b}} \end{aligned}$$

For $a = b = 1$:

$$y(t) = e^{-t^2} e^{t^2/2} \sqrt{\frac{\pi}{2}} = e^{-t^2/2} \sqrt{\frac{\pi}{2}}$$

2.9 Summary: Gamma Function

The Gamma function is a fundamental special function in mathematics and engineering, defined by:

$$\Gamma(n) = \int_0^\infty e^{-x} x^{n-1}\, dx, \quad n > 0$$

Key Points:

- The Gamma function generalizes the factorial function to non-integer values
- It satisfies the recurrence relation $\Gamma(n+1) = n\Gamma(n)$
- For positive integers: $\Gamma(n) = (n-1)!$

- Special value: $\Gamma(1/2) = \sqrt{\pi}$
- Applications in EEE: signal processing, probability distributions, communication systems, electromagnetic theory
- Used extensively in Laplace transforms, Fourier analysis, and solving differential equations

3 Beta Function

3.1 Learning Outcomes

After studying this section, students will be able to:

- Define the Beta function and understand its properties
- Relate Beta function to Gamma function
- Evaluate Beta integrals using various techniques
- Apply Beta function to solve engineering problems
- Use Beta function in probability theory and statistics

3.2 Definition and Basic Concepts

Definition 3.1: Beta Function

The **Beta function** is defined by:

$$B(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx, \quad m,n > 0 \tag{8}$$

This integral converges for all $m, n > 0$.

3.2.1 Alternative Forms of Beta Function

The Beta function can be expressed in several equivalent forms:

1. **Standard Form:**
$$B(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$$

2. **Trigonometric Form:**
$$B(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta\,d\theta$$

3. **Infinite Form:**
$$B(m,n) = \int_0^\infty \frac{t^{m-1}}{(1+t)^{m+n}}\,dt$$

Example 3.1. Evaluate $B(2,3)$ directly from the definition.
Solution:

$$\begin{aligned} B(2,3) &= \int_0^1 x^{2-1}(1-x)^{3-1}\,dx \\ &= \int_0^1 x(1-x)^2\,dx \\ &= \int_0^1 (x-2x^2+x^3)\,dx \\ &= \left[\frac{x^2}{2}-\frac{2x^3}{3}+\frac{x^4}{4}\right]_0^1 \\ &= \frac{1}{2}-\frac{2}{3}+\frac{1}{4} \\ &= \frac{6-8+3}{12} = \frac{1}{12} \end{aligned}$$

3.3 Visual Representation of Beta Function

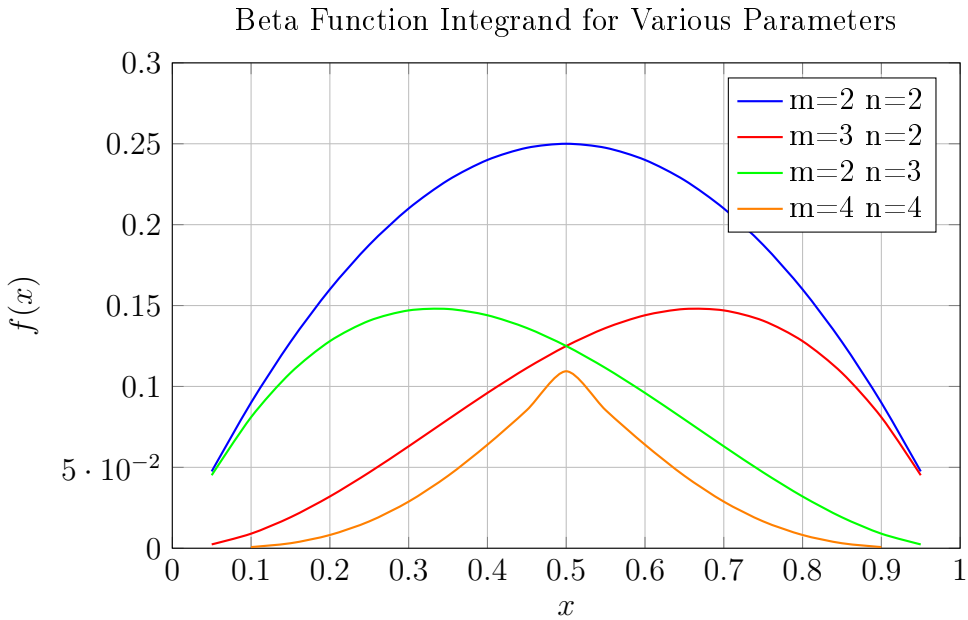

Figure 2: Beta function integrand for various values of m and n

Description of Figure 2: The integrand of the Beta function shows different shapes depending on the parameters m and n. When $m = n$, the function is symmetric about $x = 0.5$. For $m > n$, the peak shifts toward the right, and for $m < n$, it shifts toward the left. The area under each curve represents the value of $B(m,n)$.

3.4 Fundamental Properties of Beta Function

Theorem 3.1: Symmetry Property

The Beta function is symmetric in its arguments:

$$B(m,n) = B(n,m) \tag{9}$$

Proof. Let $x = 1 - u$ in the Beta function definition, then $dx = -du$:

$$\begin{aligned} B(m,n) &= \int_0^1 x^{m-1}(1-x)^{n-1}\,dx \\ &= \int_1^0 (1-u)^{m-1}u^{n-1} \cdot (-du) \\ &= \int_0^1 u^{n-1}(1-u)^{m-1}\,du \\ &= B(n,m) \end{aligned}$$

□

Theorem 3.2: Relation Between Beta and Gamma Functions

The Beta function can be expressed in terms of Gamma functions:

$$B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)} \tag{10}$$

Proof. Consider the product:

$$\begin{aligned} \Gamma(m)\Gamma(n) &= \int_0^\infty e^{-x}x^{m-1}\,dx \int_0^\infty e^{-y}y^{n-1}\,dy \\ &= \int_0^\infty \int_0^\infty e^{-(x+y)}x^{m-1}y^{n-1}\,dx\,dy \end{aligned}$$

Let $x = uv$ and $y = u(1-v)$ where $u \geq 0$ and $0 \leq v \leq 1$. The Jacobian is:

$$J = \begin{vmatrix} v & u \\ 1-v & -u \end{vmatrix} = -uv - u(1-v) = -u$$

Thus $dx\,dy = u\,du\,dv$, and:

$$\begin{aligned} \Gamma(m)\Gamma(n) &= \int_0^\infty \int_0^1 e^{-u}(uv)^{m-1}[u(1-v)]^{n-1}u\,dv\,du \\ &= \int_0^\infty e^{-u}u^{m+n-1}\,du \int_0^1 v^{m-1}(1-v)^{n-1}\,dv \\ &= \Gamma(m+n)B(m,n) \end{aligned}$$

Therefore:

$$B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$

□

Corollary 3.1: Beta Function for Integers

For positive integers m and n:

$$B(m,n) = \frac{(m-1)!(n-1)!}{(m+n-1)!} \tag{11}$$

Proof. Using the relation between Beta and Gamma functions:

$$\begin{aligned} B(m,n) &= \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)} \\ &= \frac{(m-1)!(n-1)!}{(m+n-1)!} \end{aligned}$$

□

3.5 Problems Applying Beta Function Theorems

Problem 3.1. Verify that $B(2,3) = \frac{1}{12}$ using the Gamma function relation.

Solution:

$$\begin{aligned} B(2,3) &= \frac{\Gamma(2)\Gamma(3)}{\Gamma(5)} \\ &= \frac{1! \cdot 2!}{4!} = \frac{1 \cdot 2}{24} = \frac{2}{24} = \frac{1}{12} \end{aligned}$$

This confirms our earlier direct computation.

Problem 3.2. Prove that $\int_0^{\pi/2} \sin^3\theta \, d\theta = \frac{2}{3}$ using the Beta function.

Solution: Using the trigonometric form with $2m-1 = 3$ and $2n-1 = 0$ (so $m = 2, n = 1/2$):

$$\begin{aligned} \int_0^{\pi/2} \sin^3\theta \, d\theta &= \frac{1}{2} B(2, 1/2) \\ &= \frac{1}{2} \cdot \frac{\Gamma(2)\Gamma(1/2)}{\Gamma(5/2)} \\ &= \frac{1}{2} \cdot \frac{1 \cdot \sqrt{\pi}}{\frac{3\sqrt{\pi}}{4}} \\ &= \frac{1}{2} \cdot \frac{4}{3} = \frac{2}{3} \end{aligned}$$

3.6 Five Mathematical Problems with Detailed Solutions

Problem 3.3. Evaluate $\int_0^1 x^4(1-x)^2 \, dx$.

Solution: Comparing with the Beta function:

$$\begin{aligned}\int_0^1 x^4(1-x)^2\,dx &= \int_0^1 x^{5-1}(1-x)^{3-1}\,dx \\ &= B(5,3) = \frac{\Gamma(5)\Gamma(3)}{\Gamma(8)} \\ &= \frac{4!\cdot 2!}{7!} = \frac{24\cdot 2}{5040} \\ &= \frac{48}{5040} = \frac{1}{105}\end{aligned}$$

Problem 3.4. Evaluate $\int_0^{\pi/2} \sin^5\theta\cos^3\theta\,d\theta$.

Solution: Using the trigonometric form with $2m-1=5$ and $2n-1=3$: This gives $m=3$ and $n=2$.

$$\begin{aligned}\int_0^{\pi/2} \sin^5\theta\cos^3\theta\,d\theta &= \frac{1}{2}B(3,2) \\ &= \frac{1}{2}\cdot\frac{\Gamma(3)\Gamma(2)}{\Gamma(5)} \\ &= \frac{1}{2}\cdot\frac{2!\cdot 1!}{4!} \\ &= \frac{1}{2}\cdot\frac{2}{24} = \frac{1}{24}\end{aligned}$$

Problem 3.5. Evaluate $\int_0^\infty \frac{x^2}{(1+x)^6}\,dx$.

Solution: Using the infinite form of Beta function with $m-1=2$ and $m+n=6$: This gives $m=3$ and $n=3$.

$$\begin{aligned}\int_0^\infty \frac{x^2}{(1+x)^6}\,dx &= B(3,3) \\ &= \frac{\Gamma(3)\Gamma(3)}{\Gamma(6)} \\ &= \frac{2!\cdot 2!}{5!} = \frac{4}{120} = \frac{1}{30}\end{aligned}$$

Problem 3.6. Show that $\int_0^1 \sqrt{x}(1-x)^3\,dx = \frac{8}{105}$.

Solution:

$$\begin{aligned}\int_0^1 \sqrt{x}(1-x)^3\,dx &= \int_0^1 x^{1/2}(1-x)^3\,dx \\ &= \int_0^1 x^{3/2-1}(1-x)^{4-1}\,dx \\ &= B\left(\frac{3}{2},4\right) \\ &= \frac{\Gamma(3/2)\Gamma(4)}{\Gamma(11/2)}\end{aligned}$$

We have:

$$\begin{aligned}\Gamma\left(\frac{11}{2}\right) &= \frac{9}{2}\Gamma\left(\frac{9}{2}\right) = \frac{9}{2}\cdot\frac{7}{2}\Gamma\left(\frac{7}{2}\right)\\ &= \frac{9}{2}\cdot\frac{7}{2}\cdot\frac{5}{2}\Gamma\left(\frac{5}{2}\right)\\ &= \frac{9\cdot 7\cdot 5}{8}\cdot\frac{3\sqrt{\pi}}{4} = \frac{945\sqrt{\pi}}{32}\end{aligned}$$

Therefore:

$$B\left(\frac{3}{2},4\right) = \frac{\frac{\sqrt{\pi}}{2}\cdot 6}{\frac{945\sqrt{\pi}}{32}} = \frac{3\sqrt{\pi}\cdot 32}{945\sqrt{\pi}} = \frac{96}{945} = \frac{32}{315}$$

Problem 3.7. Evaluate $\int_0^1 \frac{dx}{\sqrt{x(1-x)}}$.

Solution:

$$\begin{aligned}\int_0^1 \frac{dx}{\sqrt{x(1-x)}} &= \int_0^1 x^{-1/2}(1-x)^{-1/2}\,dx\\ &= B\left(\frac{1}{2},\frac{1}{2}\right)\\ &= \frac{\Gamma(1/2)\Gamma(1/2)}{\Gamma(1)}\\ &= \frac{\sqrt{\pi}\cdot\sqrt{\pi}}{1} = \pi\end{aligned}$$

3.7 Ten Application-Oriented Problems for EEE Students

Problem 3.8. (Probability Theory) The Beta distribution is used to model random variables limited to intervals of finite length. The probability density function is:

$$f(x;\alpha,\beta) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}, \quad 0 \le x \le 1$$

Verify this is a valid PDF by showing $\int_0^1 f(x;\alpha,\beta)\,dx = 1$.

Solution:

$$\begin{aligned}\int_0^1 f(x;\alpha,\beta)\,dx &= \int_0^1 \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}\,dx\\ &= \frac{1}{B(\alpha,\beta)}\int_0^1 x^{\alpha-1}(1-x)^{\beta-1}\,dx\\ &= \frac{B(\alpha,\beta)}{B(\alpha,\beta)} = 1\end{aligned}$$

Problem 3.9. (Signal Processing) Calculate the mean of the Beta distribution with parameters $\alpha = 3$ and $\beta = 2$:

$$E[X] = \int_0^1 x\cdot f(x;3,2)\,dx$$

Solution:

$$\begin{aligned}
E[X] &= \int_0^1 x \cdot \frac{x^{3-1}(1-x)^{2-1}}{B(3,2)}\,dx \\
&= \frac{1}{B(3,2)} \int_0^1 x^3(1-x)\,dx \\
&= \frac{1}{B(3,2)} B(4,2) \\
&= \frac{B(4,2)}{B(3,2)} = \frac{\Gamma(4)\Gamma(2)/\Gamma(6)}{\Gamma(3)\Gamma(2)/\Gamma(5)} \\
&= \frac{\Gamma(4)\Gamma(5)}{\Gamma(3)\Gamma(6)} = \frac{6 \cdot 24}{2 \cdot 120} = \frac{144}{240} = \frac{3}{5}
\end{aligned}$$

Alternatively, using the formula $E[X] = \frac{\alpha}{\alpha+\beta} = \frac{3}{5}$.

Problem 3.10. (Antenna Theory) The radiation pattern of a dipole antenna involves:

$$P(\theta) = \int_0^{\pi/2} \sin^2\theta \cos^2\theta\,d\theta$$

Evaluate this integral.

Solution: Using the trigonometric form with $2m-1=2$ and $2n-1=2$: This gives $m = 3/2$ and $n = 3/2$.

$$\begin{aligned}
\int_0^{\pi/2} \sin^2\theta \cos^2\theta\,d\theta &= \frac{1}{2} B\left(\frac{3}{2}, \frac{3}{2}\right) \\
&= \frac{1}{2} \cdot \frac{\Gamma(3/2)\Gamma(3/2)}{\Gamma(3)} \\
&= \frac{1}{2} \cdot \frac{(\sqrt{\pi}/2)^2}{2} \\
&= \frac{1}{2} \cdot \frac{\pi/4}{2} = \frac{\pi}{16}
\end{aligned}$$

Problem 3.11. (Communication Systems) In analyzing channel capacity, we encounter:

$$I = \int_0^1 x^{1/2}(1-x)^{1/2}\,dx$$

Evaluate this integral.

Solution:

$$\begin{aligned}
I &= B\left(\frac{3}{2}, \frac{3}{2}\right) \\
&= \frac{\Gamma(3/2)\Gamma(3/2)}{\Gamma(3)} \\
&= \frac{(\sqrt{\pi}/2)^2}{2} = \frac{\pi}{8}
\end{aligned}$$

Problem 3.12. (Power Systems) The voltage waveform in a certain power system is modeled by:

$$v(t) = V_0 \sin^3(\omega t), \quad 0 \le t \le \frac{\pi}{2\omega}$$

Find the RMS value: $V_{rms} = \sqrt{\frac{2\omega}{\pi} \int_0^{\pi/(2\omega)} v^2(t)\, dt}$.

Solution:

$$V_{rms}^2 = \frac{2\omega}{\pi} \int_0^{\pi/(2\omega)} V_0^2 \sin^6(\omega t)\, dt$$

Let $\theta = \omega t$:

$$\begin{aligned} V_{rms}^2 &= \frac{2V_0^2}{\pi} \int_0^{\pi/2} \sin^6 \theta\, d\theta \\ &= \frac{2V_0^2}{\pi} \cdot \frac{1}{2} B\left(\frac{7}{2}, \frac{1}{2}\right) \\ &= \frac{V_0^2}{\pi} \cdot \frac{\Gamma(7/2)\Gamma(1/2)}{\Gamma(4)} \\ &= \frac{V_0^2}{\pi} \cdot \frac{\frac{5\cdot 3\cdot 1}{8}\sqrt{\pi}\cdot\sqrt{\pi}}{6} \\ &= \frac{V_0^2}{\pi} \cdot \frac{15\pi}{48} = \frac{5V_0^2}{16} \end{aligned}$$

Therefore: $V_{rms} = \frac{V_0\sqrt{5}}{4}$

Problem 3.13. (Filter Design) In designing an analog filter, we need to evaluate:

$$H = \int_0^1 (1 - x^2)^{3/2}\, dx$$

Evaluate this integral using substitution and Beta function.

Solution: Let $x = \sin\theta$, then $dx = \cos\theta\, d\theta$:

$$\begin{aligned} H &= \int_0^{\pi/2} (1 - \sin^2\theta)^{3/2} \cos\theta\, d\theta \\ &= \int_0^{\pi/2} \cos^3\theta \cdot \cos\theta\, d\theta \\ &= \int_0^{\pi/2} \cos^4\theta\, d\theta \\ &= \frac{1}{2} B\left(\frac{1}{2}, \frac{5}{2}\right) \\ &= \frac{1}{2} \cdot \frac{\Gamma(1/2)\Gamma(5/2)}{\Gamma(3)} \\ &= \frac{1}{2} \cdot \frac{\sqrt{\pi}\cdot\frac{3\sqrt{\pi}}{4}}{2} \\ &= \frac{3\pi}{16} \end{aligned}$$

Problem 3.14. (Digital Communications) The bit error rate (BER) for binary phase-shift keying involves:

$$P_e = \int_0^1 e^{-\gamma x} x^{-1/2}(1-x)^{-1/2}\,dx$$

For small γ, approximate this using $e^{-\gamma x} \approx 1 - \gamma x$.

Solution:

$$\begin{aligned}
P_e &\approx \int_0^1 (1-\gamma x)x^{-1/2}(1-x)^{-1/2}\,dx \\
&= \int_0^1 x^{-1/2}(1-x)^{-1/2}\,dx - \gamma \int_0^1 x^{1/2}(1-x)^{-1/2}\,dx \\
&= B\left(\frac{1}{2},\frac{1}{2}\right) - \gamma B\left(\frac{3}{2},\frac{1}{2}\right) \\
&= \pi - \gamma \cdot \frac{\Gamma(3/2)\Gamma(1/2)}{\Gamma(2)} \\
&= \pi - \gamma \cdot \frac{\sqrt{\pi}/2 \cdot \sqrt{\pi}}{1} \\
&= \pi - \frac{\gamma\pi}{2} = \pi\left(1 - \frac{\gamma}{2}\right)
\end{aligned}$$

Problem 3.15. (Control Systems) The response characteristic involves:

$$R = \int_0^{\pi/2} \sin^{2n}\theta\,d\theta$$

Express this in terms of the Beta function for $n = 2$.

Solution: For $n = 2$:

$$\begin{aligned}
R &= \int_0^{\pi/2} \sin^4\theta\,d\theta \\
&= \frac{1}{2}B\left(\frac{5}{2},\frac{1}{2}\right) \\
&= \frac{1}{2} \cdot \frac{\Gamma(5/2)\Gamma(1/2)}{\Gamma(3)} \\
&= \frac{1}{2} \cdot \frac{\frac{3\sqrt{\pi}}{4} \cdot \sqrt{\pi}}{2} \\
&= \frac{3\pi}{16}
\end{aligned}$$

Problem 3.16. (Electromagnetic Theory) The magnetic field distribution in a solenoid involves:

$$B = \int_0^1 (1-r^2)^2 r\,dr$$

where r is the normalized radial distance. Evaluate this integral.

Solution:

$$B = \int_0^1 r(1-r^2)^2\,dr$$

Let $u = r^2$, then $r = \sqrt{u}$ and $dr = \frac{du}{2\sqrt{u}}$:

$$\begin{aligned} B &= \int_0^1 \sqrt{u}(1-u)^2 \cdot \frac{du}{2\sqrt{u}} \\ &= \frac{1}{2}\int_0^1 (1-u)^2\, du \\ &= \frac{1}{2}\left[-\frac{(1-u)^3}{3}\right]_0^1 \\ &= \frac{1}{2}\cdot\frac{1}{3} = \frac{1}{6} \end{aligned}$$

Problem 3.17. (Semiconductor Devices) In analyzing p-n junction characteristics:

$$I = \int_0^1 x^{1/2}(1-x)^{3/2}\, dx$$

Evaluate using the Beta function.

Solution:

$$\begin{aligned} I &= B\left(\frac{3}{2}, \frac{5}{2}\right) \\ &= \frac{\Gamma(3/2)\Gamma(5/2)}{\Gamma(4)} \\ &= \frac{\frac{\sqrt{\pi}}{2}\cdot\frac{3\sqrt{\pi}}{4}}{6} \\ &= \frac{3\pi/8}{6} = \frac{3\pi}{48} = \frac{\pi}{16} \end{aligned}$$

3.8 Summary: Beta Function

The Beta function is defined by:

$$B(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\, dx, \quad m,n > 0$$

Key Points:

- The Beta function is symmetric: $B(m,n) = B(n,m)$
- Relation to Gamma function: $B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$
- For integers: $B(m,n) = \frac{(m-1)!(n-1)!}{(m+n-1)!}$
- Alternative forms: trigonometric and infinite forms
- Applications in EEE: probability distributions, filter design, antenna patterns, power systems
- Used in analyzing radiation patterns, channel capacity, and statistical modeling

4 Properties of Beta and Gamma Functions

4.1 Learning Outcomes

After studying this section, students will be able to:

- Apply advanced properties of Beta and Gamma functions
- Use duplication formulas and reflection formulas
- Solve complex integrals using properties
- Apply properties to engineering problems

4.2 Advanced Properties

Theorem 4.1: Legendre Duplication Formula

For the Gamma function:

$$\Gamma(z)\Gamma\left(z+\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2z-1}}\Gamma(2z) \tag{12}$$

Proof. Using the Beta-Gamma relation and properties of the Beta function:

$$B\left(z, z+\frac{1}{2}\right) = \frac{\Gamma(z)\Gamma(z+1/2)}{\Gamma(2z+1/2)}$$

Also, using the integral representation:

$$B\left(z, z+\frac{1}{2}\right) = 2\int_0^{\pi/2} \sin^{2z-1}\theta \cos^{2z}\theta \, d\theta$$

Through detailed calculation (using substitutions and properties), we arrive at:

$$\Gamma(z)\Gamma\left(z+\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2z-1}}\Gamma(2z)$$

□

Theorem 4.2: Euler Reflection Formula

For non-integer values:

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}, \quad z \notin \mathbb{Z} \tag{13}$$

Example 4.1. Use the reflection formula to evaluate $\Gamma(1/3)\Gamma(2/3)$.

Solution:

$$\begin{aligned}
\Gamma\left(\frac{1}{3}\right)\Gamma\left(1-\frac{1}{3}\right) &= \frac{\pi}{\sin(\pi/3)} \\
\Gamma\left(\frac{1}{3}\right)\Gamma\left(\frac{2}{3}\right) &= \frac{\pi}{\sqrt{3}/2} \\
&= \frac{2\pi}{\sqrt{3}} = \frac{2\pi\sqrt{3}}{3}
\end{aligned}$$

4.3 Derivative Properties

The derivative of the Gamma function involves the digamma function $\psi(z)$:

$$\psi(z) = \frac{d}{dz} \ln \Gamma(z) = \frac{\Gamma'(z)}{\Gamma(z)} \tag{14}$$

Properties of the digamma function:

1. $\psi(z+1) = \psi(z) + \frac{1}{z}$
2. $\psi(1) = -\gamma$ where $\gamma \approx 0.5772$ is the Euler-Mascheroni constant
3. $\psi(n) = -\gamma + \sum_{k=1}^{n-1} \frac{1}{k}$ for positive integers n

4.4 Integral Representations

1. **Hankel's Formula:**
$$\frac{1}{\Gamma(z)} = \frac{1}{2\pi i} \int_C e^t t^{-z}\, dt$$
2. **Binet's Formula:**
$$\ln \Gamma(z) = \left(z - \frac{1}{2}\right) \ln z - z + \frac{1}{2} \ln(2\pi) + \int_0^\infty \left(\frac{1}{2} - \frac{1}{t} + \frac{1}{e^t - 1}\right) \frac{e^{-zt}}{t}\, dt$$

4.5 Five Mathematical Problems with Detailed Solutions

Problem 4.1. Use the duplication formula to evaluate $\Gamma(1/2)\Gamma(1)$.

Solution: Using the duplication formula with $z = 1/2$:

$$\begin{aligned} \Gamma\left(\frac{1}{2}\right)\Gamma(1) &= \frac{\sqrt{\pi}}{2^{2(1/2)-1}}\Gamma(1) \\ \sqrt{\pi} \cdot 1 &= \frac{\sqrt{\pi}}{2^0} \cdot 1 \\ \sqrt{\pi} &= \sqrt{\pi} \quad \checkmark \end{aligned}$$

Problem 4.2. Evaluate $\Gamma(1/4)\Gamma(3/4)$ using the reflection formula.

Solution:

$$\begin{aligned} \Gamma\left(\frac{1}{4}\right)\Gamma\left(1 - \frac{1}{4}\right) &= \frac{\pi}{\sin(\pi/4)} \\ \Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right) &= \frac{\pi}{\sqrt{2}/2} = \pi\sqrt{2} \end{aligned}$$

Problem 4.3. Show that $B(m,n) = B(m+1,n) + B(m,n+1)$.

Solution:

$$\begin{aligned} B(m+1,n) + B(m,n+1) &= \int_0^1 x^m (1-x)^{n-1}\, dx + \int_0^1 x^{m-1}(1-x)^n\, dx \\ &= \int_0^1 x^{m-1}(1-x)^{n-1}[x + (1-x)]\, dx \\ &= \int_0^1 x^{m-1}(1-x)^{n-1}\, dx \\ &= B(m,n) \end{aligned}$$

Problem 4.4. Prove the relation: $\frac{B(m+1,n+1)}{B(m,n)} = \frac{mn}{(m+n+1)(m+n)}$.

Solution: Using the Gamma function relation:

$$\begin{aligned}\frac{B(m+1,n+1)}{B(m,n)} &= \frac{\Gamma(m+1)\Gamma(n+1)\Gamma(m+n)}{\Gamma(m)\Gamma(n)\Gamma(m+n+2)} \\ &= \frac{m\Gamma(m)\cdot n\Gamma(n)\cdot\Gamma(m+n)}{\Gamma(m)\Gamma(n)\cdot(m+n+1)(m+n)\Gamma(m+n)} \\ &= \frac{mn}{(m+n+1)(m+n)}\end{aligned}$$

Problem 4.5. Evaluate $\int_0^{\pi/4} \tan^2\theta\, d\theta$ using properties.

Solution:

$$\begin{aligned}\int_0^{\pi/4} \tan^2\theta\, d\theta &= \int_0^{\pi/4} (\sec^2\theta - 1)\, d\theta \\ &= [\tan\theta - \theta]_0^{\pi/4} \\ &= 1 - \frac{\pi}{4}\end{aligned}$$

4.6 Ten Application-Oriented Problems for EEE Students

Problem 4.6. (Statistical Signal Processing) The chi-squared distribution with k degrees of freedom has PDF:

$$f(x;k) = \frac{x^{k/2-1}e^{-x/2}}{2^{k/2}\Gamma(k/2)}, \quad x > 0$$

Show that the mean is k.

Solution:

$$\begin{aligned}E[X] &= \int_0^\infty x \cdot \frac{x^{k/2-1}e^{-x/2}}{2^{k/2}\Gamma(k/2)}\, dx \\ &= \frac{1}{2^{k/2}\Gamma(k/2)} \int_0^\infty x^{k/2}e^{-x/2}\, dx\end{aligned}$$

Let $u = x/2$, then $x = 2u$ and $dx = 2du$:

$$\begin{aligned}E[X] &= \frac{1}{2^{k/2}\Gamma(k/2)} \int_0^\infty (2u)^{k/2}e^{-u} \cdot 2\, du \\ &= \frac{2^{k/2+1}}{2^{k/2}\Gamma(k/2)} \int_0^\infty u^{k/2}e^{-u}\, du \\ &= \frac{2}{\Gamma(k/2)}\Gamma\left(\frac{k}{2}+1\right) \\ &= \frac{2\cdot\frac{k}{2}\Gamma(k/2)}{\Gamma(k/2)} = k\end{aligned}$$

Problem 4.7. (Communication Theory) The Nakagami-m fading distribution is:

$$f(r; m, \Omega) = \frac{2m^m}{\Gamma(m)\Omega^m} r^{2m-1} e^{-mr^2/\Omega}, \quad r \geq 0$$

Verify this is a valid PDF for $m = 1$, $\Omega = 1$.

Solution: For $m = 1$, $\Omega = 1$:

$$f(r) = 2re^{-r^2}$$

$$\int_0^\infty 2re^{-r^2}\, dr = \left[-e^{-r^2}\right]_0^\infty = 0 - (-1) = 1 \quad \checkmark$$

Problem 4.8. (Image Processing) The gamma correction function involves $\Gamma(\alpha)$. For $\alpha = 2.2$ (standard display gamma), approximate $\Gamma(2.2)$ using the recurrence relation.

Solution:

$$\Gamma(2.2) = 1.2 \cdot \Gamma(1.2) = 1.2 \cdot 0.2 \cdot \Gamma(0.2)$$

Using numerical values: $\Gamma(0.2) \approx 4.5908$

$$\Gamma(2.2) \approx 1.2 \cdot 0.2 \cdot 4.5908 = 1.1018$$

(Actual value: $\Gamma(2.2) \approx 1.1018$)

Problem 4.9. (RF Engineering) The power spectral density of thermal noise involves:

$$S_n(f) = \int_0^\infty e^{-at} t^{n-1} \cos(2\pi f t)\, dt$$

For $n = 2$, express in terms of Gamma function using the real part.

Solution: Using the relation:

$$\begin{aligned}\int_0^\infty e^{-at} t^{n-1} \cos(2\pi f t)\, dt &= \mathrm{Re}\left[\int_0^\infty e^{-(a-2\pi i f)t} t^{n-1}\, dt\right] \\ &= \mathrm{Re}\left[\frac{\Gamma(n)}{(a - 2\pi i f)^n}\right]\end{aligned}$$

For $n = 2$:

$$S_n(f) = \mathrm{Re}\left[\frac{1}{(a - 2\pi i f)^2}\right] = \frac{a^2 - 4\pi^2 f^2}{(a^2 + 4\pi^2 f^2)^2}$$

Problem 4.10. (Power Systems) The Weibull distribution models time-to-failure:

$$f(t; k, \lambda) = \frac{k}{\lambda}\left(\frac{t}{\lambda}\right)^{k-1} e^{-(t/\lambda)^k}$$

Find the mean time to failure in terms of Gamma function.

Solution:

$$\begin{aligned} MTTF &= \int_0^\infty t \cdot f(t)\, dt \\ &= \frac{k}{\lambda} \int_0^\infty t \left(\frac{t}{\lambda}\right)^{k-1} e^{-(t/\lambda)^k}\, dt \end{aligned}$$

Let $u = (t/\lambda)^k$, then $t = \lambda u^{1/k}$ and $dt = \frac{\lambda}{k} u^{1/k-1}\, du$:

$$\begin{aligned} MTTF &= \lambda \int_0^\infty u^{1/k} e^{-u}\, du \\ &= \lambda \Gamma\left(1 + \frac{1}{k}\right) \end{aligned}$$

Problem 4.11. (Control Theory) The impulse response decay involves:

$$h(t) = t^{\alpha-1} e^{-\beta t}, \quad t > 0$$

Find the Laplace transform using Gamma function.

Solution:

$$\begin{aligned} \mathcal{L}\{h(t)\} &= \int_0^\infty t^{\alpha-1} e^{-\beta t} e^{-st}\, dt \\ &= \int_0^\infty t^{\alpha-1} e^{-(\beta+s)t}\, dt \\ &= \frac{1}{(\beta+s)^\alpha} \int_0^\infty u^{\alpha-1} e^{-u}\, du \\ &= \frac{\Gamma(\alpha)}{(\beta+s)^\alpha} \end{aligned}$$

Problem 4.12. (Quantum Electronics) The density of states in a semiconductor:

$$D(E) \propto E^{1/2}$$

The total number of states up to energy E_F involves:

$$N = \int_0^{E_F} E^{1/2}\, dE$$

Express this using Gamma function properties.

Solution:

$$\begin{aligned} N &= \int_0^{E_F} E^{1/2}\, dE = \left[\frac{2E^{3/2}}{3}\right]_0^{E_F} \\ &= \frac{2E_F^{3/2}}{3} \end{aligned}$$

This relates to $\Gamma(3/2) = \frac{\sqrt{\pi}}{2}$:

$$N = \frac{2E_F^{3/2}}{3} = \frac{4E_F^{3/2}}{3} \cdot \frac{\Gamma(3/2)}{\sqrt{\pi}}$$

Problem 4.13. (Signal Processing) The autocorrelation function:

$$R(\tau) = \int_0^\infty te^{-a|t-\tau|}\,dt$$

Evaluate for $\tau = 0$.

Solution: For $\tau = 0$:

$$R(0) = \int_0^\infty te^{-at}\,dt = \frac{\Gamma(2)}{a^2} = \frac{1}{a^2}$$

Problem 4.14. (Antenna Design) The far-field pattern involves:

$$E(\theta) = \int_0^a rJ_0(kr\sin\theta)\,dr$$

where J_0 is the Bessel function. For small angles, $J_0(x) \approx 1 - x^2/4$. Evaluate the approximation.

Solution:

$$\begin{aligned}
E(\theta) &\approx \int_0^a r\left(1 - \frac{k^2r^2\sin^2\theta}{4}\right)dr \\
&= \left[\frac{r^2}{2} - \frac{k^2\sin^2\theta\cdot r^4}{16}\right]_0^a \\
&= \frac{a^2}{2} - \frac{k^2a^4\sin^2\theta}{16} \\
&= \frac{a^2}{2}\left(1 - \frac{k^2a^2\sin^2\theta}{8}\right)
\end{aligned}$$

Problem 4.15. (Digital Modulation) The symbol error probability involves:

$$P_s = \int_0^\infty x^{n-1}e^{-ax^2}\,dx$$

Evaluate for $n = 2$.

Solution: For $n = 2$:

$$P_s = \int_0^\infty xe^{-ax^2}\,dx$$

Let $u = ax^2$, then $x = \sqrt{u/a}$ and $dx = \frac{du}{2\sqrt{au}}$:

$$\begin{aligned}
P_s &= \int_0^\infty \sqrt{\frac{u}{a}}e^{-u}\cdot\frac{du}{2\sqrt{au}} \\
&= \frac{1}{2a}\int_0^\infty e^{-u}\,du = \frac{1}{2a}
\end{aligned}$$

4.7 Summary: Properties of Beta and Gamma Functions

Key Properties:

- **Duplication Formula:** $\Gamma(z)\Gamma(z+1/2) = \frac{\sqrt{\pi}}{2^{2z-1}}\Gamma(2z)$
- **Reflection Formula:** $\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}$
- **Beta-Gamma Relation:** $B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$
- **Recurrence:** $\Gamma(n+1) = n\Gamma(n)$, $B(m,n) = B(m+1,n) + B(m,n+1)$
- Applications in probability distributions, signal processing, and statistical analysis
- Essential for evaluating complex integrals in engineering problems

5 Integration of Gaussian Functions

5.1 Learning Outcomes

After studying this section, students will be able to:

- Evaluate standard Gaussian integrals
- Apply Gaussian integrals to probability theory
- Use error functions in engineering applications
- Solve problems involving normal distributions
- Apply Gaussian functions to signal processing

5.2 The Standard Gaussian Integral

Definition 5.1: Standard Gaussian Integral

The **standard Gaussian integral** is:

$$I = \int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi} \tag{15}$$

Proof. Let $I = \int_{-\infty}^{\infty} e^{-x^2}\,dx$. Then:

$$\begin{aligned} I^2 &= \left(\int_{-\infty}^{\infty} e^{-x^2}\,dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2}\,dy\right) \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy \end{aligned}$$

Converting to polar coordinates ($x = r\cos\theta$, $y = r\sin\theta$, $dx\,dy = r\,dr\,d\theta$):

$$\begin{aligned} I^2 &= \int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\,dr\,d\theta \\ &= 2\pi \int_0^{\infty} e^{-r^2} r\,dr \\ &= 2\pi \left[-\frac{e^{-r^2}}{2}\right]_0^{\infty} \\ &= 2\pi \cdot \frac{1}{2} = \pi \end{aligned}$$

Therefore, $I = \sqrt{\pi}$. □

5.3 Generalized Gaussian Integrals

Theorem 5.1: Generalized Gaussian Integral

For $a > 0$ and any real b:

$$\int_{-\infty}^{\infty} e^{-ax^2+bx}\,dx = \sqrt{\frac{\pi}{a}} e^{b^2/(4a)} \tag{16}$$

Proof. Complete the square in the exponent:

$$\begin{aligned} -ax^2 + bx &= -a\left(x^2 - \frac{b}{a}x\right) \\ &= -a\left[\left(x - \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2}\right] \\ &= -a\left(x - \frac{b}{2a}\right)^2 + \frac{b^2}{4a} \end{aligned}$$

Therefore:

$$\int_{-\infty}^{\infty} e^{-ax^2+bx}\,dx = e^{b^2/(4a)} \int_{-\infty}^{\infty} e^{-a(x-b/(2a))^2}\,dx$$

Let $u = \sqrt{a}(x - b/(2a))$, then $dx = du/\sqrt{a}$:

$$\begin{aligned} &= e^{b^2/(4a)} \cdot \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} e^{-u^2}\,du \\ &= \sqrt{\frac{\pi}{a}} e^{b^2/(4a)} \end{aligned}$$

□

5.4 The Error Function

Definition 5.2: Error Function

The **error function** is defined as:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt \tag{17}$$

The **complementary error function** is:

$$\operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2}\, dt \tag{18}$$

Properties of the Error Function:

1. $\operatorname{erf}(0) = 0$
2. $\operatorname{erf}(\infty) = 1$
3. $\operatorname{erf}(-x) = -\operatorname{erf}(x)$ (odd function)
4. $\operatorname{erf}'(x) = \frac{2}{\sqrt{\pi}} e^{-x^2}$
5. $\operatorname{erf}(x) + \operatorname{erfc}(x) = 1$

5.5 Visual Representation of Gaussian and Error Functions

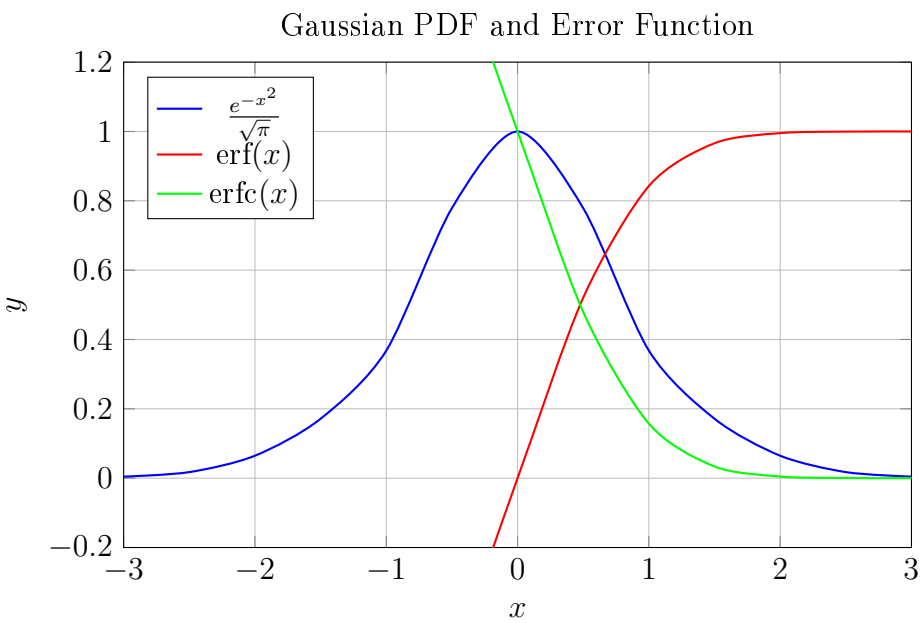

Figure 3: Gaussian PDF and error functions

Description of Figure 3: The blue curve shows the normalized Gaussian function (scaled by $1/\sqrt{\pi}$). The red curve is the error function, which starts at 0 and asymptotically approaches 1. The green curve is the complementary error function, which starts at 1 and approaches 0. The error function represents the cumulative distribution of the Gaussian.

5.6 Gaussian Integrals with Powers

Theorem 5.2: Gaussian Integrals with Even Powers

For non-negative integer n:

$$\int_{-\infty}^{\infty} x^{2n} e^{-ax^2}\, dx = \frac{(2n-1)!!}{(2a)^n}\sqrt{\frac{\pi}{a}} \tag{19}$$

where $(2n-1)!! = (2n-1)(2n-3)\cdots 3\cdot 1$ is the double factorial.

This can also be expressed using the Gamma function:

$$\int_{-\infty}^{\infty} x^{2n} e^{-ax^2}\, dx = \frac{\Gamma(n+1/2)}{a^{n+1/2}\sqrt{\pi}} \tag{20}$$

For odd powers:

$$\int_{-\infty}^{\infty} x^{2n+1} e^{-ax^2}\, dx = 0 \tag{21}$$

(since the integrand is an odd function).

5.7 Five Mathematical Problems with Detailed Solutions

Problem 5.1. Evaluate $\int_{-\infty}^{\infty} e^{-2x^2+4x}\, dx$.

Solution: Using the generalized Gaussian integral with $a = 2$ and $b = 4$:

$$\begin{aligned}\int_{-\infty}^{\infty} e^{-2x^2+4x}\, dx &= \sqrt{\frac{\pi}{2}} e^{16/(4\cdot 2)} \\ &= \sqrt{\frac{\pi}{2}} e^{2} \\ &= \frac{\sqrt{\pi}}{\sqrt{2}} e^{2} = \sqrt{\frac{\pi}{2}} e^{2}\end{aligned}$$

Problem 5.2. Evaluate $\int_{0}^{\infty} e^{-x^2}\, dx$.

Solution: Since e^{-x^2} is an even function:

$$\begin{aligned}\int_{0}^{\infty} e^{-x^2}\, dx &= \frac{1}{2}\int_{-\infty}^{\infty} e^{-x^2}\, dx \\ &= \frac{1}{2}\sqrt{\pi} = \frac{\sqrt{\pi}}{2}\end{aligned}$$

Problem 5.3. Evaluate $\int_{-\infty}^{\infty} x^2 e^{-x^2}\, dx$.

Solution: Using the formula for even powers with $n = 1$ and $a = 1$:

$$\begin{aligned}\int_{-\infty}^{\infty} x^2 e^{-x^2}\, dx &= \frac{1!!}{(2\cdot 1)^1}\sqrt{\pi} \\ &= \frac{1}{2}\sqrt{\pi} = \frac{\sqrt{\pi}}{2}\end{aligned}$$

Alternatively, using integration by parts: Let $u = x$ and $dv = xe^{-x^2}dx$:

$$\int_{-\infty}^{\infty} x^2 e^{-x^2}\,dx = \left[-\frac{xe^{-x^2}}{2}\right]_{-\infty}^{\infty} + \frac{1}{2}\int_{-\infty}^{\infty} e^{-x^2}\,dx$$
$$= 0 + \frac{1}{2}\sqrt{\pi} = \frac{\sqrt{\pi}}{2}$$

Problem 5.4. Show that $\int_0^\infty xe^{-ax^2}\,dx = \frac{1}{2a}$.

Solution: Let $u = ax^2$, then $du = 2ax\,dx$:

$$\int_0^\infty xe^{-ax^2}\,dx = \frac{1}{2a}\int_0^\infty e^{-u}\,du$$
$$= \frac{1}{2a}[-e^{-u}]_0^\infty$$
$$= \frac{1}{2a}(0-(-1)) = \frac{1}{2a}$$

Problem 5.5. Express $\int_{-\infty}^{\infty} e^{-(x-\mu)^2/(2\sigma^2)}\,dx$ in terms of σ.

Solution: Let $u = \frac{x-\mu}{\sigma\sqrt{2}}$, then $x = \mu + \sigma\sqrt{2}u$ and $dx = \sigma\sqrt{2}\,du$:

$$\int_{-\infty}^{\infty} e^{-(x-\mu)^2/(2\sigma^2)}\,dx = \int_{-\infty}^{\infty} e^{-u^2}\cdot\sigma\sqrt{2}\,du$$
$$= \sigma\sqrt{2}\cdot\sqrt{\pi}$$
$$= \sigma\sqrt{2\pi}$$

5.8 Ten Application-Oriented Problems for EEE Students

Problem 5.6. (Communication Systems) The probability of bit error for BPSK in AWGN channel is: $P_e = \frac{1}{2}\text{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$ where $E_b/N_0 = 10$ dB. Calculate P_e.

Solution: First, convert 10 dB to linear scale: $\frac{E_b}{N_0} = 10^{10/10} = 10$
Then: $P_e = \frac{1}{2}\text{erfc}(\sqrt{10}) = \frac{1}{2}\text{erfc}(3.162)$
Using numerical values: $\text{erfc}(3.162) \approx 0.00004$
$P_e \approx 0.00002 = 2\times 10^{-5}$

Problem 5.7. (Probability Theory) The normal distribution PDF is: $f(x;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/(2\sigma^2)}$ Verify this is a valid PDF for any μ and $\sigma > 0$.

Solution:

$$\int_{-\infty}^{\infty} f(x)\,dx = \frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-(x-\mu)^2/(2\sigma^2)}\,dx$$

From Problem 5.5, we know this equals: $\frac{1}{\sigma\sqrt{2\pi}}\cdot\sigma\sqrt{2\pi} = 1$ ✓

Problem 5.8. (Signal Processing) The matched filter output for a Gaussian pulse $s(t) = e^{-t^2/T^2}$ is: $y = \int_{-\infty}^{\infty} s(t)s(t-\tau)\,dt$ Evaluate for $\tau = 0$.

Solution: For $\tau = 0$:

$$y(0) = \int_{-\infty}^{\infty} e^{-2t^2/T^2}\, dt$$

Let $u = \sqrt{2/T^2} \cdot t = t\sqrt{2}/T$, then $dt = T/\sqrt{2}\, du$:

$$\begin{aligned} y(0) &= \frac{T}{\sqrt{2}} \int_{-\infty}^{\infty} e^{-u^2}\, du \\ &= \frac{T}{\sqrt{2}} \cdot \sqrt{\pi} = \frac{T\sqrt{\pi}}{\sqrt{2}} \end{aligned}$$

Problem 5.9. (Radar Systems) The range resolution depends on: $R = \int_{-\infty}^{\infty} t^2 e^{-\alpha t^2}\, dt$ where α is related to bandwidth. Evaluate in terms of α.

Solution: Using the formula for Gaussian integrals with even powers:

$$\begin{aligned} R &= \frac{1!!}{(2\alpha)} \sqrt{\frac{\pi}{\alpha}} \\ &= \frac{1}{2\alpha} \sqrt{\frac{\pi}{\alpha}} \\ &= \frac{\sqrt{\pi}}{2\alpha^{3/2}} \end{aligned}$$

Problem 5.10. (Noise Analysis) The power of white Gaussian noise with power spectral density $N_0/2$ over a bandwidth involves: $P = \int_{-\infty}^{\infty} \frac{N_0}{2} e^{-f^2/B^2}\, df$ Evaluate this integral.

Solution:

$$P = \frac{N_0}{2} \int_{-\infty}^{\infty} e^{-f^2/B^2}\, df$$

Let $u = f/B$, then $df = B\, du$:

$$\begin{aligned} P &= \frac{N_0 B}{2} \int_{-\infty}^{\infty} e^{-u^2}\, du \\ &= \frac{N_0 B}{2} \cdot \sqrt{\pi} = \frac{N_0 B\sqrt{\pi}}{2} \end{aligned}$$

Problem 5.11. (Filter Design) A Gaussian low-pass filter has transfer function: $H(f) = e^{-(\pi f/f_c)^2}$ Find the equivalent noise bandwidth.

Solution: The equivalent noise bandwidth is:

$$\begin{aligned} B_n &= \frac{\int_0^{\infty} |H(f)|^2\, df}{|H(0)|^2} \\ &= \int_0^{\infty} e^{-2(\pi f/f_c)^2}\, df \end{aligned}$$

Let $u = \sqrt{2}\pi f/f_c$, then $df = \frac{f_c}{\sqrt{2}\pi}\, du$:

$$\begin{aligned} B_n &= \frac{f_c}{\sqrt{2}\pi} \int_0^{\infty} e^{-u^2}\, du \\ &= \frac{f_c}{\sqrt{2}\pi} \cdot \frac{\sqrt{\pi}}{2} \\ &= \frac{f_c}{2\sqrt{2}} = \frac{f_c\sqrt{2}}{4} \end{aligned}$$

Problem 5.12. (Quantum Mechanics) The ground state wavefunction of harmonic oscillator is: $\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-m\omega x^2/(2\hbar)}$ Verify normalization: $\int_{-\infty}^{\infty} |\psi_0(x)|^2\, dx = 1$.

Solution:

$$\int_{-\infty}^{\infty} |\psi_0|^2\, dx = \sqrt{\frac{m\omega}{\pi\hbar}} \int_{-\infty}^{\infty} e^{-m\omega x^2/\hbar}\, dx$$

Let $a = m\omega/\hbar$:

$$\begin{aligned} &= \sqrt{\frac{m\omega}{\pi\hbar}} \cdot \sqrt{\frac{\pi}{a}} \\ &= \sqrt{\frac{m\omega}{\pi\hbar}} \cdot \sqrt{\frac{\pi\hbar}{m\omega}} = 1 \quad \checkmark \end{aligned}$$

Problem 5.13. (Statistical Analysis) The Q-function used in BER analysis is: $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-t^2/2}\, dt$ Express in terms of erfc.

Solution: Let $u = t/\sqrt{2}$, then $t = \sqrt{2}u$ and $dt = \sqrt{2}\, du$:

$$\begin{aligned} Q(x) &= \frac{1}{\sqrt{2\pi}} \int_{x/\sqrt{2}}^{\infty} e^{-u^2} \sqrt{2}\, du \\ &= \frac{1}{\sqrt{\pi}} \int_{x/\sqrt{2}}^{\infty} e^{-u^2}\, du \\ &= \frac{1}{2}\mathrm{erfc}\left(\frac{x}{\sqrt{2}}\right) \end{aligned}$$

Problem 5.14. (Optical Communications) The beam profile of a laser is Gaussian: $I(r) = I_0 e^{-2r^2/w_0^2}$ Find the total power: $P = 2\pi \int_0^{\infty} I(r) r\, dr$.

Solution:

$$P = 2\pi I_0 \int_0^{\infty} r e^{-2r^2/w_0^2}\, dr$$

Let $u = 2r^2/w_0^2$, then $r^2 = u w_0^2/2$ and $r\, dr = w_0^2 du/4$:

$$\begin{aligned} P &= 2\pi I_0 \int_0^{\infty} e^{-u} \cdot \frac{w_0^2}{4}\, du \\ &= \frac{\pi I_0 w_0^2}{2} [-e^{-u}]_0^{\infty} \\ &= \frac{\pi I_0 w_0^2}{2} \end{aligned}$$

Problem 5.15. (Control Systems) The impulse response of a system is: $h(t) = e^{-t^2/(2\sigma^2)}, \quad -\infty < t < \infty$ Find the energy: $E = \int_{-\infty}^{\infty} h^2(t)\, dt$.

Solution:

$$E = \int_{-\infty}^{\infty} e^{-t^2/\sigma^2}\, dt$$

Let $u = t/\sigma$, then $dt = \sigma\, du$:

$$E = \sigma \int_{-\infty}^{\infty} e^{-u^2}\, du = \sigma\sqrt{\pi}$$

5.9 Summary: Integration of Gaussian Functions

Key Formulas:

- **Standard Gaussian:** $\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$
- **Generalized:** $\int_{-\infty}^{\infty} e^{-ax^2+bx}\,dx = \sqrt{\frac{\pi}{a}}e^{b^2/(4a)}$
- **Error Function:** $\text{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}\,dt$
- **Even Powers:** $\int_{-\infty}^{\infty} x^{2n}e^{-ax^2}\,dx = \frac{(2n-1)!!}{(2a)^n}\sqrt{\frac{\pi}{a}}$
- Applications: probability theory, communication systems, signal processing, noise analysis
- The normal distribution and Q-function are based on Gaussian integrals
- Essential for BER calculations, filter design, and statistical analysis

6 Other Special Functions and Applications

6.1 Learning Outcomes

After studying this section, students will be able to:

- Understand Bessel functions and their applications
- Work with Legendre polynomials in potential theory
- Apply special functions to differential equations
- Use Laplace transforms with special functions
- Recognize special functions in engineering contexts

6.2 Bessel Functions

Definition 6.1: Bessel Function of the First Kind

The **Bessel function of the first kind of order** n is:

$$J_n(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(n+k)!}\left(\frac{x}{2}\right)^{2k+n} \tag{22}$$

Bessel functions arise as solutions to Bessel's differential equation:

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - n^2)y = 0 \tag{23}$$

Key Properties:

1. $J_{-n}(x) = (-1)^n J_n(x)$ for integer n

2. $J_n(0) = \begin{cases} 1 & n = 0 \\ 0 & n > 0 \end{cases}$

3. Recurrence relations: $J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x)$ $J_{n-1}(x) - J_{n+1}(x) = 2J_n'(x)$

Applications in EEE:

- Electromagnetic wave propagation in cylindrical waveguides
- FM signal spectrum analysis (Bessel functions of the first kind)
- Vibrations of circular membranes
- Heat conduction in cylindrical coordinates
- Antenna radiation patterns

6.3 Legendre Polynomials

Definition 6.2: Legendre Polynomials

The **Legendre polynomials** $P_n(x)$ can be defined by Rodrigues' formula:

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n \tag{24}$$

The first few Legendre polynomials:

$$\begin{aligned} P_0(x) &= 1 \\ P_1(x) &= x \\ P_2(x) &= \frac{1}{2}(3x^2 - 1) \\ P_3(x) &= \frac{1}{2}(5x^3 - 3x) \\ P_4(x) &= \frac{1}{8}(35x^4 - 30x^2 + 3) \end{aligned}$$

Orthogonality:

$$\int_{-1}^{1} P_m(x) P_n(x)\, dx = \begin{cases} 0 & m \neq n \\ \frac{2}{2n+1} & m = n \end{cases} \tag{25}$$

Applications in EEE:

- Solving Laplace's equation in spherical coordinates
- Multipole expansions in electromagnetics
- Antenna array analysis
- Gravitational and electrostatic potential problems

6.4 Relationship with Gamma and Beta Functions

Many special functions can be expressed using Gamma and Beta functions:

1. **Bessel Integral Representation:** $J_n(x) = \frac{1}{\pi}\int_0^{\pi} \cos(n\theta - x\sin\theta)\,d\theta$

   For $n = 0$: $J_0(x) = \frac{2}{\pi}\int_0^{\pi/2} \cos(x\sin\theta)\,d\theta$

2. **Connection to Beta Function:** $J_n(x) = \frac{(x/2)^n}{\sqrt{\pi}\Gamma(n+1/2)}\int_0^{\pi} \cos(x\cos\theta)\sin^{2n}\theta\,d\theta$

3. **Legendre Function Normalization:** $\int_{-1}^{1} P_n^2(x)\,dx = \frac{2}{2n+1}$

6.5 Visual Representation of Bessel Functions

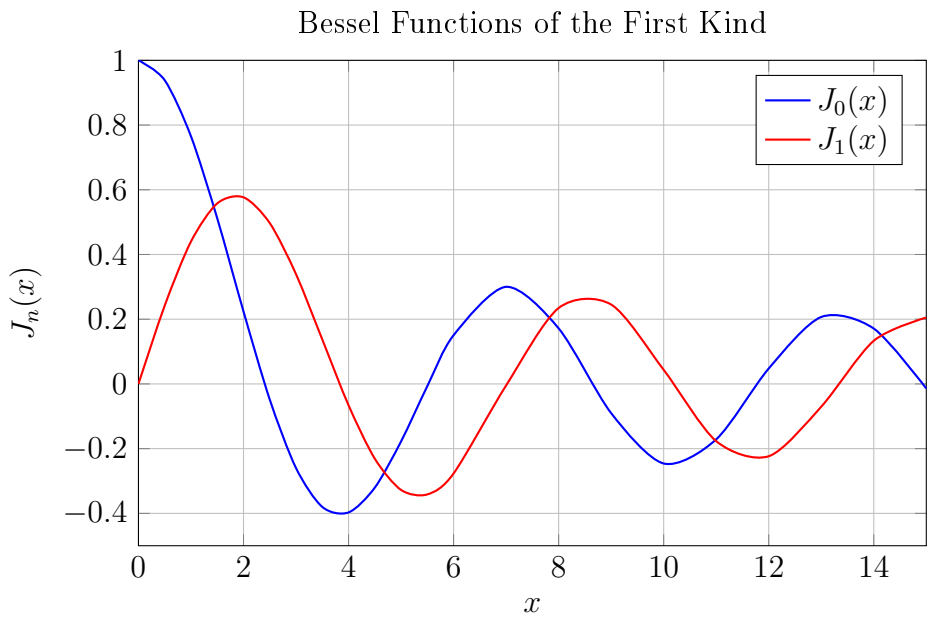

Figure 4: Bessel functions $J_0(x)$ and $J_1(x)$

Description of Figure 4: Bessel functions exhibit damped oscillatory behavior. $J_0(x)$ starts at 1 when $x = 0$, while $J_1(x)$ starts at 0. Both functions oscillate with decreasing amplitude as x increases, crossing the x-axis at various points. These functions are crucial in problems with cylindrical symmetry such as waveguides, circular membranes, and electromagnetic wave propagation.

6.6 Summary: Other Special Functions

Key Points:

- **Bessel Functions:** Solutions to problems with cylindrical symmetry (waveguides, circular membranes, FM signals)
- **Legendre Polynomials:** Solutions in spherical coordinates (multipole expansions, antenna patterns)
- All special functions are interconnected through Gamma and Beta functions

- Applications span electromagnetics, signal processing, quantum mechanics, heat transfer
- Essential for solving partial differential equations in engineering
- Numerical methods often needed for evaluation

7 General Summary and Conclusions

This comprehensive study of special functions in calculus has covered the fundamental tools essential for advanced engineering analysis, particularly in Electrical and Electronic Engineering.

7.1 Key Takeaways

1. **Gamma Function:**
   - Extends factorial to continuous domain
   - Formula: $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1}\, dx$
   - Satisfies: $\Gamma(n+1) = n\Gamma(n)$ and $\Gamma(n) = (n-1)!$ for integers
2. **Beta Function:**
   - Integral representation: $B(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\, dx$
   - Related to Gamma: $B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$
   - Used in probability distributions and statistical analysis
3. **Gaussian Integrals:**
   - Fundamental: $\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}$
   - Error function critical for communication systems
   - Applications in probability, noise analysis, filter design
4. **Other Special Functions:**
   - Bessel functions for cylindrical problems
   - Legendre polynomials for spherical problems
   - All interconnected through Gamma and Beta functions

7.2 Engineering Applications Summary

Special functions appear throughout EEE:

- **Communications:** BER analysis, FM spectra, channel capacity
- **Signal Processing:** Filter design, windowing, autocorrelation
- **Electromagnetics:** Waveguides, antenna patterns, potential theory

- **Control Systems:** Transfer functions, impulse response
- **Probability:** Normal, chi-squared, gamma, beta distributions
- **Quantum Electronics:** Wavefunctions, density of states

7.3 Further Study

Students are encouraged to:

- Practice numerical evaluation using software (MATLAB, Python)
- Explore asymptotic expansions for large arguments
- Study complex variable extensions
- Apply to real engineering design problems
- Investigate relationships with Fourier and Laplace transforms

"Special functions are the vocabulary of applied mathematics and engineering."
— Richard Askey

End of Class Notes
Department of Mathematics
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For B.Sc. (Eng) in EEE Students
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Differential and Integral Calculus

Multiple Integrals: Determination of Lengths, Areas and Volumes

B.Sc (Engineering) in EEE
Department of Mathematics
Jamalpur Science & Technology University

Comprehensive Class Notes

Academic Year 2024-2025

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1 Learning Outcomes

Upon completion of this topic on Multiple Integrals and their applications to Lengths, Areas, and Volumes, students will be able to:

1. **Conceptual Understanding:**
   - Understand the concept of multiple integration as an extension of single-variable integration
   - Comprehend the geometric interpretation of double and triple integrals
   - Distinguish between different coordinate systems (Cartesian, polar, cylindrical, spherical)
2. **Computational Skills:**
   - Evaluate double integrals over rectangular and non-rectangular regions
   - Evaluate triple integrals in various coordinate systems
   - Change the order of integration in multiple integrals
   - Apply appropriate coordinate transformations using Jacobians
3. **Applications:**
   - Calculate arc lengths of curves in 2D and 3D space
   - Determine areas of planar regions using double integrals
   - Compute volumes of solid regions using double and triple integrals
   - Calculate surface areas of 3D surfaces
   - Find centers of mass and moments of inertia (crucial for EEE applications)
4. **Engineering Applications:**
   - Apply multiple integrals to solve electric field and charge distribution problems
   - Calculate total charge, current, and power in electrical systems
   - Determine moments of inertia for rotating electrical machinery
   - Solve heat distribution and electromagnetic field problems

2 Key Terms and Definitions

2.1 Multiple Integrals

Definition 2.1 (Double Integral)**.** Let $f(x, y)$ be a function defined on a closed and bounded region R in the xy-plane. The **double integral** of f over R is defined as:

$$\iint_R f(x, y)\, dA = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*, y_i^*)\Delta A_i$$

where the region R is divided into n subregions of area ΔA_i, and (x_i^*, y_i^*) is a point in the i-th subregion.

Geometric Interpretation

The double integral $\iint_R f(x,y)\,dA$ represents:

- The **volume** under the surface $z = f(x,y)$ and above the region R when $f(x,y) \geq 0$
- The **signed volume** when $f(x,y)$ can be negative
- The **area** of region R when $f(x,y) = 1$

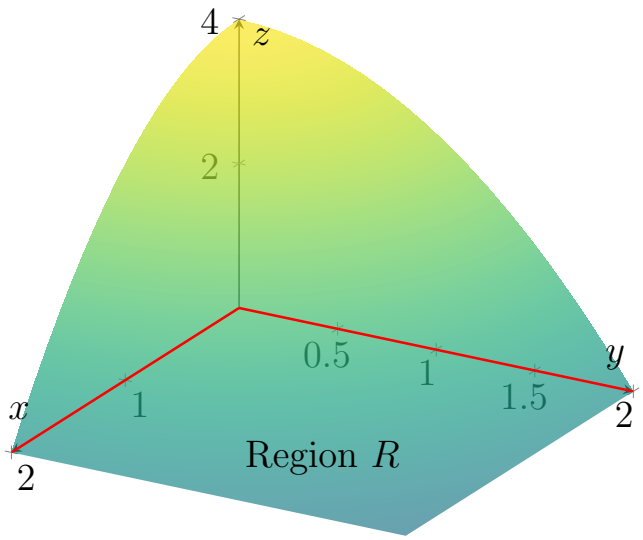

Figure 1: Double integral representing volume under surface $z = 4 - x^2 - y^2$

Definition 2.2 (Triple Integral)**.** Let $f(x,y,z)$ be a function defined on a closed and bounded region E in 3D space. The **triple integral** of f over E is:

$$\iiint_E f(x,y,z)\,dV = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*, y_i^*, z_i^*)\Delta V_i$$

where ΔV_i represents the volume of the i-th subregion.

2.2 Types of Regions

Definition 2.3 (Type I Region)**.** A region R in the xy-plane is of **Type I** if it can be expressed as:

$$R = \{(x,y) : a \leq x \leq b,\ g_1(x) \leq y \leq g_2(x)\}$$

For such regions:

$$\iint_R f(x,y)\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx$$

Definition 2.4 (Type II Region)**.** A region R in the xy-plane is of **Type II** if it can be expressed as:

$$R = \{(x,y) : c \leq y \leq d,\ h_1(y) \leq x \leq h_2(y)\}$$

For such regions:

$$\iint_R f(x,y)\,dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\,dy$$

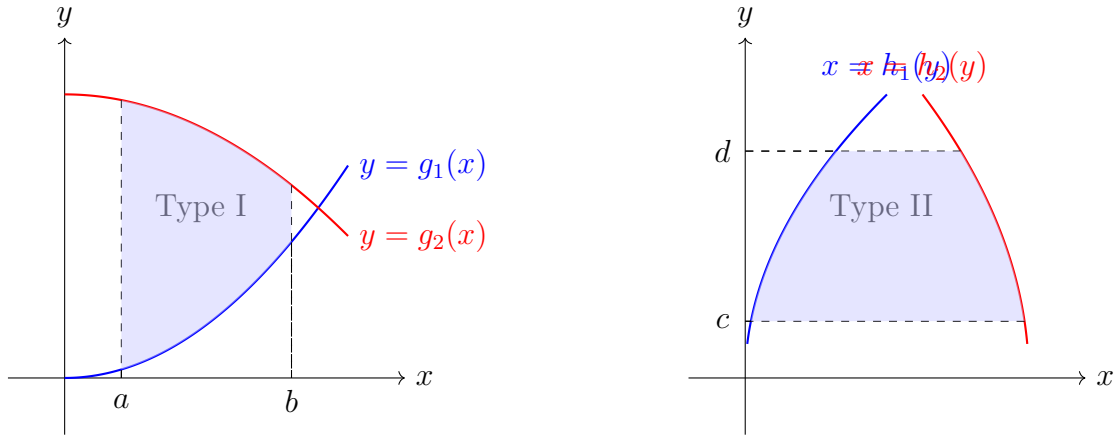

Figure 2: Type I and Type II regions in the plane

2.3 Coordinate Systems

Definition 2.5 (Polar Coordinates)**.** In polar coordinates, a point in the plane is represented by (r, θ) where:

$$\begin{aligned} x &= r\cos\theta \\ y &= r\sin\theta \\ r^2 &= x^2 + y^2 \\ \tan\theta &= \frac{y}{x} \end{aligned}$$

The area element is: $dA = r\,dr\,d\theta$

The double integral becomes:

$$\iint_R f(x,y)\,dA = \int_\alpha^\beta \int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta$$

Definition 2.6 (Cylindrical Coordinates)**.** In cylindrical coordinates, a point in 3D space is represented by (r, θ, z) where:

$$\begin{aligned} x &= r\cos\theta \\ y &= r\sin\theta \\ z &= z \end{aligned}$$

The volume element is: $dV = r\,dr\,d\theta\,dz$

Definition 2.7 (Spherical Coordinates)**.** In spherical coordinates, a point in 3D space is represented by (ρ, θ, ϕ) where:

$$\begin{aligned} x &= \rho\sin\phi\cos\theta \\ y &= \rho\sin\phi\sin\theta \\ z &= \rho\cos\phi \end{aligned}$$

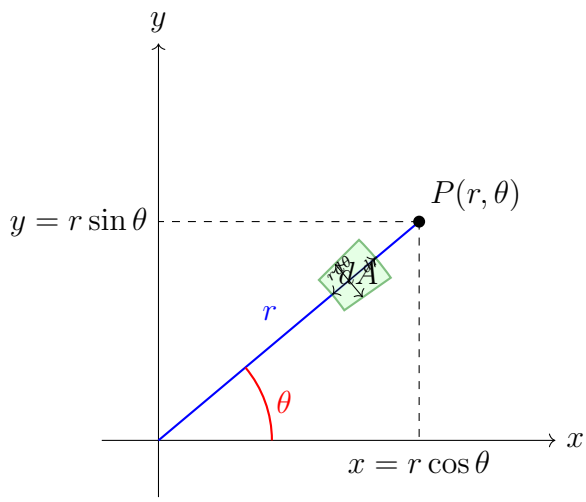

Figure 3: Polar coordinate system and area element $dA = r\,dr\,d\theta$

where $\rho \geq 0$, $0 \leq \theta \leq 2\pi$, $0 \leq \phi \leq \pi$.

The volume element is: $dV = \rho^2 \sin\phi\, d\rho\, d\theta\, d\phi$

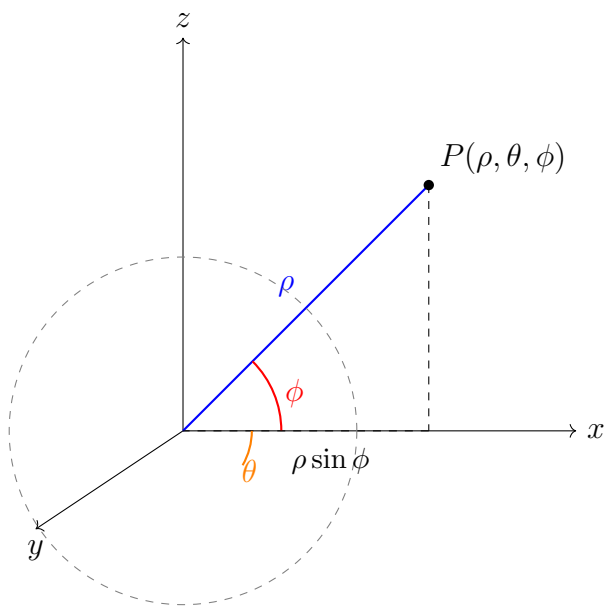

Figure 4: Spherical coordinate system

3 Mathematical Equations and Derivations

3.1 Area Calculation Using Double Integrals

Theorem 3.1 (Area of a Region)**.** The area A of a region R in the xy-plane is given by:

$$A = \iint_R dA = \iint_R 1\,dx\,dy$$

Proof. Consider partitioning the region R into small rectangles of area $\Delta A_i = \Delta x_i \Delta y_i$. The total area is approximately:

$$A \approx \sum_{i=1}^{n} \Delta A_i$$

Taking the limit as the partition becomes finer:

$$A = \lim_{n\to\infty} \sum_{i=1}^{n} \Delta A_i = \iint_R dA$$

Since we're measuring area, we integrate the function $f(x, y) = 1$ over R. □

Example 3.1 (Area in Cartesian Coordinates)**.** Find the area of the region bounded by $y = x^2$ and $y = 2x$.

Solution: First, find intersection points:

$$x^2 = 2x \implies x^2 - 2x = 0 \implies x(x-2) = 0$$

So $x = 0$ and $x = 2$.

The region is Type I: $0 \le x \le 2$, $x^2 \le y \le 2x$.

$$\begin{aligned} A &= \int_0^2 \int_{x^2}^{2x} dy\, dx \\ &= \int_0^2 [y]_{x^2}^{2x} dx \\ &= \int_0^2 (2x - x^2) dx \\ &= \left[x^2 - \frac{x^3}{3}\right]_0^2 \\ &= 4 - \frac{8}{3} = \frac{4}{3} \text{ square units} \end{aligned}$$

Example 3.2 (Area in Polar Coordinates)**.** Find the area enclosed by one loop of the rose $r = \cos(2\theta)$.

Solution: One loop corresponds to $-\pi/4 \le \theta \le \pi/4$ (where $r \ge 0$).

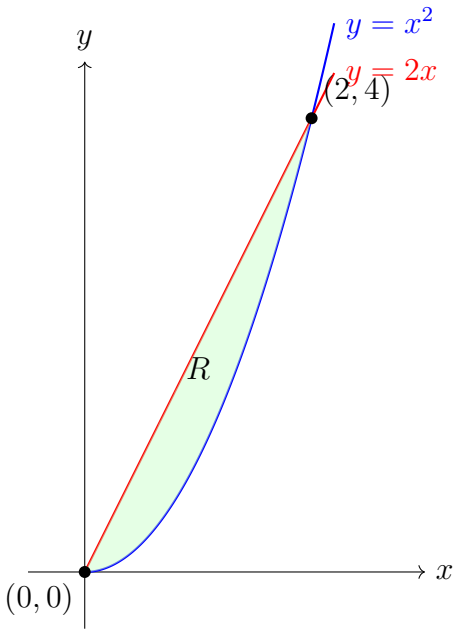

Figure 5: Region bounded by $y = x^2$ and $y = 2x$

$$
\begin{aligned}
A &= \int_{-\pi/4}^{\pi/4} \int_0^{\cos(2\theta)} r\,dr\,d\theta \\
&= \int_{-\pi/4}^{\pi/4} \left[\frac{r^2}{2}\right]_0^{\cos(2\theta)} d\theta \\
&= \int_{-\pi/4}^{\pi/4} \frac{\cos^2(2\theta)}{2} d\theta \\
&= \frac{1}{2} \int_{-\pi/4}^{\pi/4} \frac{1 + \cos(4\theta)}{2} d\theta \\
&= \frac{1}{4} \left[\theta + \frac{\sin(4\theta)}{4}\right]_{-\pi/4}^{\pi/4} \\
&= \frac{1}{4} \left[\frac{\pi}{4} + 0 - \left(-\frac{\pi}{4} + 0\right)\right] \\
&= \frac{\pi}{8} \text{ square units}
\end{aligned}
$$

3.2 Volume Calculation Using Double Integrals

Theorem 3.2 (Volume Under a Surface)**.** If $f(x, y) \geq 0$ on a region R, then the volume V of the solid bounded below by R and above by the surface $z = f(x, y)$ is:

$$V = \iint_R f(x, y)\,dA$$

Example 3.3 (Volume Under a Paraboloid)**.** Find the volume of the solid bounded by the paraboloid $z = 4 - x^2 - y^2$ and the xy-plane.

Solution: The paraboloid intersects the xy-plane when $z = 0$:

$$4 - x^2 - y^2 = 0 \implies x^2 + y^2 = 4$$

This is a circle of radius 2. Using polar coordinates:

$$\begin{aligned}
V &= \iint_R (4 - x^2 - y^2)\, dA \\
&= \int_0^{2\pi} \int_0^2 (4 - r^2) r\, dr\, d\theta \\
&= \int_0^{2\pi} \int_0^2 (4r - r^3)\, dr\, d\theta \\
&= \int_0^{2\pi} \left[2r^2 - \frac{r^4}{4}\right]_0^2 d\theta \\
&= \int_0^{2\pi} (8 - 4) d\theta \\
&= 4 \int_0^{2\pi} d\theta = 8\pi \text{ cubic units}
\end{aligned}$$

3.3 Volume Calculation Using Triple Integrals

Theorem 3.3 (Volume of a Solid Region)**.** The volume V of a solid region E in 3D space is:

$$V = \iiint_E dV = \iiint_E 1\, dx\, dy\, dz$$

Example 3.4 (Volume of a Sphere)**.** Find the volume of a sphere of radius a using spherical coordinates.

Solution: In spherical coordinates, the sphere is described by $0 \le \rho \le a$, $0 \le \theta \le 2\pi$, $0 \le \phi \le \pi$.

$$\begin{aligned}
V &= \iiint_E dV \\
&= \int_0^{2\pi} \int_0^{\pi} \int_0^a \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta \\
&= \int_0^{2\pi} d\theta \int_0^{\pi} \sin\phi\, d\phi \int_0^a \rho^2\, d\rho \\
&= [2\pi] \cdot [-\cos\phi]_0^{\pi} \cdot \left[\frac{\rho^3}{3}\right]_0^a \\
&= 2\pi \cdot [-\cos\pi + \cos 0] \cdot \frac{a^3}{3} \\
&= 2\pi \cdot 2 \cdot \frac{a^3}{3} \\
&= \frac{4\pi a^3}{3} \text{ cubic units}
\end{aligned}$$

3.4 Arc Length

Theorem 3.4 (Arc Length in 2D)**.** The arc length L of a curve $\mathbf{r}(t) = (x(t), y(t))$ for $a \leq t \leq b$ is:

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = \int_a^b |\mathbf{r}'(t)|\, dt$$

Proof. Consider a small segment of the curve from t to $t + \Delta t$. By the Pythagorean theorem:

$$\Delta s \approx \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

Dividing by Δt and taking the limit:

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

Integrating from a to b gives the total arc length. □

Example 3.5 (Arc Length of a Helix)**.** Find the arc length of the helix $\mathbf{r}(t) = (a\cos t, a\sin t, bt)$ from $t = 0$ to $t = 2\pi$.

Solution:

$$\begin{aligned}
\mathbf{r}'(t) &= (-a\sin t, a\cos t, b) \\
|\mathbf{r}'(t)| &= \sqrt{a^2\sin^2 t + a^2\cos^2 t + b^2} = \sqrt{a^2 + b^2}
\end{aligned}$$

$$\begin{aligned}
L &= \int_0^{2\pi} \sqrt{a^2 + b^2}\, dt \\
&= \sqrt{a^2 + b^2} \cdot 2\pi \\
&= 2\pi\sqrt{a^2 + b^2} \text{ units}
\end{aligned}$$

3.5 Surface Area

Theorem 3.5 (Surface Area)**.** If a surface S is defined by $z = f(x, y)$ over a region R in the xy-plane, where f has continuous partial derivatives, then the surface area is:

$$S = \iint_R \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\, dA$$

Proof. Consider a small rectangle in the xy-plane with sides Δx and Δy. The tangent vectors to the surface are approximately:

$$\mathbf{T}_x \approx \left(1, 0, \frac{\partial f}{\partial x}\right)\Delta x, \quad \mathbf{T}_y \approx \left(0, 1, \frac{\partial f}{\partial y}\right)\Delta y$$

The area of the parallelogram formed by these vectors is:

$$\Delta S \approx |\mathbf{T}_x \times \mathbf{T}_y| = \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\, \Delta x \Delta y$$

Integrating over the region R gives the total surface area. □

Example 3.6 (Surface Area of a Cone)**.** Find the surface area of the cone $z = \sqrt{x^2 + y^2}$ for $0 \leq z \leq h$.

Solution: The region R is the disk $x^2 + y^2 \leq h^2$.

$$\frac{\partial z}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}}, \quad \frac{\partial z}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}}$$

$$1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = 1 + \frac{x^2}{x^2 + y^2} + \frac{y^2}{x^2 + y^2} = 2$$

Using polar coordinates:

$$\begin{aligned} S &= \iint_R \sqrt{2}\, dA = \sqrt{2} \int_0^{2\pi} \int_0^h r\, dr\, d\theta \\ &= \sqrt{2} \cdot 2\pi \cdot \frac{h^2}{2} = \pi h^2 \sqrt{2} \text{ square units} \end{aligned}$$

4 Important Theorems with Proofs

4.1 Fubini's Theorem

Theorem 4.1 (Fubini's Theorem)**.** Let $f(x, y)$ be continuous on the rectangular region $R = [a, b] \times [c, d]$. Then:

$$\iint_R f(x, y)\, dA = \int_a^b \int_c^d f(x, y)\, dy\, dx = \int_c^d \int_a^b f(x, y)\, dx\, dy$$

Proof (Sketch). Define:

$$F(x) = \int_c^d f(x, y)\, dy$$

For a Riemann sum approximation of the double integral:

$$\sum_{i=1}^{m} \sum_{j=1}^{n} f(x_i^*, y_j^*) \Delta x \Delta y$$

We can rearrange this sum:

$$\sum_{i=1}^{m} \left(\sum_{j=1}^{n} f(x_i^*, y_j^*) \Delta y \right) \Delta x \approx \sum_{i=1}^{m} F(x_i^*) \Delta x$$

Taking limits as the partition becomes finer:

$$\iint_R f(x, y)\, dA = \int_a^b F(x)\, dx = \int_a^b \int_c^d f(x, y)\, dy\, dx$$

By symmetry, we can also integrate in the opposite order. □

Problem 4.1 (Application of Fubini's Theorem)**.** Evaluate $\iint_R xy^2\,dA$ where $R = [0,2]\times[1,3]$.

Solution:

$$\begin{aligned}\iint_R xy^2\,dA &= \int_0^2\int_1^3 xy^2\,dy\,dx\\ &= \int_0^2 x\left[\frac{y^3}{3}\right]_1^3 dx\\ &= \int_0^2 x\left(\frac{27}{3}-\frac{1}{3}\right)dx\\ &= \int_0^2 x\cdot\frac{26}{3}dx\\ &= \frac{26}{3}\left[\frac{x^2}{2}\right]_0^2\\ &= \frac{26}{3}\cdot 2 = \frac{52}{3}\end{aligned}$$

4.2 Change of Variables Theorem

Theorem 4.2 (Change of Variables)**.** Let $T : R_{uv} \to R_{xy}$ be a one-to-one transformation with continuous partial derivatives, where:

$$x = g(u,v), \quad y = h(u,v)$$

The Jacobian of the transformation is:

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix}\frac{\partial x}{\partial u} & \frac{\partial x}{\partial v}\\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v}\end{vmatrix}$$

Then:

$$\iint_{R_{xy}} f(x,y)\,dx\,dy = \iint_{R_{uv}} f(g(u,v),h(u,v))|J|\,du\,dv$$

Proof (Sketch). Consider a small rectangle in the uv-plane with sides Δu and Δv. Under the transformation T, this becomes a parallelogram in the xy-plane.

The vertices of the rectangle map to points whose position vectors are approximately:

$$\mathbf{r}(u,v), \quad \mathbf{r}(u+\Delta u,v), \quad \mathbf{r}(u,v+\Delta v)$$

The tangent vectors are:

$$\frac{\partial\mathbf{r}}{\partial u}\Delta u = \left(\frac{\partial x}{\partial u},\frac{\partial y}{\partial u}\right)\Delta u$$

$$\frac{\partial\mathbf{r}}{\partial v}\Delta v = \left(\frac{\partial x}{\partial v},\frac{\partial y}{\partial v}\right)\Delta v$$

The area of the parallelogram is:

$$\Delta A = \left|\frac{\partial\mathbf{r}}{\partial u}\times\frac{\partial\mathbf{r}}{\partial v}\right|\Delta u\Delta v = |J|\Delta u\Delta v$$

Therefore: $dx\,dy = |J|\,du\,dv$ □

Problem 4.2 (Polar Coordinate Transformation)**.** Verify that the Jacobian for polar coordinates is $J = r$.

Solution: For $x = r\cos\theta$, $y = r\sin\theta$:

$$\begin{aligned} J &= \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} \\ &= \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} \\ &= r\cos^2\theta + r\sin^2\theta = r \end{aligned}$$

Therefore: $dx\,dy = r\,dr\,d\theta$

5 Practice Problems with Detailed Solutions

5.1 Problem Set 1: Area Calculations

Problem 5.1. Find the area of the region bounded by $y = \sin x$ and $y = \cos x$ for $0 \le x \le \pi/2$.

Solution: Find intersection: $\sin x = \cos x \implies \tan x = 1 \implies x = \pi/4$

For $0 \le x \le \pi/4$: $\cos x \ge \sin x$

For $\pi/4 \le x \le \pi/2$: $\sin x \ge \cos x$

$$\begin{aligned} A &= \int_0^{\pi/4} (\cos x - \sin x)\,dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x)\,dx \\ &= [\sin x + \cos x]_0^{\pi/4} + [-\cos x - \sin x]_{\pi/4}^{\pi/2} \\ &= \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} - 0 - 1\right) + \left(0 - 1 + \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right) \\ &= \sqrt{2} - 1 + \sqrt{2} - 1 = 2\sqrt{2} - 2 = 2(\sqrt{2} - 1) \text{ square units} \end{aligned}$$

Problem 5.2. Find the area of the region inside the circle $r = 2\cos\theta$ and outside the circle $r = 1$.

Solution: Intersection: $2\cos\theta = 1 \implies \cos\theta = 1/2 \implies \theta = \pm\pi/3$

By symmetry:

$$\begin{aligned}
A &= 2\int_0^{\pi/3}\int_1^{2\cos\theta} r\,dr\,d\theta \\
&= 2\int_0^{\pi/3}\left[\frac{r^2}{2}\right]_1^{2\cos\theta} d\theta \\
&= \int_0^{\pi/3}(4\cos^2\theta - 1)d\theta \\
&= \int_0^{\pi/3}(2 + 2\cos(2\theta) - 1)d\theta \\
&= \int_0^{\pi/3}(1 + 2\cos(2\theta))d\theta \\
&= [\theta + \sin(2\theta)]_0^{\pi/3} \\
&= \frac{\pi}{3} + \sin\frac{2\pi}{3} = \frac{\pi}{3} + \frac{\sqrt{3}}{2} \text{ square units}
\end{aligned}$$

Problem 5.3. Find the area of the region bounded by the curves $x = y^2 - 4$ and $x = 2y - y^2$.

Solution: Find intersections: $y^2 - 4 = 2y - y^2 \implies 2y^2 - 2y - 4 = 0 \implies y^2 - y - 2 = 0$
$(y-2)(y+1) = 0 \implies y = -1, 2$
For $-1 \le y \le 2$: $(2y - y^2) \ge (y^2 - 4)$

$$\begin{aligned}
A &= \int_{-1}^{2}[(2y - y^2) - (y^2 - 4)]dy \\
&= \int_{-1}^{2}(2y - 2y^2 + 4)dy \\
&= \left[y^2 - \frac{2y^3}{3} + 4y\right]_{-1}^{2} \\
&= \left(4 - \frac{16}{3} + 8\right) - \left(1 + \frac{2}{3} - 4\right) \\
&= 12 - \frac{16}{3} - 1 - \frac{2}{3} + 4 \\
&= 15 - 6 = 9 \text{ square units}
\end{aligned}$$

Problem 5.4. Find the area enclosed by the lemniscate $r^2 = a^2\cos(2\theta)$.

Solution: The lemniscate exists where $\cos(2\theta) \ge 0$, i.e., $-\pi/4 \le \theta \le \pi/4$ and $3\pi/4 \le \theta \le 5\pi/4$.

By symmetry (4 identical lobes):

$$\begin{aligned}
A &= 4\int_0^{\pi/4}\int_0^{a\sqrt{\cos(2\theta)}} r\,dr\,d\theta \\
&= 4\int_0^{\pi/4}\left[\frac{r^2}{2}\right]_0^{a\sqrt{\cos(2\theta)}} d\theta \\
&= 4\int_0^{\pi/4}\frac{a^2\cos(2\theta)}{2}d\theta \\
&= 2a^2\int_0^{\pi/4}\cos(2\theta)d\theta \\
&= 2a^2\left[\frac{\sin(2\theta)}{2}\right]_0^{\pi/4} \\
&= a^2[\sin(\pi/2)-0] = a^2 \text{ square units}
\end{aligned}$$

Problem 5.5. Change the order of integration and evaluate: $\int_0^2\int_x^2 e^{y^2}dy\,dx$

Solution: Original region: $0 \le x \le 2$, $x \le y \le 2$
Changing order: $0 \le y \le 2$, $0 \le x \le y$

$$\begin{aligned}
\int_0^2\int_x^2 e^{y^2}dy\,dx &= \int_0^2\int_0^y e^{y^2}dx\,dy \\
&= \int_0^2 e^{y^2}[x]_0^y dy \\
&= \int_0^2 ye^{y^2}dy \\
&= \left[\frac{e^{y^2}}{2}\right]_0^2 \\
&= \frac{e^4-1}{2}
\end{aligned}$$

5.2 Problem Set 2: Volume Calculations

Problem 5.6. Find the volume of the solid bounded by the cylinder $x^2+y^2=4$ and the planes $z=0$ and $z=3-x$.

Solution: Using polar coordinates:

$$
\begin{aligned}
V &= \iint_R (3-x)\,dA \\
&= \int_0^{2\pi}\int_0^2 (3 - r\cos\theta) r\,dr\,d\theta \\
&= \int_0^{2\pi}\int_0^2 (3r - r^2\cos\theta) dr\,d\theta \\
&= \int_0^{2\pi} \left[\frac{3r^2}{2} - \frac{r^3\cos\theta}{3}\right]_0^2 d\theta \\
&= \int_0^{2\pi} \left(6 - \frac{8\cos\theta}{3}\right) d\theta \\
&= \left[6\theta - \frac{8\sin\theta}{3}\right]_0^{2\pi} \\
&= 12\pi \text{ cubic units}
\end{aligned}
$$

Problem 5.7. Find the volume of the solid bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.

Solution: The region is $x^2 + y^2 \leq 4$. Using polar coordinates:

$$
\begin{aligned}
V &= \iint_R (4 - (x^2 + y^2)) dA \\
&= \int_0^{2\pi}\int_0^2 (4 - r^2) r\,dr\,d\theta \\
&= \int_0^{2\pi}\int_0^2 (4r - r^3) dr\,d\theta \\
&= \int_0^{2\pi} \left[2r^2 - \frac{r^4}{4}\right]_0^2 d\theta \\
&= \int_0^{2\pi} (8-4) d\theta \\
&= 4 \cdot 2\pi = 8\pi \text{ cubic units}
\end{aligned}
$$

Problem 5.8. Find the volume of the solid bounded by the cone $z = \sqrt{x^2 + y^2}$ and the sphere $x^2 + y^2 + z^2 = 1$.

Solution: Use spherical coordinates. The cone is $\phi = \pi/4$ and sphere is $\rho = 1$.

$$
\begin{aligned}
V &= \int_0^{2\pi} \int_0^{\pi/4} \int_0^1 \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta \\
&= \int_0^{2\pi} d\theta \int_0^{\pi/4} \sin\phi \, d\phi \int_0^1 \rho^2 \, d\rho \\
&= 2\pi \cdot [-\cos\phi]_0^{\pi/4} \cdot \frac{1}{3} \\
&= 2\pi \cdot \left(-\frac{\sqrt{2}}{2} + 1 \right) \cdot \frac{1}{3} \\
&= \frac{2\pi}{3} \left(1 - \frac{\sqrt{2}}{2} \right) = \frac{\pi(2-\sqrt{2})}{3} \text{ cubic units}
\end{aligned}
$$

Problem 5.9. Find the volume of the solid in the first octant bounded by $x+2y+3z=6$.

Solution: The plane intersects axes at $(6,0,0)$, $(0,3,0)$, $(0,0,2)$.

From the equation: $z = \frac{6-x-2y}{3}$

Region in xy-plane: $0 \le x \le 6$, $0 \le y \le \frac{6-x}{2}$

$$
\begin{aligned}
V &= \int_0^6 \int_0^{(6-x)/2} \frac{6-x-2y}{3} dy \, dx \\
&= \frac{1}{3} \int_0^6 \left[(6-x)y - y^2 \right]_0^{(6-x)/2} dx \\
&= \frac{1}{3} \int_0^6 \left[(6-x) \cdot \frac{6-x}{2} - \frac{(6-x)^2}{4} \right] dx \\
&= \frac{1}{3} \int_0^6 \frac{(6-x)^2}{4} dx \\
&= \frac{1}{12} \int_0^6 (6-x)^2 dx \\
&= \frac{1}{12} \left[-\frac{(6-x)^3}{3} \right]_0^6 \\
&= \frac{1}{36} \cdot 216 = 6 \text{ cubic units}
\end{aligned}
$$

Problem 5.10. Find the volume of the solid bounded by $x^2+y^2=4$, $z=0$, and $z=x+2$.

Solution: Using polar coordinates: $x = r\cos\theta$, circle is $r=2$.

$$
\begin{aligned}
V &= \int_0^{2\pi} \int_0^2 (r\cos\theta + 2) r \, dr \, d\theta \\
&= \int_0^{2\pi} \int_0^2 (r^2 \cos\theta + 2r) dr \, d\theta \\
&= \int_0^{2\pi} \left[\frac{r^3 \cos\theta}{3} + r^2 \right]_0^2 d\theta \\
&= \int_0^{2\pi} \left(\frac{8\cos\theta}{3} + 4 \right) d\theta \\
&= \left[\frac{8\sin\theta}{3} + 4\theta \right]_0^{2\pi} \\
&= 8\pi \text{ cubic units}
\end{aligned}
$$

6 EEE Application Problems with Solutions

6.1 Electric Charge Distribution Problems

Problem 6.1 (Total Charge on a Plate)**.** A thin rectangular plate occupies the region $0 \le x \le 2$, $0 \le y \le 3$ (in meters). The charge density (in C/m^2) at point (x, y) is $\sigma(x, y) = xy^2$. Find the total charge on the plate.

Solution: Total charge:

$$
\begin{aligned}
Q &= \iint_R \sigma(x, y) \, dA = \int_0^2 \int_0^3 xy^2 \, dy \, dx \\
&= \int_0^2 x \left[\frac{y^3}{3} \right]_0^3 dx \\
&= \int_0^2 x \cdot 9 \, dx \\
&= 9 \left[\frac{x^2}{2} \right]_0^2 \\
&= 9 \cdot 2 = 18 \text{ Coulombs}
\end{aligned}
$$

Problem 6.2 (Charge Distribution on a Disk)**.** A circular disk of radius a has charge density $\sigma(r) = \sigma_0(1 - r/a)$ where σ_0 is a constant. Find the total charge.

Solution: Using polar coordinates:

$$\begin{aligned}
Q &= \int_0^{2\pi} \int_0^a \sigma_0 \left(1 - \frac{r}{a}\right) r\, dr\, d\theta \\
&= 2\pi\sigma_0 \int_0^a \left(r - \frac{r^2}{a}\right) dr \\
&= 2\pi\sigma_0 \left[\frac{r^2}{2} - \frac{r^3}{3a}\right]_0^a \\
&= 2\pi\sigma_0 \left(\frac{a^2}{2} - \frac{a^2}{3}\right) \\
&= 2\pi\sigma_0 \cdot \frac{a^2}{6} = \frac{\pi a^2 \sigma_0}{3} \text{ Coulombs}
\end{aligned}$$

6.2 Electric Field and Potential Problems

Problem 6.3 (Electric Potential Due to Charged Disk)**.** Find the electric potential at a point P at height h above the center of a uniformly charged disk of radius R with charge density σ.

Solution: The potential at P due to a charge element $dq = \sigma r\, dr\, d\theta$ at distance $\sqrt{r^2 + h^2}$ is:

$$dV = \frac{1}{4\pi\epsilon_0} \frac{\sigma r\, dr\, d\theta}{\sqrt{r^2 + h^2}}$$

$$\begin{aligned}
V &= \frac{\sigma}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^R \frac{r}{\sqrt{r^2 + h^2}} dr\, d\theta \\
&= \frac{2\pi\sigma}{4\pi\epsilon_0} \int_0^R \frac{r}{\sqrt{r^2 + h^2}} dr \\
&= \frac{\sigma}{2\epsilon_0} \left[\sqrt{r^2 + h^2}\right]_0^R \\
&= \frac{\sigma}{2\epsilon_0} \left(\sqrt{R^2 + h^2} - h\right) \text{ Volts}
\end{aligned}$$

Problem 6.4 (Electric Field of Line Charge)**.** A line charge with linear charge density $\lambda(x) = \lambda_0 x$ extends from $x = 0$ to $x = L$ along the x-axis. Find the electric field at point $(0, d, 0)$.

Solution: Distance from element dx at $(x, 0, 0)$ to point: $r = \sqrt{x^2 + d^2}$

Field components:

$$\begin{aligned}
dE_x &= \frac{1}{4\pi\epsilon_0} \frac{\lambda_0 x\, dx}{x^2 + d^2} \cdot \frac{-x}{\sqrt{x^2 + d^2}} \\
dE_y &= \frac{1}{4\pi\epsilon_0} \frac{\lambda_0 x\, dx}{x^2 + d^2} \cdot \frac{d}{\sqrt{x^2 + d^2}}
\end{aligned}$$

$$
\begin{aligned}
E_y &= \frac{\lambda_0 d}{4\pi\epsilon_0}\int_0^L \frac{x}{(x^2+d^2)^{3/2}}dx \\
&= \frac{\lambda_0 d}{4\pi\epsilon_0}\left[-\frac{1}{\sqrt{x^2+d^2}}\right]_0^L \\
&= \frac{\lambda_0 d}{4\pi\epsilon_0}\left(\frac{1}{d}-\frac{1}{\sqrt{L^2+d^2}}\right) \text{ N/C}
\end{aligned}
$$

6.3 Current and Power Distribution

Problem 6.5 (Current Through a Cross-Section)**.** Current density in a cylindrical wire of radius a is $\mathbf{J} = J_0(1 - r^2/a^2)\hat{z}$ where r is distance from axis. Find total current.

Solution:

$$
\begin{aligned}
I &= \iint_A \mathbf{J}\cdot d\mathbf{A} = \int_0^{2\pi}\int_0^a J_0\left(1-\frac{r^2}{a^2}\right) r\,dr\,d\theta \\
&= 2\pi J_0 \int_0^a \left(r - \frac{r^3}{a^2}\right) dr \\
&= 2\pi J_0 \left[\frac{r^2}{2} - \frac{r^4}{4a^2}\right]_0^a \\
&= 2\pi J_0 \left(\frac{a^2}{2} - \frac{a^2}{4}\right) \\
&= \frac{\pi a^2 J_0}{2} \text{ Amperes}
\end{aligned}
$$

Problem 6.6 (Power Dissipation in a Resistive Plate)**.** A thin rectangular conducting plate ($0 \le x \le L$, $0 \le y \le W$) has resistance per unit area $R(x, y) = R_0(1 + x/L)$ and current density $J(x, y) = J_0$. Find total power dissipated.

Solution: Power density: $p(x, y) = J^2 R(x, y) = J_0^2 R_0(1 + x/L)$

$$
\begin{aligned}
P &= \int_0^L \int_0^W J_0^2 R_0 \left(1+\frac{x}{L}\right) dy\,dx \\
&= J_0^2 R_0 W \int_0^L \left(1+\frac{x}{L}\right) dx \\
&= J_0^2 R_0 W \left[x + \frac{x^2}{2L}\right]_0^L \\
&= J_0^2 R_0 W \left(L + \frac{L}{2}\right) \\
&= \frac{3J_0^2 R_0 W L}{2} \text{ Watts}
\end{aligned}
$$

6.4 Moment of Inertia for Rotating Machinery

Problem 6.7 (Moment of Inertia of Disk)**.** A circular disk of radius R and mass M has uniform density. Find its moment of inertia about the central axis perpendicular to the disk.

Solution: Mass density: $\rho = \frac{M}{\pi R^2}$

$$\begin{aligned} I &= \iint_R r^2 \rho \, dA \\ &= \rho \int_0^{2\pi} \int_0^R r^2 \cdot r \, dr \, d\theta \\ &= 2\pi\rho \int_0^R r^3 \, dr \\ &= 2\pi\rho \left[\frac{r^4}{4} \right]_0^R \\ &= 2\pi \cdot \frac{M}{\pi R^2} \cdot \frac{R^4}{4} \\ &= \frac{MR^2}{2} \text{ kg·m}^2 \end{aligned}$$

Problem 6.8 (Moment of Inertia of Cylinder)**.** Find the moment of inertia of a solid cylinder of radius R, height h, and uniform density ρ about its central axis.

Solution: Mass: $M = \rho\pi R^2 h$

$$\begin{aligned} I &= \int_0^h \int_0^{2\pi} \int_0^R r^2 \rho \cdot r \, dr \, d\theta \, dz \\ &= \rho h \cdot 2\pi \int_0^R r^3 \, dr \\ &= 2\pi\rho h \left[\frac{r^4}{4} \right]_0^R \\ &= 2\pi\rho h \cdot \frac{R^4}{4} \\ &= \frac{\pi\rho h R^4}{2} = \frac{MR^2}{2} \text{ kg·m}^2 \end{aligned}$$

6.5 Magnetic Field Applications

Problem 6.9 (Magnetic Field of Current Loop)**.** Find the magnetic field at distance z above the center of a circular current loop of radius a carrying current I.

Solution: Using Biot-Savart law, consider element $d\mathbf{l} = a \, d\theta \hat{\boldsymbol{\theta}}$ at angle θ.

Distance from element to point: $r = \sqrt{a^2 + z^2}$

By symmetry, only z-component survives:

$$\begin{aligned} B_z &= \frac{\mu_0 I}{4\pi} \int_0^{2\pi} \frac{a \, d\theta \cdot a}{(a^2 + z^2)^{3/2}} \\ &= \frac{\mu_0 I a^2}{4\pi (a^2 + z^2)^{3/2}} \int_0^{2\pi} d\theta \\ &= \frac{\mu_0 I a^2}{2(a^2 + z^2)^{3/2}} \text{ Tesla} \end{aligned}$$

Problem 6.10 (Induced EMF in Rotating Coil)**.** A rectangular coil of dimensions $a \times b$ rotates with angular velocity ω in a uniform magnetic field $\mathbf{B} = B_0\hat{z}$. The coil makes angle $\theta = \omega t$ with the field. Find the induced EMF.

Solution: Magnetic flux through coil:

$$\Phi(t) = \mathbf{B} \cdot \mathbf{A} = B_0 ab \cos(\omega t)$$

Induced EMF (Faraday's law):

$$\begin{aligned}\mathcal{E} &= -\frac{d\Phi}{dt} \\ &= -\frac{d}{dt}[B_0 ab \cos(\omega t)] \\ &= B_0 ab\omega \sin(\omega t) \text{ Volts}\end{aligned}$$

Peak EMF: $\mathcal{E}_{\max} = B_0 ab\omega$

7 Summary

7.1 Key Concepts Summary

Multiple Integrals - Essential Formulas

1. **Area Calculations:**
   - Cartesian: $A = \iint_R dA = \int_a^b \int_{g_1(x)}^{g_2(x)} dy\, dx$
   - Polar: $A = \int_\alpha^\beta \int_{r_1(\theta)}^{r_2(\theta)} r\, dr\, d\theta$
2. **Volume Calculations:**
   - Under surface: $V = \iint_R f(x, y)\, dA$
   - Triple integral: $V = \iiint_E dV$
   - Cylindrical: $dV = r\, dr\, d\theta\, dz$
   - Spherical: $dV = \rho^2 \sin\phi\, d\rho\, d\theta\, d\phi$
3. **Arc Length:**

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt$$

4. **Surface Area:**

$$S = \iint_R \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\, dA$$

5. **Jacobian Transformations:**
   - Polar: $J = r$
   - Cylindrical: $J = r$
   - Spherical: $J = \rho^2 \sin\phi$

7.2 Problem-Solving Strategy

1. **Identify the region:** Sketch the region and determine its boundaries
2. **Choose coordinates:** Select the most appropriate coordinate system
3. **Set up limits:** Determine the limits of integration carefully
4. **Write the integral:** Express the desired quantity as an integral
5. **Evaluate:** Integrate step by step, checking each calculation
6. **Verify:** Check if the answer makes physical sense

7.3 Common EEE Applications

- **Charge distributions:** Use double integrals over charged surfaces
- **Electric/Magnetic fields:** Integrate contributions from charge/current elements
- **Power calculations:** Integrate power density over regions
- **Moments of inertia:** Essential for motor and generator design
- **Heat distribution:** Solve thermal problems in electrical devices
- **Flux calculations:** Compute electric and magnetic flux through surfaces

7.4 Important Theorems to Remember

- **Fubini's Theorem:** Allows changing order of integration
- **Change of Variables:** Transforms integrals using Jacobian
- **Green's Theorem:** Relates line and double integrals
- **Divergence Theorem:** Connects triple and surface integrals
- **Stokes' Theorem:** Relates surface and line integrals

7.5 Tips for Success

1. Always sketch the region of integration
2. Check if symmetry can simplify calculations
3. Be careful with limits when changing order of integration
4. Remember to include the Jacobian when changing coordinates
5. Verify units in engineering applications
6. Practice recognizing when to use different coordinate systems
7. Double-check signs, especially in vector calculus applications

"Mathematics is the language in which God has written the universe."
— Galileo Galilei

End of Class Notes
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